

Chapter 4

TRANSIENT HEAT CONDUCTION

Lumped System Analysis

4-1C In heat transfer analysis, some bodies are observed to behave like a "lump" whose entire body temperature remains essentially uniform at all times during a heat transfer process. The temperature of such bodies can be taken to be a function of time only. Heat transfer analysis which utilizes this idealization is known as the lumped system analysis. It is applicable when the Biot number (the ratio of conduction resistance within the body to convection resistance at the surface of the body) is less than or equal to 0.1.

4-2C The lumped system analysis is more likely to be applicable for the body cooled naturally since the Biot number is proportional to the convection heat transfer coefficient, which is proportional to the air velocity. Therefore, the Biot number is more likely to be less than 0.1 for the case of natural convection.

4-3C The lumped system analysis is more likely to be applicable for the body allowed to cool in the air since the Biot number is proportional to the convection heat transfer coefficient, which is larger in water than it is in air because of the larger thermal conductivity of water. Therefore, the Biot number is more likely to be less than 0.1 for the case of the solid cooled in the air

4-4C The temperature drop of the potato during the second minute will be less than 4°C since the temperature of a body approaches the temperature of the surrounding medium asymptotically, and thus it changes rapidly at the beginning, but slowly later on.

4-5C The temperature rise of the potato during the second minute will be less than 5°C since the temperature of a body approaches the temperature of the surrounding medium asymptotically, and thus it changes rapidly at the beginning, but slowly later on.

4-6C Biot number represents the ratio of conduction resistance within the body to convection resistance at the surface of the body. The Biot number is more likely to be larger for poorly conducting solids since such bodies have larger resistances against heat conduction.

4-7C The heat transfer is proportional to the surface area. Two half pieces of the roast have a much larger surface area than the single piece and thus a higher rate of heat transfer. As a result, the two half pieces will cook much faster than the single large piece.

4-8C The cylinder will cool faster than the sphere since heat transfer rate is proportional to the surface area, and the sphere has the smallest area for a given volume.

4-9C The lumped system analysis is more likely to be applicable in air than in water since the convection heat transfer coefficient and thus the Biot number is much smaller in air.

4-10C The lumped system analysis is more likely to be applicable for a golden apple than for an actual apple since the thermal conductivity is much larger and thus the Biot number is much smaller for gold.

4-11C The lumped system analysis is more likely to be applicable to slender bodies than the well-rounded bodies since the characteristic length (ratio of volume to surface area) and thus the Biot number is much smaller for slender bodies.

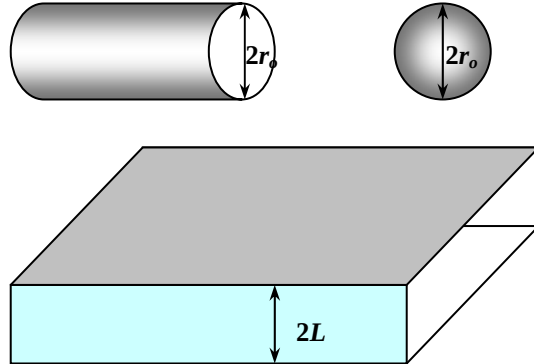
4-12 Relations are to be obtained for the characteristic lengths of a large plane wall of thickness $2L$, a very long cylinder of radius r_o and a sphere of radius r_o

Analysis Relations for the characteristic lengths of a large plane wall of thickness $2L$, a very long cylinder of radius r_o and a sphere of radius r_o are

$$L_{c,wall} = \frac{V}{A_{surface}} = \frac{2LA}{2A} = L$$

$$L_{c,cylinder} = \frac{V}{A_{surface}} = \frac{\pi r_o^2 h}{2\pi r_o h} = \frac{r_o}{2}$$

$$L_{c,sphere} = \frac{V}{A_{surface}} = \frac{4\pi r_o^3 / 3}{4\pi r_o^2} = \frac{r_o}{3}$$

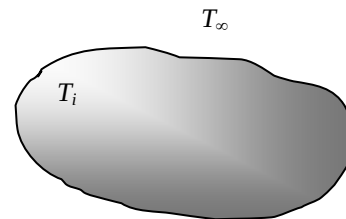


4-13 A relation for the time period for a lumped system to reach the average temperature $(T_i + T_\infty) / 2$ is to be obtained.

Analysis The relation for time period for a lumped system to reach the average temperature $(T_i + T_\infty) / 2$ can be determined as

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{\frac{T_i + T_\infty}{2} - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{T_i - T_\infty}{2(T_i - T_\infty)} = e^{-bt} \longrightarrow \frac{1}{2} = e^{-bt}$$

$$-bt = -\ln 2 \longrightarrow t = \frac{\ln 2}{b} = \frac{0.693}{b}$$



4-14 The temperature of a gas stream is to be measured by a thermocouple. The time it takes to register 99 percent of the initial ΔT is to be determined.

Assumptions **1** The junction is spherical in shape with a diameter of $D = 0.0012$ m. **2** The thermal properties of the junction are constant. **3** The heat transfer coefficient is constant and uniform over the entire surface. **4** Radiation effects are negligible. **5** The Biot number is $Bi < 0.1$ so that the lumped system analysis is applicable (this assumption will be verified).

Properties The properties of the junction are given to be $k = 35$ W/m.°C, $\rho = 8500$ kg/m³, and $C_p = 320$ J/kg.°C.

Analysis The characteristic length of the junction and the Biot number are

$$L_c = \frac{V}{A_{\text{surface}}} = \frac{\pi D^3 / 6}{\pi D^2} = \frac{D}{6} = \frac{0.0012 \text{ m}}{6} = 0.0002 \text{ m}$$

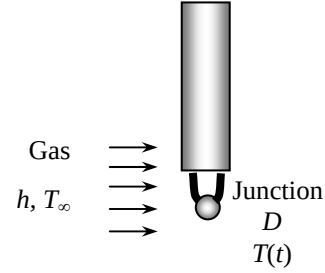
$$Bi = \frac{hL_c}{k} = \frac{(65 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0002 \text{ m})}{(35 \text{ W/m} \cdot ^\circ\text{C})} = 0.00037 < 0.1$$

Since $Bi < 0.1$, the lumped system analysis is applicable. Then the time period for the thermocouple to read 99% of the initial temperature difference is determined from

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = 0.01$$

$$b = \frac{hA}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{65 \text{ W/m}^2 \cdot ^\circ\text{C}}{(8500 \text{ kg/m}^3)(320 \text{ J/kg} \cdot ^\circ\text{C})(0.0002 \text{ m})} = 0.1195 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow 0.01 = e^{-(0.1195 \text{ s}^{-1})t} \longrightarrow t = 38.5 \text{ s}$$



4-15E A number of brass balls are to be quenched in a water bath at a specified rate. The temperature of the balls after quenching and the rate at which heat needs to be removed from the water in order to keep its temperature constant are to be determined.

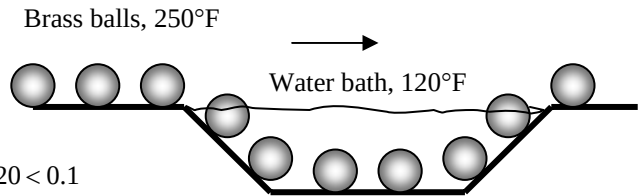
Assumptions 1 The balls are spherical in shape with a radius of $r_0 = 1$ in. 2 The thermal properties of the balls are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Biot number is $Bi < 0.1$ so that the lumped system analysis is applicable (this assumption will be verified).

Properties The thermal conductivity, density, and specific heat of the brass balls are given to be $k = 64.1$ Btu/h.ft.°F, $\rho = 532$ lbm/ft³, and $C_p = 0.092$ Btu/lbm.°F.

Analysis (a) The characteristic length and the Biot number for the brass balls are

$$L_c = \frac{V}{A_s} = \frac{\pi D^3 / 6}{\pi D^2} = \frac{D}{6} = \frac{2 / 12 \text{ ft}}{6} = 0.02778 \text{ ft}$$

$$Bi = \frac{hL_c}{k} = \frac{(42 \text{ Btu/h.ft}^2 \cdot \text{°F})(0.02778 \text{ ft})}{(64.1 \text{ Btu/h.ft.°F})} = 0.01820 < 0.1$$



The lumped system analysis is applicable since $Bi < 0.1$. Then the temperature of the balls after quenching becomes

$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{42 \text{ Btu/h.ft}^2 \cdot \text{°F}}{(532 \text{ lbm/ft}^3)(0.092 \text{ Btu/lbm.°F})(0.02778 \text{ ft})} = 30.9 \text{ h}^{-1} = 0.00858 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{T(t) - 120}{250 - 120} = e^{-(0.00858 \text{ s}^{-1})(120 \text{ s})} \longrightarrow T(t) = \mathbf{166 \text{ °F}}$$

(b) The total amount of heat transfer from a ball during a 2-minute period is

$$m = \rho V = \rho \frac{\pi D^3}{6} = (532 \text{ lbm/ft}^3) \frac{\pi (2 / 12 \text{ ft})^3}{6} = 1.290 \text{ lbm}$$

$$Q = mC_p [T_i - T(t)] = (1.29 \text{ lbm})(0.092 \text{ Btu/lbm.°F})(250 - 166) \text{ °F} = 9.97 \text{ Btu}$$

Then the rate of heat transfer from the balls to the water becomes

$$\dot{Q}_{total} = \dot{n}_{ball} Q_{ball} = (120 \text{ balls/min}) \times (9.97 \text{ Btu}) = \mathbf{1196 \text{ Btu/min}}$$

Therefore, heat must be removed from the water at a rate of 1196 Btu/min in order to keep its temperature constant at 120 °F.

4-16E A number of aluminum balls are to be quenched in a water bath at a specified rate. The temperature of balls after quenching and the rate at which heat needs to be removed from the water in order to keep its temperature constant are to be determined.

Assumptions 1 The balls are spherical in shape with a radius of $r_0 = 1$ in. 2 The thermal properties of the balls are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Biot number is $Bi < 0.1$ so that the lumped system analysis is applicable (this assumption will be verified).

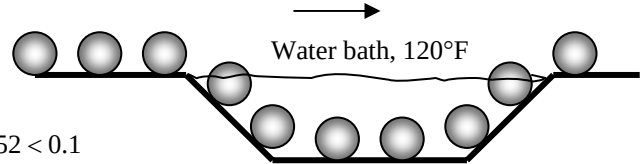
Properties The thermal conductivity, density, and specific heat of the aluminum balls are $k = 137$ Btu/h.ft.°F, $\rho = 168$ lbm/ft³, and $C_p = 0.216$ Btu/lbm.°F (Table A-3E).

Analysis (a) The characteristic length and the Biot number for the aluminum balls are

$$L_c = \frac{V}{A} = \frac{\pi D^3 / 6}{\pi D^2} = \frac{D}{6} = \frac{2 / 12 \text{ ft}}{6} = 0.02778 \text{ ft}$$

$$Bi = \frac{hL_c}{k} = \frac{(42 \text{ Btu/h.ft}^2 \cdot \text{°F})(0.02778 \text{ ft})}{(137 \text{ Btu/h.ft.°F})} = 0.00852 < 0.1$$

Aluminum balls, 250°F



The lumped system analysis is applicable since $Bi < 0.1$. Then the temperature of the balls after quenching becomes

$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{42 \text{ Btu/h.ft}^2 \cdot \text{°F}}{(168 \text{ lbm/ft}^3)(0.216 \text{ Btu/lbm.°F})(0.02778 \text{ ft})} = 41.66 \text{ h}^{-1} = 0.01157 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{T(t) - 120}{250 - 120} = e^{-(0.01157 \text{ s}^{-1})(120 \text{ s})} \longrightarrow T(t) = 152^\circ\text{F}$$

(b) The total amount of heat transfer from a ball during a 2-minute period is

$$m = \rho V = \rho \frac{\pi D^3}{6} = (168 \text{ lbm/ft}^3) \frac{\pi (2 / 12 \text{ ft})^3}{6} = 0.4072 \text{ lbm}$$

$$Q = mC_p [T_i - T(t)] = (0.4072 \text{ lbm})(0.216 \text{ Btu/lbm.°F})(250 - 152)^\circ\text{F} = 8.62 \text{ Btu}$$

Then the rate of heat transfer from the balls to the water becomes

$$\dot{Q}_{total} = \dot{n}_{ball} Q_{ball} = (120 \text{ balls/min}) \times (8.62 \text{ Btu}) = \mathbf{1034 \text{ Btu/min}}$$

Therefore, heat must be removed from the water at a rate of 1034 Btu/min in order to keep its temperature constant at 120°F.

4-17 Milk in a thin-walled glass container is to be warmed up by placing it into a large pan filled with hot water. The warming time of the milk is to be determined.

Assumptions 1 The glass container is cylindrical in shape with a radius of $r_0 = 3$ cm. 2 The thermal properties of the milk are taken to be the same as those of water. 3 Thermal properties of the milk are constant at room temperature. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Biot number in this case is large (much larger than 0.1). However, the lumped system analysis is still applicable since the milk is stirred constantly, so that its temperature remains uniform at all times.

Properties The thermal conductivity, density, and specific heat of the milk at 20°C are $k = 0.607$ W/m.°C, $\rho = 998$ kg/m³, and $C_p = 4.182$ kJ/kg.°C (Table A-9).

Analysis The characteristic length and Biot number for the glass of milk are

$$L_c = \frac{V}{A_s} = \frac{\pi r_o^2 L}{2\pi r_o L + 2\pi r_o^2} = \frac{\pi (0.03 \text{ m})^2 (0.07 \text{ m})}{2\pi (0.03 \text{ m})(0.07 \text{ m}) + 2\pi (0.03 \text{ m})^2} = 0.01050 \text{ m}$$

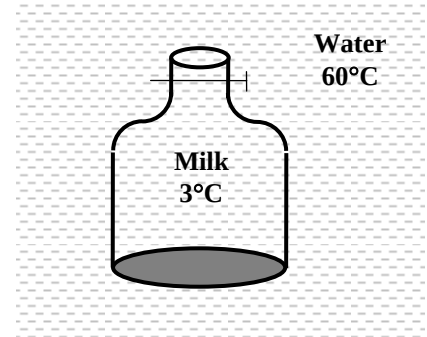
$$Bi = \frac{hL_c}{k} = \frac{(120 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0105 \text{ m})}{(0.607 \text{ W/m} \cdot ^\circ\text{C})} = 2.076 > 0.1$$

For the reason explained above we can use the lumped system analysis to determine how long it will take for the milk to warm up to 38°C:

$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{120 \text{ W/m}^2 \cdot ^\circ\text{C}}{(998 \text{ kg/m}^3)(4182 \text{ J/kg} \cdot ^\circ\text{C})(0.0105 \text{ m})} = 0.002738 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{38 - 60}{3 - 60} = e^{-(0.002738 \text{ s}^{-1})t} \longrightarrow t = 348 \text{ s} = 5.8 \text{ min}$$

Therefore, it will take about 6 minutes to warm the milk from 3 to 38°C.



4-18 A thin-walled glass containing milk is placed into a large pan filled with hot water to warm up the milk. The warming time of the milk is to be determined.

Assumptions 1 The glass container is cylindrical in shape with a radius of $r_0 = 3$ cm. 2 The thermal properties of the milk are taken to be the same as those of water. 3 Thermal properties of the milk are constant and uniform over the entire surface. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Biot number in this case is large (much larger than 0.1). However, the lumped system analysis is still applicable since the milk is stirred constantly, so that its temperature remains uniform at all times.

Properties The thermal conductivity, density, and specific heat of the milk at 20°C are $k = 0.607$ W/m·°C, $\rho = 998$ kg/m³, and $C_p = 4.182$ kJ/kg·°C (Table A-9).

Analysis The characteristic length and Biot number for the glass of milk are

$$L_c = \frac{V}{A_s} = \frac{\pi r_o^2 L}{2\pi r_o L + 2\pi r_o^2} = \frac{\pi (0.03 \text{ m})^2 (0.07 \text{ m})}{2\pi (0.03 \text{ m})(0.07 \text{ m}) + 2\pi (0.03 \text{ m})^2} = 0.01050 \text{ m}$$

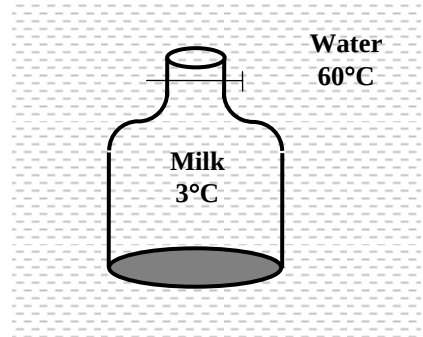
$$Bi = \frac{hL_c}{k} = \frac{(240 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0105 \text{ m})}{(0.607 \text{ W/m} \cdot ^\circ\text{C})} = 4.15 > 0.1$$

For the reason explained above we can use the lumped system analysis to determine how long it will take for the milk to warm up to 38°C:

$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{240 \text{ W/m}^2 \cdot ^\circ\text{C}}{(998 \text{ kg/m}^3)(4182 \text{ J/kg} \cdot ^\circ\text{C})(0.0105 \text{ m})} = 0.005477 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{38 - 60}{3 - 60} = e^{-(0.005477 \text{ s}^{-1})t} \longrightarrow t = 174 \text{ s} = 2.9 \text{ min}$$

Therefore, it will take about 3 minutes to warm the milk from 3 to 38°C.

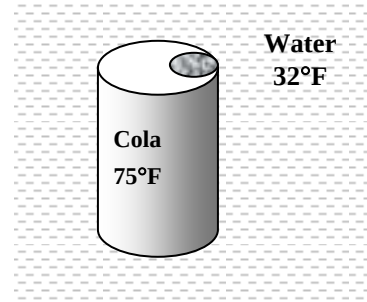


4-19E A person shakes a can of drink in a iced water to cool it. The cooling time of the drink is to be determined.

Assumptions 1 The can containing the drink is cylindrical in shape with a radius of $r_0 = 1.25$ in. 2 The thermal properties of the milk are taken to be the same as those of water. 3 Thermal properties of the milk are constant at room temperature. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Biot number in this case is large (much larger than 0.1). However, the lumped system analysis is still applicable since the milk is stirred constantly, so that its temperature remains uniform at all times.

Properties The density and specific heat of water at room temperature are $\rho = 62.22$ lbm/ft³, and $C_p = 0.999$ Btu/lbm.°F (Table A-9E).

Analysis Application of lumped system analysis in this case gives



$$L_c = \frac{V}{A_s} = \frac{\pi r_o^2 L}{2\pi r_o L + 2\pi r_o^2} = \frac{\pi (1.25/12 \text{ ft})^2 (5/12 \text{ ft})}{2\pi (1.25/12 \text{ ft})(5/12 \text{ ft}) + 2\pi (1.25/12 \text{ ft})^2} = 0.04167 \text{ ft}$$

$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{30 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F}}{(62.22 \text{ lbm/ft}^3)(0.999 \text{ Btu/lbm.}^\circ\text{F})(0.04167 \text{ ft})} = 11.583 \text{ h}^{-1} = 0.00322 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{45 - 32}{80 - 32} = e^{-(0.00322 \text{ s}^{-1})t} \longrightarrow t = \mathbf{406 \text{ s}}$$

Therefore, it will take 7 minutes and 46 seconds to cool the canned drink to 45°F.

4-20 An iron whose base plate is made of an aluminum alloy is turned on. The time for the plate temperature to reach 140°C and whether it is realistic to assume the plate temperature to be uniform at all times are to be determined.

Assumptions 1 85 percent of the heat generated in the resistance wires is transferred to the plate. 2 The thermal properties of the plate are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface.

Properties The density, specific heat, and thermal diffusivity of the aluminum alloy plate are given to be $\rho = 2770 \text{ kg/m}^3$, $C_p = 875 \text{ kJ/kg}\cdot^\circ\text{C}$, and $\alpha = 7.3 \times 10^{-5} \text{ m}^2/\text{s}$. The thermal conductivity of the plate can be determined from $\alpha = k/(\rho C_p) = 177 \text{ W/m}\cdot^\circ\text{C}$ (or it can be read from Table A-3).

Analysis The mass of the iron's base plate is

$$m = \rho V = \rho LA = (2770 \text{ kg/m}^3)(0.005 \text{ m})(0.03 \text{ m}^2) = 0.4155 \text{ kg}$$

Noting that only 85 percent of the heat generated is transferred to the plate, the rate of heat transfer to the iron's base plate is

$$\dot{Q}_{\text{in}} = 0.85 \times 1000 \text{ W} = 850 \text{ W}$$

The temperature of the plate, and thus the rate of heat transfer from the plate, changes during the process. Using the average plate temperature, the average rate of heat loss from the plate is determined from

$$\dot{Q}_{\text{loss}} = hA(T_{\text{plate, ave}} - T_{\infty}) = (12 \text{ W/m}^2\cdot^\circ\text{C})(0.03 \text{ m}^2) \left(\frac{140 + 22}{2} - 22 \right)^\circ\text{C} = 21.2 \text{ W}$$

Energy balance on the plate can be expressed as

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{plate}} \rightarrow \dot{Q}_{\text{in}} \Delta t - \dot{Q}_{\text{out}} \Delta t = \Delta E_{\text{plate}} = mC_p \Delta T_{\text{plate}}$$

Solving for Δt and substituting,

$$\Delta t = \frac{mC_p \Delta T_{\text{plate}}}{\dot{Q}_{\text{in}} - \dot{Q}_{\text{out}}} = \frac{(0.4155 \text{ kg})(875 \text{ J/kg}\cdot^\circ\text{C})(140 - 22)^\circ\text{C}}{(850 - 21.2) \text{ J/s}} = 51.8 \text{ s}$$

which is the time required for the plate temperature to reach 140°C. To determine whether it is realistic to assume the plate temperature to be uniform at all times, we need to calculate the Biot number,

$$L_c = \frac{V}{A_s} = \frac{LA}{A} = L = 0.005 \text{ m}$$

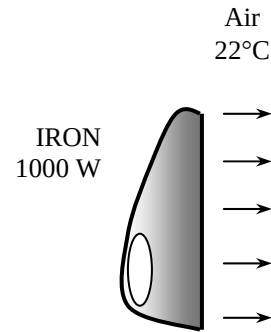
$$Bi = \frac{hL_c}{k} = \frac{(12 \text{ W/m}^2\cdot^\circ\text{C})(0.005 \text{ m})}{(177.0 \text{ W/m}\cdot^\circ\text{C})} = 0.00034 < 0.1$$

It is realistic to assume uniform temperature for the plate since $Bi < 0.1$.

Discussion This problem can also be solved by obtaining the differential equation from an energy balance on the plate for a differential time interval, and solving the differential equation. It gives

$$T(t) = T_{\infty} + \frac{\dot{Q}_{\text{in}}}{hA} \left(1 - \exp\left(-\frac{hA}{mC_p} t\right) \right)$$

Substituting the known quantities and solving for t again gives 51.8 s.



4-21 "PROBLEM 4-21"

"GIVEN"

$$\dot{E} = 1000 \text{ [W]}$$

$$L = 0.005 \text{ [m]}$$

$$A = 0.03 \text{ [m}^2\text{]}$$

$$T_{\infty} = 22 \text{ [}^{\circ}\text{C]}$$

$$T_i = T_{\infty}$$

$$h = 12 \text{ [W/m}^2\text{-}^{\circ}\text{C}], \text{ parameter to be varied}$$

$$f_{\text{heat}} = 0.85$$

$$T_f = 140 \text{ [}^{\circ}\text{C}], \text{ parameter to be varied}$$

"PROPERTIES"

$$\rho = 2770 \text{ [kg/m}^3\text{]}$$

$$C_p = 875 \text{ [J/kg-}^{\circ}\text{C]}$$

$$\alpha = 7.3 \times 10^{-5} \text{ [m}^2\text{/s]}$$

"ANALYSIS"

$$V = L \cdot A$$

$$m = \rho \cdot V$$

$$\dot{Q}_{\text{in}} = f_{\text{heat}} \cdot \dot{E}$$

$$\dot{Q}_{\text{out}} = h \cdot A \cdot (T_{\text{ave}} - T_{\infty})$$

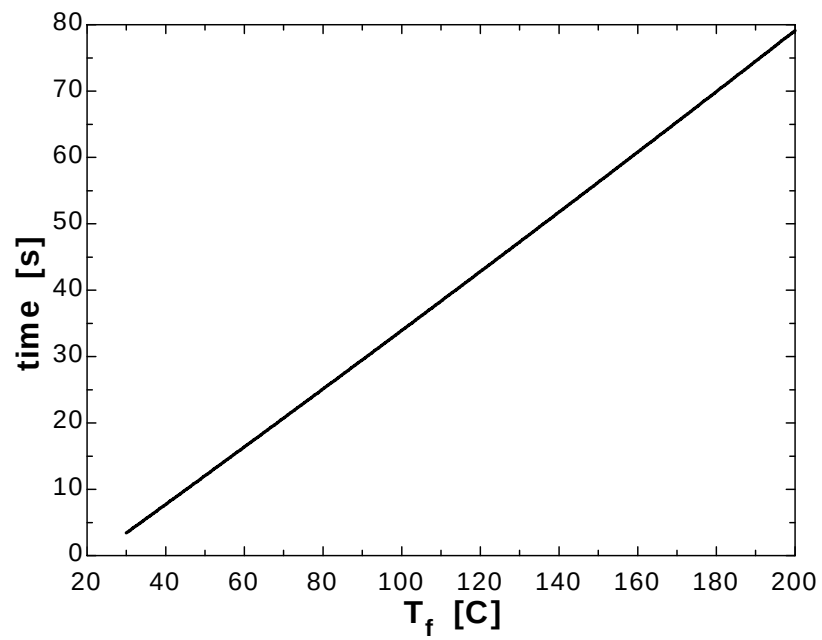
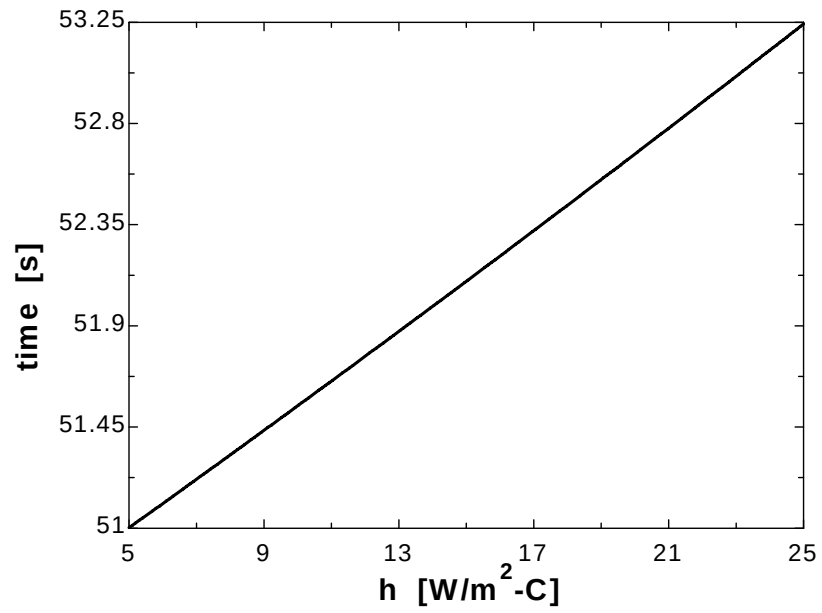
$$T_{\text{ave}} = \frac{1}{2} (T_i + T_f)$$

$$(\dot{Q}_{\text{in}} - \dot{Q}_{\text{out}}) \cdot \text{time} = m \cdot C_p \cdot (T_f - T_i) \text{ "energy balance on the plate"}$$

$h \text{ [W/m}^2\text{-}^{\circ}\text{C]}$	time [s]
5	51
7	51.22
9	51.43
11	51.65
13	51.88
15	52.1
17	52.32
19	52.55
21	52.78
23	53.01
25	53.24

$T_f \text{ [}^{\circ}\text{C]}$	time [s]
30	3.428
40	7.728
50	12.05
60	16.39
70	20.74
80	25.12
90	29.51
100	33.92
110	38.35
120	42.8
130	47.28
140	51.76
150	56.27
160	60.8
170	65.35
180	69.92
190	74.51

200	79.12
-----	-------



4-22 Ball bearings leaving the oven at a uniform temperature of 900°C are exposed to air for a while before they are dropped into the water for quenching. The time they can stand in the air before their temperature falls below 850°C is to be determined.

Assumptions **1** The bearings are spherical in shape with a radius of $r_0 = 0.6$ cm. **2** The thermal properties of the bearings are constant. **3** The heat transfer coefficient is constant and uniform over the entire surface. **4** The Biot number is $Bi < 0.1$ so that the lumped system analysis is applicable (this assumption will be verified).

Properties The thermal conductivity, density, and specific heat of the bearings are given to be $k = 15.1$ W/m.°C, $\rho = 8085$ kg/m³, and $C_p = 0.480$ kJ/kg.°F.

Analysis The characteristic length of the steel ball bearings and Biot number are

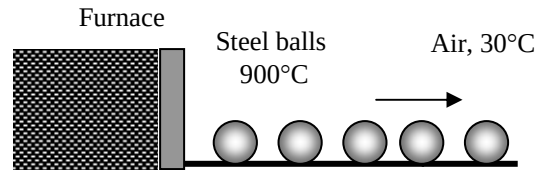
$$L_c = \frac{V}{A_s} = \frac{\pi D^3 / 6}{\pi D^2} = \frac{D}{6} = \frac{0.012 \text{ m}}{6} = 0.002 \text{ m}$$

$$Bi = \frac{hL_c}{k} = \frac{(125 \text{ W/m}^2 \cdot ^\circ\text{C})(0.002 \text{ m})}{(15.1 \text{ W/m} \cdot ^\circ\text{C})} = 0.0166 < 0.1$$

Therefore, the lumped system analysis is applicable. Then the allowable time is determined to be

$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{125 \text{ W/m}^2 \cdot ^\circ\text{C}}{(8085 \text{ kg/m}^3)(480 \text{ J/kg} \cdot ^\circ\text{C})(0.002 \text{ m})} = 0.01610 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{850 - 30}{900 - 30} = e^{-(0.0161 \text{ s}^{-1})t} \longrightarrow t = \mathbf{3.68 \text{ s}}$$



The result indicates that the ball bearing can stay in the air about 4 s before being dropped into the water.

4-23 A number of carbon steel balls are to be annealed by heating them first and then allowing them to cool slowly in ambient air at a specified rate. The time of annealing and the total rate of heat transfer from the balls to the ambient air are to be determined.

Assumptions 1 The balls are spherical in shape with a radius of $r_0 = 4$ mm. 2 The thermal properties of the balls are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Biot number is $Bi < 0.1$ so that the lumped system analysis is applicable (this assumption will be verified).

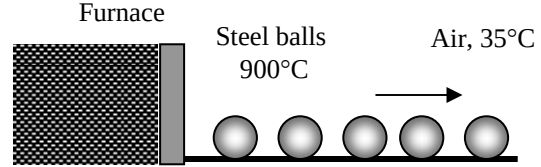
Properties The thermal conductivity, density, and specific heat of the balls are given to be $k = 54$ W/m.°C, $\rho = 7833$ kg/m³, and $C_p = 0.465$ kJ/kg.°C.

Analysis The characteristic length of the balls and the Biot number are

$$L_c = \frac{V}{A_s} = \frac{\pi D^3 / 6}{\pi D^2} = \frac{D}{6} = \frac{0.008 \text{ m}}{6} = 0.0013 \text{ m}$$

$$Bi = \frac{hL_c}{k} = \frac{(75 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0013 \text{ m})}{(54 \text{ W/m} \cdot ^\circ\text{C})} = 0.0018 < 0.1$$

Therefore, the lumped system analysis is applicable. Then the time for the annealing process is determined to be



$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{75 \text{ W/m}^2 \cdot ^\circ\text{C}}{(7833 \text{ kg/m}^3)(465 \text{ J/kg} \cdot ^\circ\text{C})(0.0013 \text{ m})} = 0.01584 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{100 - 35}{900 - 35} = e^{-(0.01584 \text{ s}^{-1})t} \longrightarrow t = 163 \text{ s} = 2.7 \text{ min}$$

The amount of heat transfer from a single ball is

$$m = \rho V = \rho \frac{\pi D^3}{6} = (7833 \text{ kg/m}^3) \frac{\pi (0.008 \text{ m})^3}{6} = 0.0021 \text{ kg}$$

$$Q = mC_p [T_f - T_i] = (0.0021 \text{ kg})(465 \text{ J/kg} \cdot ^\circ\text{C})(900 - 100)^\circ\text{C} = 781 \text{ J} = 0.781 \text{ kJ (per ball)}$$

Then the total rate of heat transfer from the balls to the ambient air becomes

$$\dot{Q} = \dot{n}_{\text{ball}} Q = (2500 \text{ balls/h}) \times (0.781 \text{ kJ/ball}) = 1,953 \text{ kJ/h} = 543 \text{ W}$$

4-24

"!PROBLEM 4-24"

"GIVEN"

D=0.008 "[m]"

"T_i=900 [C], parameter to be varied"

T_f=100 "[C]"

T_infinity=35 "[C]"

h=75 "[W/m^2-C]"

n_dot_ball=2500 "[1/h]"

"PROPERTIES"

rho=7833 "[kg/m^3]"

k=54 "[W/m-C]"

C_p=465 "[J/kg-C]"

alpha=1.474E-6 "[m^2/s]"

"ANALYSIS"

A=pi*D^2

V=pi*D^3/6

L_c=V/A

Bi=(h*L_c)/k "if Bi < 0.1, the lumped sytem analysis is applicable"

b=(h*A)/(rho*C_p*V)

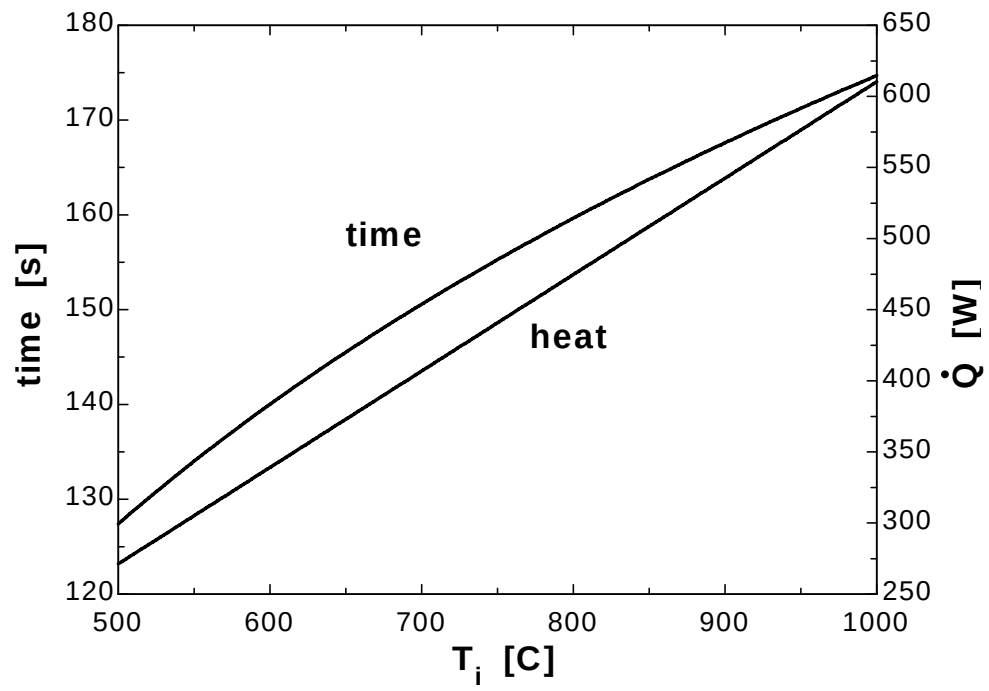
(T_f-T_infinity)/(T_i-T_infinity)=exp(-b*time)

m=rho*V

Q=m*C_p*(T_i-T_f)

Q_dot=n_dot_ball*Q*Convert(J/h, W)

T _i [C]	time [s]	Q [W]
500	127.4	271.2
550	134	305.1
600	140	339
650	145.5	372.9
700	150.6	406.9
750	155.3	440.8
800	159.6	474.7
850	163.7	508.6
900	167.6	542.5
950	171.2	576.4
1000	174.7	610.3



4-25 An electronic device is on for 5 minutes, and off for several hours. The temperature of the device at the end of the 5-min operating period is to be determined for the cases of operation with and without a heat sink.

Assumptions 1 The device and the heat sink are isothermal. 2 The thermal properties of the device and of the sink are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface.

Properties The specific heat of the device is given to be $C_p = 850 \text{ J/kg}\cdot^\circ\text{C}$. The specific heat of the aluminum sink is $903 \text{ J/kg}\cdot^\circ\text{C}$ (Table A-19), but can be taken to be $850 \text{ J/kg}\cdot^\circ\text{C}$ for simplicity in analysis.

Analysis (a) Approximate solution

This problem can be solved approximately by using an average temperature for the device when evaluating the heat loss. An energy balance on the device can be expressed as

$$E_{\text{in}} - E_{\text{out}} + E_{\text{generation}} = \Delta E_{\text{device}} \longrightarrow -\dot{Q}_{\text{out}} \Delta t + \dot{E}_{\text{generation}} \Delta t = m C_p \Delta T_{\text{device}}$$

$$\text{or,} \quad \dot{E}_{\text{generation}} \Delta t - h A_s \left(\frac{T + T_\infty}{2} - T_\infty \right) \Delta t = m C_p (T - T_\infty)$$

Substituting the given values,

$$(30 \text{ J/s})(5 \times 60 \text{ s}) - (12 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0005 \text{ m}^2) \left(\frac{T - 25}{2} \right)^\circ\text{C}(5 \times 60 \text{ s}) = (0.02 \text{ kg})(850 \text{ J/kg}\cdot^\circ\text{C})(T - 25)^\circ\text{C}$$

which gives $T = 527.8^\circ\text{C}$

If the device were attached to an aluminum heat sink, the temperature of the device would be

$$(30 \text{ J/s})(5 \times 60 \text{ s}) - (12 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0085 \text{ m}^2) \left(\frac{T - 25}{2} \right)^\circ\text{C}(5 \times 60 \text{ s}) = (0.20 + 0.02) \text{ kg} \times (850 \text{ J/kg}\cdot^\circ\text{C})(T - 25)^\circ\text{C}$$

which gives $T = 69.5^\circ\text{C}$

Note that the temperature of the electronic device drops considerably as a result of attaching it to a heat sink.

(b) Exact solution

This problem can be solved exactly by obtaining the differential equation from an energy balance on the device for a differential time interval dt . We will get

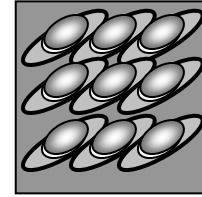
$$\frac{d(T - T_\infty)}{dt} + \frac{h A_s}{m C_p} (T - T_\infty) = \frac{\dot{E}_{\text{generation}}}{m C_p}$$

It can be solved to give

$$T(t) = T_\infty + \frac{\dot{E}_{\text{generation}}}{h A_s} \left(1 - \exp\left(-\frac{h A_s}{m C_p} t\right) \right)$$

Substituting the known quantities and solving for t gives 527.3°C for the first case and 69.4°C for the second case, which are practically identical to the results obtained from the approximate analysis.

Electronic
device
30 W



Transient Heat Conduction in Large Plane Walls, Long Cylinders, and Spheres

4-26C A cylinder whose diameter is small relative to its length can be treated as an infinitely long cylinder. When the diameter and length of the cylinder are comparable, it is not proper to treat the cylinder as being infinitely long. It is also not proper to use this model when finding the temperatures near the bottom or top surfaces of a cylinder since heat transfer at those locations can be two-dimensional.

4-27C Yes. A plane wall whose one side is insulated is equivalent to a plane wall that is twice as thick and is exposed to convection from both sides. The midplane in the latter case will behave like an insulated surface because of thermal symmetry.

4-28C The solution for determination of the one-dimensional transient temperature distribution involves many variables that make the graphical representation of the results impractical. In order to reduce the number of parameters, some variables are grouped into dimensionless quantities.

4-29C The Fourier number is a measure of heat conducted through a body relative to the heat stored. Thus a large value of Fourier number indicates faster propagation of heat through body. Since Fourier number is proportional to time, doubling the time will also double the Fourier number.

4-30C This case can be handled by setting the heat transfer coefficient h to infinity ∞ since the temperature of the surrounding medium in this case becomes equivalent to the surface temperature.

4-31C The maximum possible amount of heat transfer will occur when the temperature of the body reaches the temperature of the medium, and can be determined from $Q_{\max} = mC_p(T_{\infty} - T_i)$.

4-32C When the Biot number is less than 0.1, the temperature of the sphere will be nearly uniform at all times. Therefore, it is more convenient to use the lumped system analysis in this case.

4-33 A student calculates the total heat transfer from a spherical copper ball. It is to be determined whether his/her result is reasonable.

Assumptions The thermal properties of the copper ball are constant at room temperature.

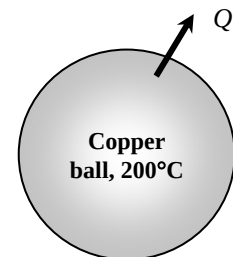
Properties The density and specific heat of the copper ball are $\rho = 8933 \text{ kg/m}^3$, and $C_p = 0.385 \text{ kJ/kg}\cdot^\circ\text{C}$ (Table A-3).

Analysis The mass of the copper ball and the maximum amount of heat transfer from the copper ball are

$$m = \rho V = \rho \left(\frac{\pi D^3}{6} \right) = (8933 \text{ kg/m}^3) \left[\frac{\pi (0.15 \text{ m})^3}{6} \right] = 15.79 \text{ kg}$$

$$Q_{\max} = mC_p [T_i - T_{\infty}] = (15.79 \text{ kg})(0.385 \text{ kJ/kg}\cdot^\circ\text{C})(200 - 25)^\circ\text{C} = 1064 \text{ kJ}$$

Discussion The student's result of 4520 kJ is **not reasonable** since it is greater than the maximum possible amount of heat transfer.



4-34 An egg is dropped into boiling water. The cooking time of the egg is to be determined. ✓

Assumptions **1** The egg is spherical in shape with a radius of $r_0 = 2.75$ cm. **2** Heat conduction in the egg is one-dimensional because of symmetry about the midpoint. **3** The thermal properties of the egg are constant. **4** The heat transfer coefficient is constant and uniform over the entire surface. **4** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal conductivity and diffusivity of the eggs are given to be $k = 0.6$ W/m.°C and $\alpha = 0.14 \times 10^{-6}$ m²/s.

Analysis The Biot number for this process is

$$Bi = \frac{hr_o}{k} = \frac{(1400 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0275 \text{ m})}{(0.6 \text{ W/m} \cdot ^\circ\text{C})} = 64.2$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

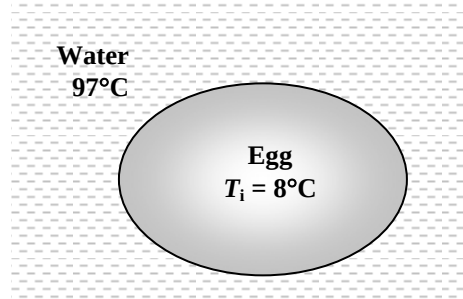
$$\lambda_1 = 3.0877 \quad \text{and} \quad A_1 = 1.9969$$

Then the Fourier number becomes

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{70 - 97}{8 - 97} = (1.9969)e^{-(3.0877)^2 \tau} \longrightarrow \tau = 0.198 \approx 0.2$$

Therefore, the one-term approximate solution (or the transient temperature charts) is applicable. Then the time required for the temperature of the center of the egg to reach 70°C is determined to be

$$t = \frac{\tau r_o^2}{\alpha} = \frac{(0.198)(0.0275 \text{ m})^2}{(0.14 \times 10^{-6} \text{ m}^2/\text{s})} = 1068 \text{ s} = \mathbf{17.8 \text{ min}}$$



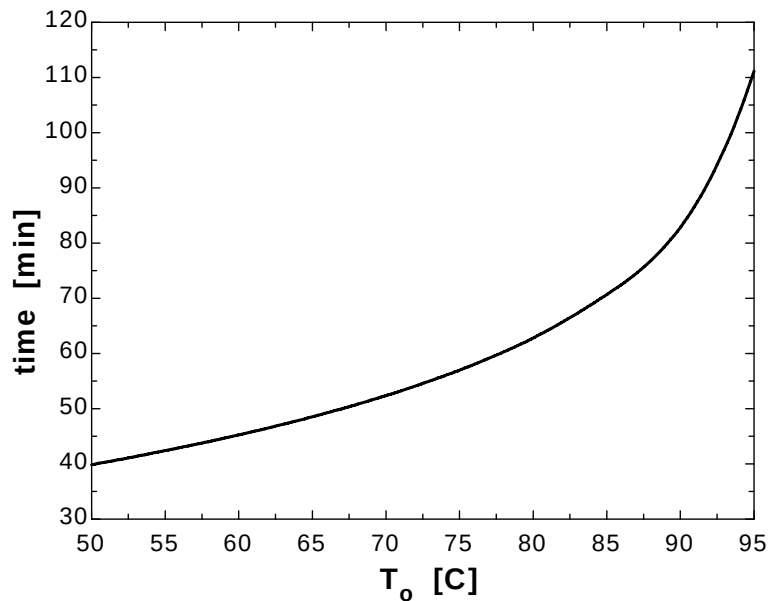
4-35

"!PROBLEM 4-35"**"GIVEN"** $D=0.055$ "[m]" $T_i=8$ "[C]" $T_o=70$ [C], parameter to be varied" $T_{\infty}=97$ "[C]" $h=1400$ "[W/m^2-C]"**"PROPERTIES"** $k=0.6$ "[W/m-C]" $\alpha=0.14E-6$ "[m^2/s]"**"ANALYSIS"** $Bi=(h \cdot r_o)/k$ $r_o=D/2$

"From Table 4-1 corresponding to this Bi number, we read"

 $\lambda_1=1.9969$ $A_1=3.0863$ $(T_o - T_{\infty})/(T_i - T_{\infty}) = A_1 \exp(-\lambda_1^2 \tau)$ $\text{time} = (\tau \cdot r_o^2) / \alpha \cdot \text{Convert}(\text{s}, \text{min})$

T_o [C]	time [min]
50	39.86
55	42.4
60	45.26
65	48.54
70	52.38
75	57
80	62.82
85	70.68
90	82.85
95	111.1



4-36 Large brass plates are heated in an oven. The surface temperature of the plates leaving the oven is to be determined.

Assumptions 1 Heat conduction in the plate is one-dimensional since the plate is large relative to its thickness and there is thermal symmetry about the center plane. 3 The thermal properties of the plate are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of brass at room temperature are given to be $k = 110 \text{ W/m} \cdot ^\circ\text{C}$, $\alpha = 33.9 \times 10^{-6} \text{ m}^2/\text{s}$

Analysis The Biot number for this process is

$$Bi = \frac{hL}{k} = \frac{(80 \text{ W/m}^2 \cdot ^\circ\text{C})(0.015 \text{ m})}{(110 \text{ W/m} \cdot ^\circ\text{C})} = 0.0109$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 0.1039 \quad \text{and} \quad A_1 = 1.0018$$

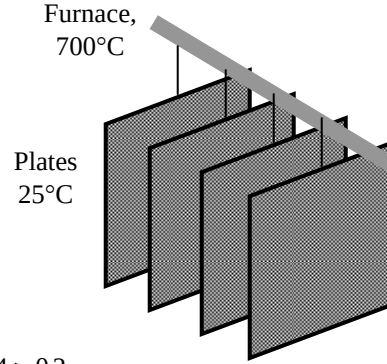
The Fourier number is

$$\tau = \frac{\alpha t}{L^2} = \frac{(33.9 \times 10^{-6} \text{ m}^2/\text{s})(10 \text{ min} \times 60 \text{ s/min})}{(0.015 \text{ m})^2} = 90.4 > 0.2$$

Therefore, the one-term approximate solution (or the transient temperature charts) is applicable. Then the temperature at the surface of the plates becomes

$$\theta(L, t)_{\text{wall}} = \frac{T(x, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 L / L) = (1.0018) e^{-(0.1039)^2 (90.4)} \cos(0.1039) = 0.378$$

$$\frac{T(L, t) - 700}{25 - 700} = 0.378 \longrightarrow T(L, t) = \mathbf{445^\circ\text{C}}$$



Discussion This problem can be solved easily using the lumped system analysis since $Bi < 0.1$, and thus the lumped system analysis is applicable. It gives

$$\alpha = \frac{k}{\rho C_p} \rightarrow \rho C_p = \frac{k}{\alpha} = \frac{110 \text{ W/m} \cdot ^\circ\text{C}}{33.9 \times 10^{-6} \text{ m}^2/\text{s}} = 3.245 \times 10^6 \text{ W} \cdot \text{s/m}^3 \cdot ^\circ\text{C}$$

$$b = \frac{hA}{\rho V C_p} = \frac{hA}{\rho(LA)C_p} = \frac{h}{\rho L C_p} = \frac{h}{L(k/\alpha)} = \frac{80 \text{ W/m}^2 \cdot ^\circ\text{C}}{(0.015 \text{ m})(3.245 \times 10^6 \text{ W} \cdot \text{s/m}^3 \cdot ^\circ\text{C})} = 0.001644 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \rightarrow T(t) = T_\infty + (T_i - T_\infty) e^{-bt} = 700^\circ\text{C} + (25 - 700^\circ\text{C}) e^{-(0.001644 \text{ s}^{-1})(600 \text{ s})} = \mathbf{448^\circ\text{C}}$$

which is almost identical to the result obtained above.

4-37 "PROBLEM 4-37"

"GIVEN" $L=0.03/2$ "[m]" $T_i=25$ "[C]" $T_{\infty}=700$ "[C], parameter to be varied"

time=10 "[min], parameter to be varied"

 $h=80$ "[W/m²-C]"**"PROPERTIES"** $k=110$ "[W/m-C]" $\alpha=33.9E-6$ "[m²/s]"**"ANALYSIS"** $Bi=(h*L)/k$

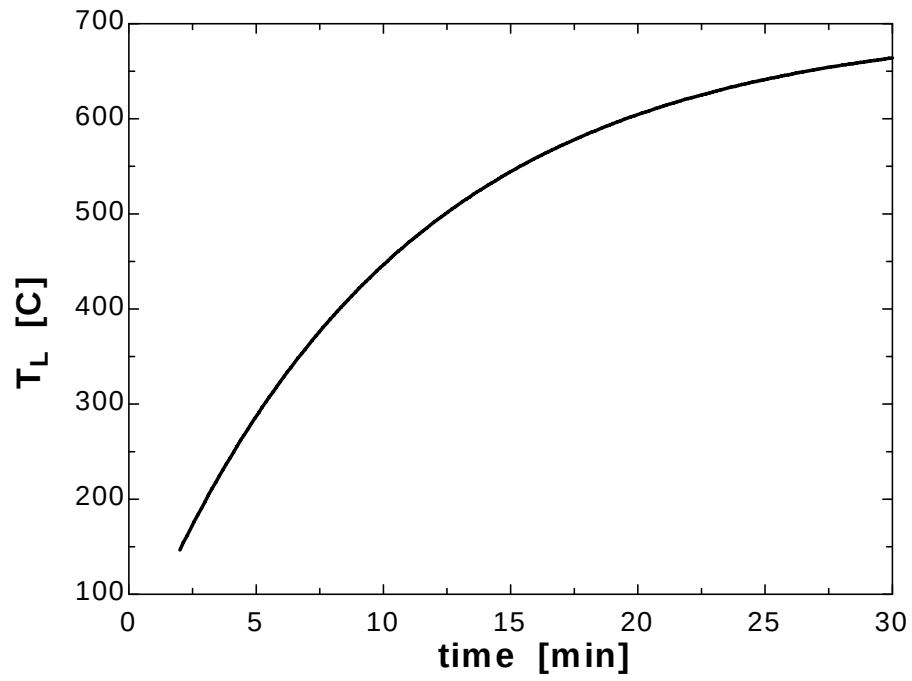
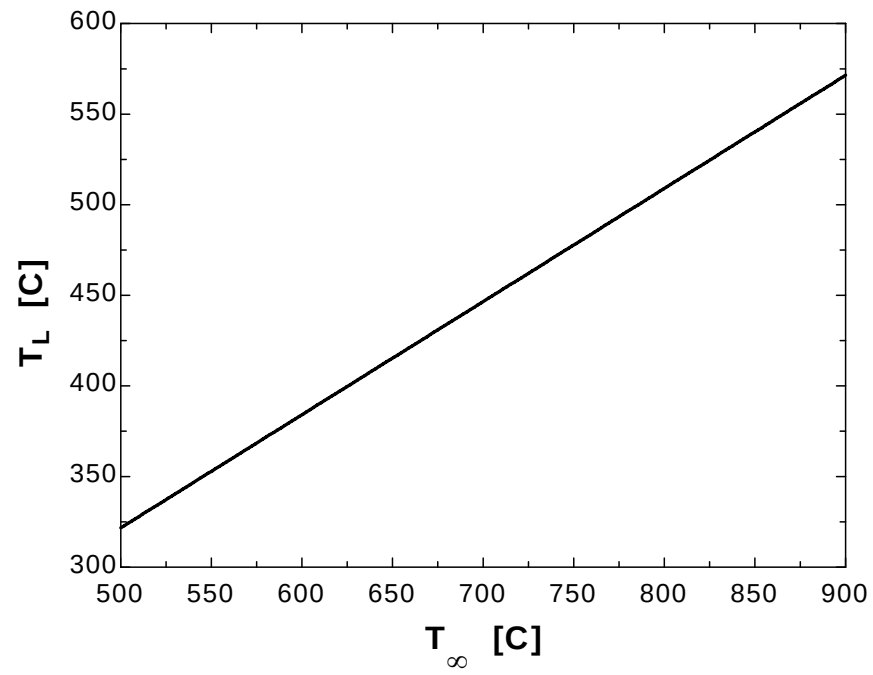
"From Table 4-1, corresponding to this Bi number, we read"

 $\lambda_1=0.1039$ $A_1=1.0018$ $\tau=(\alpha*time*Convert(min, s))/L^2$ $(T_L-T_{\infty})/(T_i-T_{\infty})=A_1*exp(-\lambda_1^2*\tau)*Cos(\lambda_1*L/L)$

T_{∞} [C]	T_L [C]
500	321.6
525	337.2
550	352.9
575	368.5
600	384.1
625	399.7
650	415.3
675	430.9
700	446.5
725	462.1
750	477.8
775	493.4
800	509
825	524.6
850	540.2
875	555.8
900	571.4

time [min]	T_L [C]
2	146.7
4	244.8
6	325.5
8	391.9
10	446.5
12	491.5
14	528.5
16	558.9
18	583.9
20	604.5
22	621.4
24	635.4
26	646.8
28	656.2

30	664
----	-----



4-38 A long cylindrical shaft at 400°C is allowed to cool slowly. The center temperature and the heat transfer per unit length of the cylinder are to be determined.

Assumptions 1 Heat conduction in the shaft is one-dimensional since it is long and it has thermal symmetry about the center line. 2 The thermal properties of the shaft are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of stainless steel 304 at room temperature are given to be $k = 14.9 \text{ W/m}\cdot^\circ\text{C}$, $\rho = 7900 \text{ kg/m}^3$, $C_p = 477 \text{ J/kg}\cdot^\circ\text{C}$, $\alpha = 3.95 \times 10^{-6} \text{ m}^2/\text{s}$

Analysis First the Biot number is calculated to be

$$Bi = \frac{hr_o}{k} = \frac{(60 \text{ W/m}^2\cdot^\circ\text{C})(0.175 \text{ m})}{(14.9 \text{ W/m}\cdot^\circ\text{C})} = 0.705$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 1.0935 \quad \text{and} \quad A_1 = 1.1558$$

The Fourier number is

$$\tau = \frac{\alpha t}{L^2} = \frac{(3.95 \times 10^{-6} \text{ m}^2/\text{s})(20 \times 60 \text{ s})}{(0.175 \text{ m})^2} = 0.1548$$

which is very close to the value of 0.2. Therefore, the one-term approximate solution (or the transient temperature charts) can still be used, with the understanding that the error involved will be a little more than 2 percent. Then the temperature at the center of the shaft becomes

$$\theta_{0,\text{cyl}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.1558) e^{-(1.0935)^2 (0.1548)} = 0.9605$$

$$\frac{T_0 - 150}{400 - 150} = 0.9605 \longrightarrow T_0 = \mathbf{390^\circ\text{C}}$$

The maximum heat can be transferred from the cylinder per meter of its length is

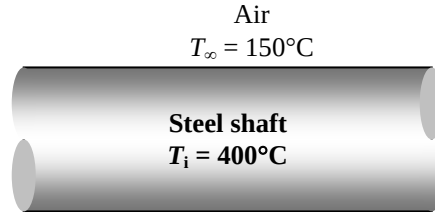
$$m = \rho V = \rho \pi r_o^2 L = (7900 \text{ kg/m}^3) [\pi (0.175 \text{ m})^2 (1 \text{ m})] = 760.1 \text{ kg}$$

$$Q_{\max} = m C_p [T_\infty - T_i] = (760.1 \text{ kg})(0.477 \text{ kJ/kg}\cdot^\circ\text{C})(400 - 150)^\circ\text{C} = 90,638 \text{ kJ}$$

Once the constant $J_1 = 0.4689$ is determined from Table 4-2 corresponding to the constant $\lambda_1 = 1.0935$, the actual heat transfer becomes

$$\left(\frac{Q}{Q_{\max}} \right)_{\text{cyl}} = 1 - 2 \left(\frac{T_0 - T_\infty}{T_i - T_\infty} \right) \frac{J_1(\lambda_1)}{\lambda_1} = 1 - 2 \left(\frac{390 - 150}{400 - 150} \right) \frac{0.4689}{1.0935} = 0.177$$

$$Q = 0.177(90,638 \text{ kJ}) = \mathbf{16,015 \text{ kJ}}$$



4-39

PROBLEM 4-39**GIVEN** $r_o = 0.35/2$ [m] $T_i = 400$ [C] $T_{\infty} = 150$ [C] $h = 60$ [W/m²-C]

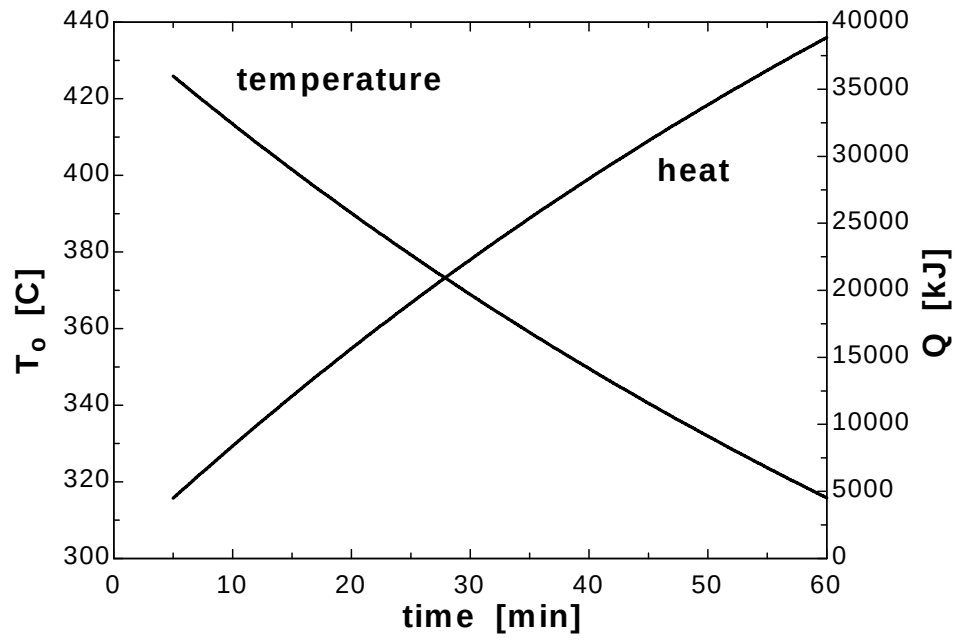
time = 20 [min], parameter to be varied

PROPERTIES $k = 14.9$ [W/m-C] $\rho = 7900$ [kg/m³] $C_p = 477$ [J/kg-C] $\alpha = 3.95 \times 10^{-6}$ [m²/s]**ANALYSIS** $Bi = (h \cdot r_o)/k$

From Table 4-1 corresponding to this Bi number, we read

 $\lambda_1 = 1.0935$ $A_1 = 1.1558$ $J_1 = 0.4709$ From Table 4-2, corresponding to λ_1 $\tau = (\alpha \cdot \text{time} \cdot \text{Convert}(\text{min}, \text{s}))/r_o^2$ $(T_o - T_{\infty})/(T_i - T_{\infty}) = A_1 \cdot \exp(-\lambda_1^2 \cdot \tau)$ $L = 1$ [m], 1 m length of the cylinder is considered $V = \pi \cdot r_o^2 \cdot L$ $m = \rho \cdot V$ $Q_{\max} = m \cdot C_p \cdot (T_i - T_{\infty}) \cdot \text{Convert}(\text{J}, \text{kJ})$ $Q/Q_{\max} = 1 - 2 \cdot (T_o - T_{\infty})/(T_i - T_{\infty}) \cdot J_1/\lambda_1$

time [min]	T_o [C]	Q [kJ]
5	425.9	4491
10	413.4	8386
15	401.5	12105
20	390.1	15656
25	379.3	19046
30	368.9	22283
35	359	25374
40	349.6	28325
45	340.5	31142
50	331.9	33832
55	323.7	36401
60	315.8	38853



4-40E Long cylindrical steel rods are heat-treated in an oven. Their centerline temperature when they leave the oven is to be determined.

Assumptions **1** Heat conduction in the rods is one-dimensional since the rods are long and they have thermal symmetry about the center line. **2** The thermal properties of the rod are constant. **3** The heat transfer coefficient is constant and uniform over the entire surface. **4** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

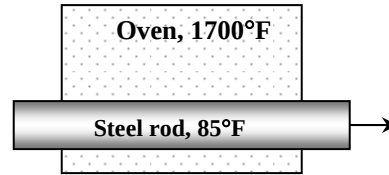
Properties The properties of AISI stainless steel rods are given to be $k = 7.74 \text{ Btu/h.ft.}^\circ\text{F}$, $\alpha = 0.135 \text{ ft}^2/\text{h}$.

Analysis The time the steel rods stays in the oven can be determined from

$$t = \frac{\text{length}}{\text{velocity}} = \frac{30 \text{ ft}}{10 \text{ ft/min}} = 3 \text{ min} = 180 \text{ s}$$

The Biot number is

$$Bi = \frac{hr_o}{k} = \frac{(20 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F})(2/12 \text{ ft})}{(7.74 \text{ Btu/h.ft.}^\circ\text{F})} = 0.4307$$



The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 0.8784 \quad \text{and} \quad A_1 = 1.0995$$

The Fourier number is

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.135 \text{ ft}^2/\text{h})(3/60 \text{ h})}{(2/12 \text{ ft})^2} = 0.243$$

Then the temperature at the center of the rods becomes

$$\theta_{0,\text{cyl}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.0995) e^{-(0.8784)^2 (0.243)} = 0.912$$

$$\frac{T_0 - 1700}{85 - 1700} = 0.912 \longrightarrow T_0 = 228^\circ\text{F}$$

4-41 Steaks are cooled by passing them through a refrigeration room. The time of cooling is to be determined.

Assumptions **1** Heat conduction in the steaks is one-dimensional since the steaks are large relative to their thickness and there is thermal symmetry about the center plane. **3** The thermal properties of the steaks are constant. **4** The heat transfer coefficient is constant and uniform over the entire surface. **5** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of steaks are given to be $k = 0.45 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 0.91 \times 10^{-7} \text{ m}^2/\text{s}$

Analysis The Biot number is

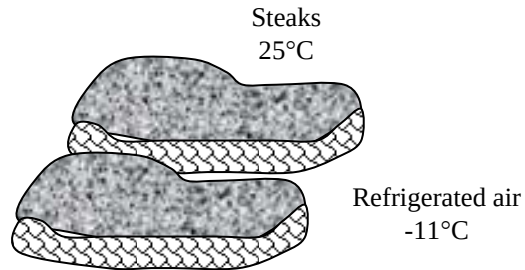
$$Bi = \frac{hL}{k} = \frac{(9 \text{ W/m}^2\cdot^\circ\text{C})(0.01 \text{ m})}{(0.45 \text{ W/m}\cdot^\circ\text{C})} = 0.200$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 0.4328 \quad \text{and} \quad A_1 = 1.0311$$

The Fourier number is

$$\begin{aligned} \frac{T(L, t) - T_\infty}{T_i - T_\infty} &= A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 L / L) \\ \frac{2 - (-11)}{25 - (-11)} &= (1.0311) e^{-(0.4328)^2 \tau} \cos(0.4328) \longrightarrow \tau = 5.601 > 0.2 \end{aligned}$$



Therefore, the one-term approximate solution (or the transient temperature charts) is applicable. Then the length of time for the steaks to be kept in the refrigerator is determined to be

$$t = \frac{\tau L^2}{\alpha} = \frac{(5.601)(0.01 \text{ m})^2}{(0.91 \times 10^{-7} \text{ m}^2/\text{s})} = 6155 \text{ s} = \mathbf{102.6 \text{ min}}$$

4-42 A long cylindrical wood log is exposed to hot gases in a fireplace. The time for the ignition of the wood is to be determined.

Assumptions **1** Heat conduction in the wood is one-dimensional since it is long and it has thermal symmetry about the center line. **2** The thermal properties of the wood are constant. **3** The heat transfer coefficient is constant and uniform over the entire surface. **4** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of wood are given to be $k = 0.17 \text{ W/m}\cdot^\circ\text{C}$, $\alpha = 1.28 \times 10^{-7} \text{ m}^2/\text{s}$

Analysis The Biot number is

$$Bi = \frac{hr_o}{k} = \frac{(13.6 \text{ W/m}^2\cdot^\circ\text{C})(0.05 \text{ m})}{(0.17 \text{ W/m}\cdot^\circ\text{C})} = 4.00$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

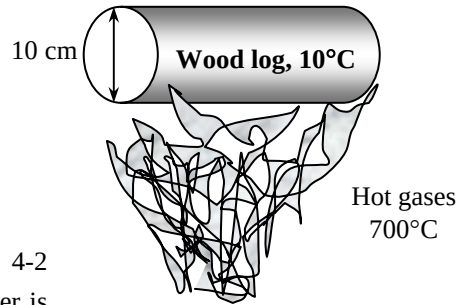
$$\lambda_1 = 1.9081 \quad \text{and} \quad A_1 = 1.4698$$

Once the constant J_0 is determined from Table 4-2 corresponding to the constant $\lambda_1 = 1.9081$, the Fourier number is determined to be

$$\begin{aligned} \frac{T(r_o, t) - T_\infty}{T_i - T_\infty} &= A_1 e^{-\lambda_1^2 \tau} J_0(\lambda_1 r_o / r_o) \\ \frac{420 - 500}{10 - 500} &= (1.4698) e^{-(1.9081)^2 \tau} (0.2771) \longrightarrow \tau = 0.251 \end{aligned}$$

which is above the value of 0.2. Therefore, the one-term approximate solution (or the transient temperature charts) can be used. Then the length of time before the log ignites is

$$t = \frac{\tau r_o^2}{\alpha} = \frac{(0.251)(0.05 \text{ m})^2}{(1.28 \times 10^{-7} \text{ m}^2/\text{s})} = 4904 \text{ s} = \mathbf{81.7 \text{ min}}$$



4-43 A rib is roasted in an oven. The heat transfer coefficient at the surface of the rib, the temperature of the outer surface of the rib and the amount of heat transfer when it is rare done are to be determined. The time it will take to roast this rib to medium level is also to be determined.

Assumptions 1 The rib is a homogeneous spherical object. 2 Heat conduction in the rib is one-dimensional because of symmetry about the midpoint. 3 The thermal properties of the rib are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of the rib are given to be $k = 0.45 \text{ W/m}\cdot^\circ\text{C}$, $\rho = 1200 \text{ kg/m}^3$, $C_p = 4.1 \text{ kJ/kg}\cdot^\circ\text{C}$, and $\alpha = 0.91 \times 10^{-7} \text{ m}^2/\text{s}$.

Analysis (a) The radius of the roast is determined to be

$$m = \rho V \longrightarrow V = \frac{m}{\rho} = \frac{3.2 \text{ kg}}{1200 \text{ kg/m}^3} = 0.002667 \text{ m}^3$$

$$V = \frac{4}{3}\pi r_o^3 \longrightarrow r_o = \sqrt[3]{\frac{3V}{4\pi}} = \sqrt[3]{\frac{3(0.002667 \text{ m}^3)}{4\pi}} = 0.08603 \text{ m}$$

The Fourier number is

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.91 \times 10^{-7} \text{ m}^2/\text{s})(2 \times 3600 + 45 \times 60) \text{ s}}{(0.08603 \text{ m})^2} = 0.1217$$

which is somewhat below the value of 0.2. Therefore, the one-term approximate solution (or the transient temperature charts) can still be used, with the understanding that the error involved will be a little more than 2 percent. Then the one-term solution can be written in the form

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{60 - 163}{4.5 - 163} = 0.65 = A_1 e^{-\lambda_1^2 (0.1217)}$$

It is determined from Table 4-1 by trial and error that this equation is satisfied when $Bi = 30$, which corresponds to $\lambda_1 = 3.0372$ and $A_1 = 1.9898$. Then the heat transfer coefficient can be determined from

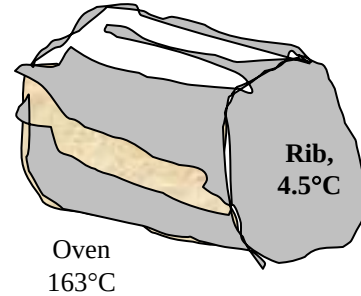
$$Bi = \frac{hr_o}{k} \longrightarrow h = \frac{kBi}{r_o} = \frac{(0.45 \text{ W/m}\cdot^\circ\text{C})(30)}{(0.08603 \text{ m})} = 156.9 \text{ W/m}^2\cdot^\circ\text{C}$$

This value seems to be larger than expected for problems of this kind. This is probably due to the Fourier number being less than 0.2.

(b) The temperature at the surface of the rib is

$$\theta(r_o, t)_{sph} = \frac{T(r_o, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r_o / r_o)}{\lambda_1 r_o / r_o} = (1.9898) e^{-(3.0372)^2 (0.1217)} \frac{\sin(3.0372 \text{ rad})}{3.0372}$$

$$\frac{T(r_o, t) - 163}{4.5 - 163} = 0.0222 \longrightarrow T(r_o, t) = 159.5^\circ\text{C}$$



(c) The maximum possible heat transfer is

$$Q_{\max} = mC_p(T_{\infty} - T_i) = (3.2 \text{ kg})(4.1 \text{ kJ/kg} \cdot ^\circ\text{C})(163 - 4.5)^\circ\text{C} = 2080 \text{ kJ}$$

Then the actual amount of heat transfer becomes

$$\frac{Q}{Q_{\max}} = 1 - 3\theta_{o,sph} \frac{\sin(\lambda_1) - \lambda_1 \cos(\lambda_1)}{\lambda_1^3} = 1 - 3(0.65) \frac{\sin(3.0372) - (3.0372) \cos(3.0372)}{(3.0372)^3} = 0.783$$

$$Q = 0.783Q_{\max} = (0.783)(2080 \text{ kJ}) = \mathbf{1629 \text{ kJ}}$$

(d) The cooking time for medium-done rib is determined to be

$$\theta_{o,sph} = \frac{T_0 - T_{\infty}}{T_i - T_{\infty}} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{71 - 163}{4.5 - 163} = (1.9898)e^{-(3.0372)^2 \tau} \longrightarrow \tau = 0.1336$$

$$t = \frac{\tau_o^2}{\alpha} = \frac{(0.1336)(0.08603 \text{ m})^2}{(0.91 \times 10^{-7} \text{ m}^2/\text{s})} = 10,866 \text{ s} = 181 \text{ min} \cong \mathbf{3 \text{ hr}}$$

This result is close to the listed value of 3 hours and 20 minutes. The difference between the two results is due to the Fourier number being less than 0.2 and thus the error in the one-term approximation.

Discussion The temperature of the outer parts of the rib is greater than that of the inner parts of the rib after it is taken out of the oven. Therefore, there will be a heat transfer from outer parts of the rib to the inner parts as a result of this temperature difference. The recommendation is logical.

4-44 A rib is roasted in an oven. The heat transfer coefficient at the surface of the rib, the temperature of the outer surface of the rib and the amount of heat transfer when it is well-done are to be determined. The time it will take to roast this rib to medium level is also to be determined.

Assumptions 1 The rib is a homogeneous spherical object. 2 Heat conduction in the rib is one-dimensional because of symmetry about the midpoint. 3 The thermal properties of the rib are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of the rib are given to be $k = 0.45 \text{ W/m} \cdot ^\circ\text{C}$, $\rho = 1200 \text{ kg/m}^3$, $C_p = 4.1 \text{ kJ/kg} \cdot ^\circ\text{C}$, and $\alpha = 0.91 \times 10^{-7} \text{ m}^2/\text{s}$

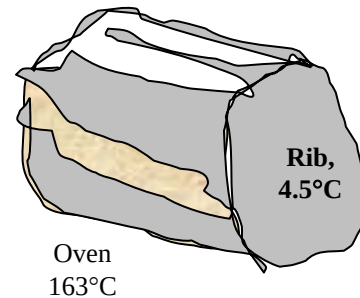
Analysis (a) The radius of the rib is determined to be

$$m = \rho V \longrightarrow V = \frac{m}{\rho} = \frac{3.2 \text{ kg}}{1200 \text{ kg/m}^3} = 0.00267 \text{ m}^3$$

$$V = \frac{4}{3} \pi r_o^3 \longrightarrow r_o = \sqrt[3]{\frac{3V}{4\pi}} = \sqrt[3]{\frac{3(0.00267 \text{ m}^3)}{4\pi}} = 0.08603 \text{ m}$$

The Fourier number is

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.91 \times 10^{-7} \text{ m}^2/\text{s})(4 \times 3600 + 15 \times 60) \text{ s}}{(0.08603 \text{ m})^2} = 0.1881$$



which is somewhat below the value of 0.2. Therefore, the one-term approximate solution (or the transient temperature charts) can still be used, with the understanding that the error involved will be a little more than 2 percent. Then the one-term solution formulation can be written in the form

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{77 - 163}{4.5 - 163} = 0.543 = A_1 e^{-\lambda_1^2 (0.1881)}$$

It is determined from Table 4-1 by trial and error that this equation is satisfied when $Bi = 4.3$, which corresponds to $\lambda_1 = 2.4900$ and $A_1 = 1.7402$. Then the heat transfer coefficient can be determined from.

$$Bi = \frac{hr_o}{k} \longrightarrow h = \frac{kBi}{r_o} = \frac{(0.45 \text{ W/m} \cdot ^\circ\text{C})(4.3)}{(0.08603 \text{ m})} = \mathbf{22.5 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

(b) The temperature at the surface of the rib is

$$\theta(r_o, t)_{sph} = \frac{T(r_o, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r_o / r_o)}{\lambda_1 r_o / r_o} = (1.7402) e^{-(2.49)^2 (0.1881)} \frac{\sin(2.49)}{2.49}$$

$$\frac{T(r_o, t) - 163}{4.5 - 163} = 0.132 \longrightarrow T(r_o, t) = \mathbf{142.1^\circ\text{C}}$$

(c) The maximum possible heat transfer is

$$Q_{\max} = mC_p (T_\infty - T_i) = (3.2 \text{ kg})(4.1 \text{ kJ/kg} \cdot ^\circ\text{C})(163 - 4.5)^\circ\text{C} = 2080 \text{ kJ}$$

Then the actual amount of heat transfer becomes

$$\frac{Q}{Q_{\max}} = 1 - 3\theta_{0,sph} \frac{\sin(\lambda_1) - \lambda_1 \cos(\lambda_1)}{\lambda_1^3} = 1 - 3(0.543) \frac{\sin(2.49) - (2.49) \cos(2.49)}{(2.49)^3} = 0.727$$

$$Q = 0.727 Q_{\max} = (0.727)(2080 \text{ kJ}) = \mathbf{1512 \text{ kJ}}$$

(d) The cooking time for medium-done rib is determined to be

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{77 - 163}{4.5 - 163} = (1.7402) e^{-(2.49)^2 \tau} \longrightarrow \tau = 0.177$$

$$t = \frac{\tau r_o^2}{\alpha} = \frac{(0.177)(0.08603 \text{ m})^2}{(0.91 \times 10^{-7} \text{ m}^2/\text{s})} = 14,403 \text{ s} = 240.0 \text{ min} = \mathbf{4 \text{ hr}}$$

This result is close to the listed value of 4 hours and 15 minutes. The difference between the two results is probably due to the Fourier number being less than 0.2 and thus the error in the one-term approximation.

Discussion The temperature of the outer parts of the rib is greater than that of the inner parts of the rib after it is taken out of the oven. Therefore, there will be a heat transfer from outer parts of the rib to the inner parts as a result of this temperature difference. The recommendation is logical.

4-45 An egg is dropped into boiling water. The cooking time of the egg is to be determined.

Assumptions **1** The egg is spherical in shape with a radius of $r_0 = 2.75$ cm. **2** Heat conduction in the egg is one-dimensional because of symmetry about the midpoint. **3** The thermal properties of the egg are constant. **4** The heat transfer coefficient is constant and uniform over the entire surface. **5** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal conductivity and diffusivity of the eggs can be approximated by those of water at room temperature to be $k = 0.607$ W/m.°C, $\alpha = k / \rho C_p = 0.146 \times 10^{-6}$ m²/s (Table A-9).

Analysis The Biot number is

$$Bi = \frac{hr_o}{k} = \frac{(800 \text{ W/m}^2 \cdot \text{°C})(0.0275 \text{ m})}{(0.607 \text{ W/m} \cdot \text{°C})} = 36.2$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

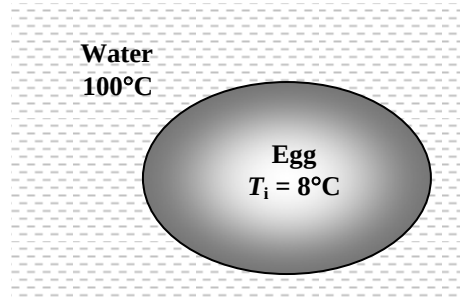
$$\lambda_1 = 3.0533 \quad \text{and} \quad A_1 = 1.9925$$

Then the Fourier number and the time period become

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{60 - 100}{8 - 100} = (1.9925) e^{-(3.0533)^2 \tau} \longrightarrow \tau = 0.1633$$

which is somewhat below the value of 0.2. Therefore, the one-term approximate solution (or the transient temperature charts) can still be used, with the understanding that the error involved will be a little more than 2 percent. Then the length of time for the egg to be kept in boiling water is determined to be

$$t = \frac{\tau_o^2}{\alpha} = \frac{(0.1633)(0.0275 \text{ m})^2}{(0.146 \times 10^{-6} \text{ m}^2/\text{s})} = 846 \text{ s} = \mathbf{14.1 \text{ min}}$$



4-46 An egg is cooked in boiling water. The cooking time of the egg is to be determined for a location at 1610-m elevation.

Assumptions **1** The egg is spherical in shape with a radius of $r_0 = 2.75$ cm. **2** Heat conduction in the egg is one-dimensional because of symmetry about the midpoint. **3** The thermal properties of the egg and heat transfer coefficient are constant. **4** The heat transfer coefficient is constant and uniform over the entire surface. **5** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal conductivity and diffusivity of the eggs can be approximated by those of water at room temperature to be $k = 0.607$ W/m.°C, $\alpha = k / \rho C_p = 0.146 \times 10^{-6}$ m²/s (Table A-9).

Analysis The Biot number is

$$Bi = \frac{hr_o}{k} = \frac{(800 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0275 \text{ m})}{(0.607 \text{ W/m} \cdot ^\circ\text{C})} = 36.2$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

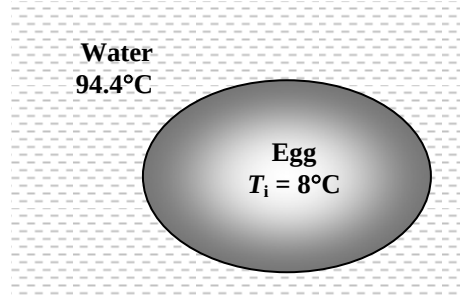
$$\lambda_1 = 3.0533 \quad \text{and} \quad A_1 = 1.9925$$

Then the Fourier number and the time period become

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{60 - 94.4}{8 - 94.4} = (1.9925) e^{-(3.0533)^2 \tau} \longrightarrow \tau = 0.1727$$

which is somewhat below the value of 0.2. Therefore, the one-term approximate solution (or the transient temperature charts) can still be used, with the understanding that the error involved will be a little more than 2 percent. Then the length of time for the egg to be kept in boiling water is determined to be

$$t = \frac{\tau_o^2}{\alpha} = \frac{(0.1727)(0.0275 \text{ m})^2}{(0.146 \times 10^{-6} \text{ m}^2/\text{s})} = 895 \text{ s} = \mathbf{14.9 \text{ min}}$$



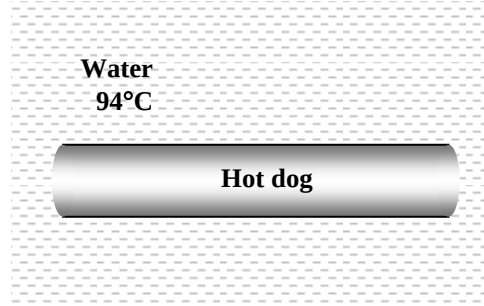
4-47 A hot dog is dropped into boiling water, and temperature measurements are taken at certain time intervals. The thermal diffusivity and thermal conductivity of the hot dog and the convection heat transfer coefficient are to be determined.

Assumptions 1 Heat conduction in the hot dog is one-dimensional since it is long and it has thermal symmetry about the center line. 2 The thermal properties of the hot dog are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of hot dog available are given to be $\rho = 980 \text{ kg/m}^3$ and $C_p = 3900 \text{ J/kg} \cdot ^\circ\text{C}$.

Analysis (a) From Fig. 4-14b we have

$$\left. \begin{aligned} \frac{T - T_\infty}{T_o - T_\infty} &= \frac{88 - 94}{59 - 94} = 0.17 \\ \frac{r}{r_o} &= \frac{r_o}{r_o} = 1 \end{aligned} \right\} \frac{1}{Bi} = \frac{k}{hr_o} = 0.15$$



The Fourier number is determined from Fig. 4-14a to be

$$\left. \begin{aligned} \frac{1}{Bi} &= \frac{k}{hr_o} = 0.15 \\ \frac{T_o - T_\infty}{T_i - T_\infty} &= \frac{59 - 94}{20 - 94} = 0.47 \end{aligned} \right\} \tau = \frac{\alpha t}{r_o^2} = 0.20$$

The thermal diffusivity of the hot dog is determined to be

$$\frac{\alpha t}{r_o^2} = 0.20 \longrightarrow \alpha = \frac{0.2 r_o^2}{t} = \frac{(0.2)(0.011 \text{ m})^2}{120 \text{ s}} = \mathbf{2.017 \times 10^{-7} \text{ m}^2/\text{s}}$$

(b) The thermal conductivity of the hot dog is determined from

$$k = \alpha \rho C_p = (2.017 \times 10^{-7} \text{ m}^2/\text{s})(980 \text{ kg/m}^3)(3900 \text{ J/kg} \cdot ^\circ\text{C}) = \mathbf{0.771 \text{ W/m} \cdot ^\circ\text{C}}$$

(c) From part (a) we have $\frac{1}{Bi} = \frac{k}{hr_o} = 0.15$. Then,

$$\frac{k}{h} = 0.15 r_o = (0.15)(0.011 \text{ m}) = 0.00165 \text{ m}$$

Therefore, the heat transfer coefficient is

$$\frac{k}{h} = 0.00165 \longrightarrow h = \frac{0.771 \text{ W/m} \cdot ^\circ\text{C}}{0.00165 \text{ m}} = \mathbf{467 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

4-48 Using the data and the answers given in Prob. 4-43, the center and the surface temperatures of the hot dog 4 min after the start of the cooking and the amount of heat transferred to the hot dog are to be determined.

Assumptions 1 Heat conduction in the hot dog is one-dimensional since it is long and it has thermal symmetry about the center line. 2 The thermal properties of the hot dog are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of hot dog and the convection heat transfer coefficient are given or obtained in P4-47 to be $k = 0.771 \text{ W/m}\cdot^\circ\text{C}$, $\rho = 980 \text{ kg/m}^3$, $C_p = 3900 \text{ J/kg}\cdot^\circ\text{C}$, $\alpha = 2.017 \times 10^{-7} \text{ m}^2/\text{s}$, and $h = 467 \text{ W/m}^2\cdot^\circ\text{C}$.

Analysis The Biot number is

$$Bi = \frac{hr_o}{k} = \frac{(467 \text{ W/m}^2\cdot^\circ\text{C})(0.011 \text{ m})}{(0.771 \text{ W/m}\cdot^\circ\text{C})} = 6.66$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 2.0785 \quad \text{and} \quad A_1 = 1.5357$$

The Fourier number is

$$\tau = \frac{\alpha t}{L^2} = \frac{(2.017 \times 10^{-7} \text{ m}^2/\text{s})(4 \text{ min} \times 60 \text{ s/min})}{(0.011 \text{ m})^2} = 0.4001 > 0.2$$

Then the temperature at the center of the hot dog is determined to be

$$\theta_{o,cyl} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.5357) e^{-(2.0785)^2 (0.4001)} = 0.2727$$

$$\frac{T_0 - 94}{20 - 94} = 0.2727 \longrightarrow T_0 = 73.8^\circ\text{C}$$

From Table 4-2 we read $J_0 = 0.2194$ corresponding to the constant $\lambda_1 = 2.0785$. Then the temperature at the surface of the hot dog becomes

$$\frac{T(r_o, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} J_0(\lambda_1 r_o / r_o) = (1.5357) e^{-(2.0785)^2 (0.4001)} (0.2194) = 0.05982$$

$$\frac{T(r_o, t) - 94}{20 - 94} = 0.05982 \longrightarrow T(r_o, t) = 89.6^\circ\text{C}$$

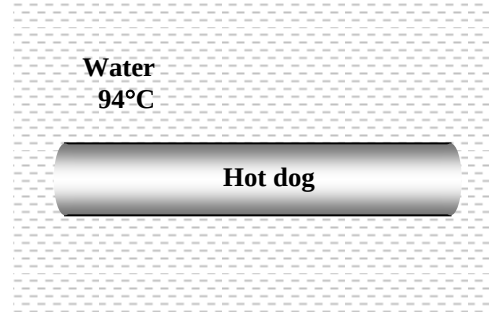
The maximum possible amount of heat transfer is

$$m = \rho V = \rho \pi r_o^2 L = (980 \text{ kg/m}^3) [\pi (0.011 \text{ m})^2 (0.125 \text{ m})] = 0.04657 \text{ kg}$$

$$Q_{\max} = m C_p (T_i - T_\infty) = (0.04657 \text{ kg})(3900 \text{ J/kg}\cdot^\circ\text{C})(94 - 20)^\circ\text{C} = 13,440 \text{ J}$$

From Table 4-2 we read $J_1 = 0.5760$ corresponding to the constant $\lambda_1 = 2.0785$. Then the actual heat transfer becomes

$$\left(\frac{Q}{Q_{\max}} \right)_{cyl} = 1 - 2\theta_{o,cyl} \frac{J_1(\lambda_1)}{\lambda_1} = 1 - 2(0.2727) \frac{0.5760}{2.0785} = 0.8489 \longrightarrow Q = 0.8489(13,440 \text{ kJ}) = 11,409 \text{ kJ}$$



4-49E Whole chickens are to be cooled in the racks of a large refrigerator. Heat transfer coefficient that will enable to meet temperature constraints of the chickens while keeping the refrigeration time to a minimum is to be determined.

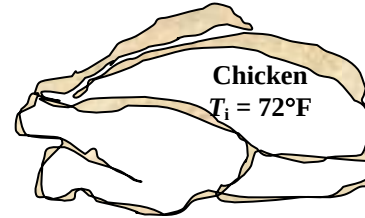
Assumptions **1** The chicken is a homogeneous spherical object. **2** Heat conduction in the chicken is one-dimensional because of symmetry about the midpoint. **3** The thermal properties of the chicken are constant. **4** The heat transfer coefficient is constant and uniform over the entire surface. **5** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of the chicken are given to be $k = 0.26 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$, $\rho = 74.9 \text{ lbm/ft}^3$, $C_p = 0.98 \text{ Btu/lbm}\cdot^\circ\text{F}$, and $\alpha = 0.0035 \text{ ft}^2/\text{h}$.

Analysis The radius of the chicken is determined to be

$$m = \rho V \longrightarrow V = \frac{m}{\rho} = \frac{5 \text{ lbm}}{74.9 \text{ lbm/ft}^3} = 0.06676 \text{ ft}^3$$

$$V = \frac{4}{3}\pi r_o^3 \longrightarrow r_o = \sqrt[3]{\frac{3V}{4\pi}} = \sqrt[3]{\frac{3(0.06676 \text{ ft}^3)}{4\pi}} = 0.2517 \text{ ft}$$



From Fig. 4-15b we have

$$\left. \begin{aligned} \frac{T - T_\infty}{T_o - T_\infty} &= \frac{35 - 5}{45 - 5} = 0.75 \\ \frac{x}{r_o} &= \frac{r_o}{r_o} = 1 \end{aligned} \right\} \frac{1}{Bi} = \frac{k}{hr_o} = 1.75$$

Then the heat transfer coefficients becomes

$$h = \frac{k}{1.75r_o} = \frac{(0.26 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F})}{1.75(0.2517 \text{ ft})} = \mathbf{0.590 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}}$$

4-50 A person puts apples into the freezer to cool them quickly. The center and surface temperatures of the apples, and the amount of heat transfer from each apple in 1 h are to be determined.

Assumptions **1** The apples are spherical in shape with a diameter of 9 cm. **2** Heat conduction in the apples is one-dimensional because of symmetry about the midpoint. **3** The thermal properties of the apples are constant. **4** The heat transfer coefficient is constant and uniform over the entire surface. **5** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of the apples are given to be $k = 0.418 \text{ W/m}\cdot^\circ\text{C}$, $\rho = 840 \text{ kg/m}^3$, $C_p = 3.81 \text{ kJ/kg}\cdot^\circ\text{C}$, and $\alpha = 1.3 \times 10^{-7} \text{ m}^2/\text{s}$.

Analysis The Biot number is

$$Bi = \frac{hr_o}{k} = \frac{(8 \text{ W/m}^2\cdot^\circ\text{C})(0.045 \text{ m})}{(0.418 \text{ W/m}\cdot^\circ\text{C})} = 0.861$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 1.476 \quad \text{and} \quad A_1 = 1.2390$$

The Fourier number is

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(1.3 \times 10^{-7} \text{ m}^2/\text{s})(1 \text{ h} \times 3600 \text{ s/h})}{(0.045 \text{ m})^2} = 0.231 > 0.2$$

Then the temperature at the center of the apples becomes

$$\theta_{o,sph} = \frac{T_o - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_o - (-15)}{20 - (-15)} = (1.239) e^{-(1.476)^2 (0.231)} = 0.749 \longrightarrow T_o = \mathbf{11.2^\circ\text{C}}$$

The temperature at the surface of the apples is

$$\theta(r_o, t)_{sph} = \frac{T(r_o, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r_o / r_o)}{\lambda_1 r_o / r_o} = (1.239) e^{-(1.476)^2 (0.231)} \frac{\sin(1.476 \text{ rad})}{1.476} = 0.505$$

$$\frac{T(r_o, t) - (-15)}{20 - (-15)} = 0.505 \longrightarrow T(r_o, t) = \mathbf{2.7^\circ\text{C}}$$

The maximum possible heat transfer is

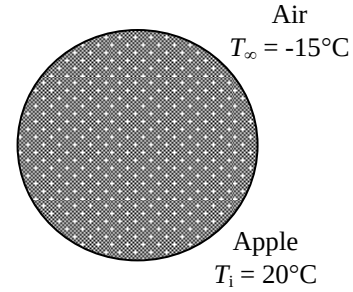
$$m = \rho V = \rho \frac{4}{3} \pi r_o^3 = (840 \text{ kg/m}^3) \left[\frac{4}{3} \pi (0.045 \text{ m})^3 \right] = 0.3206 \text{ kg}$$

$$Q_{\max} = m C_p (T_i - T_\infty) = (0.3206 \text{ kg})(3.81 \text{ kJ/kg}\cdot^\circ\text{C})[20 - (-15)]^\circ\text{C} = 42.76 \text{ kJ}$$

Then the actual amount of heat transfer becomes

$$\frac{Q}{Q_{\max}} = 1 - 3\theta_{o,sph} \frac{\sin(\lambda_1) - \lambda_1 \cos(\lambda_1)}{\lambda_1^3} = 1 - 3(0.749) \frac{\sin(1.476 \text{ rad}) - (1.476) \cos(1.476 \text{ rad})}{(1.476)^3} = 0.402$$

$$Q = 0.402 Q_{\max} = (0.402)(42.76 \text{ kJ}) = \mathbf{17.2 \text{ kJ}}$$



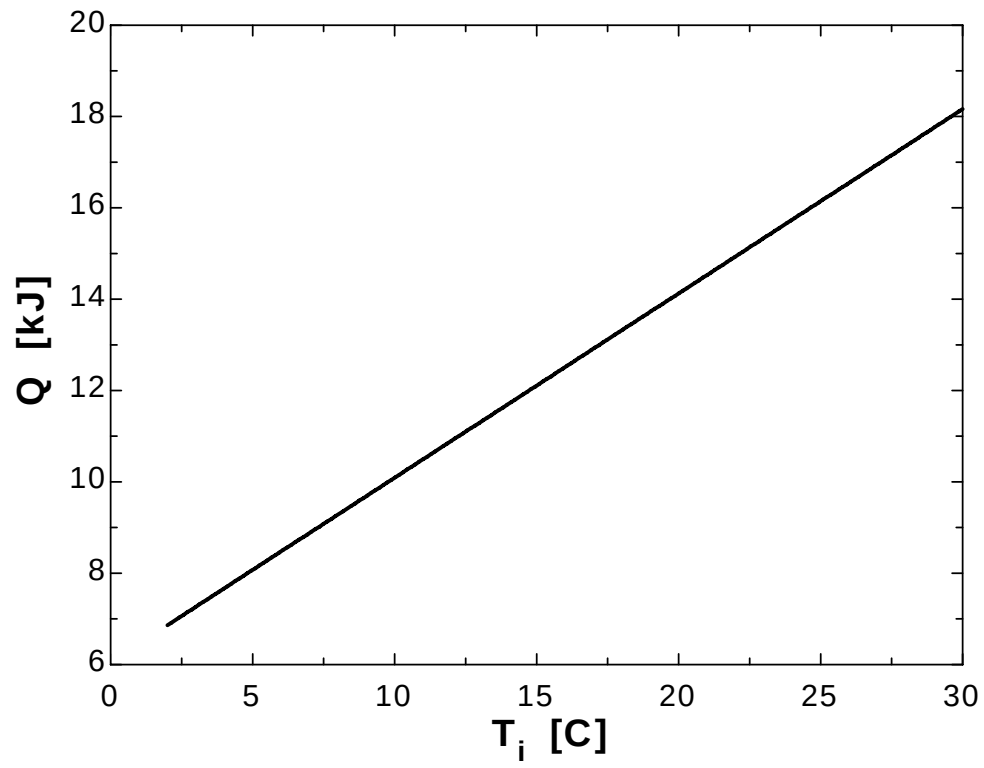
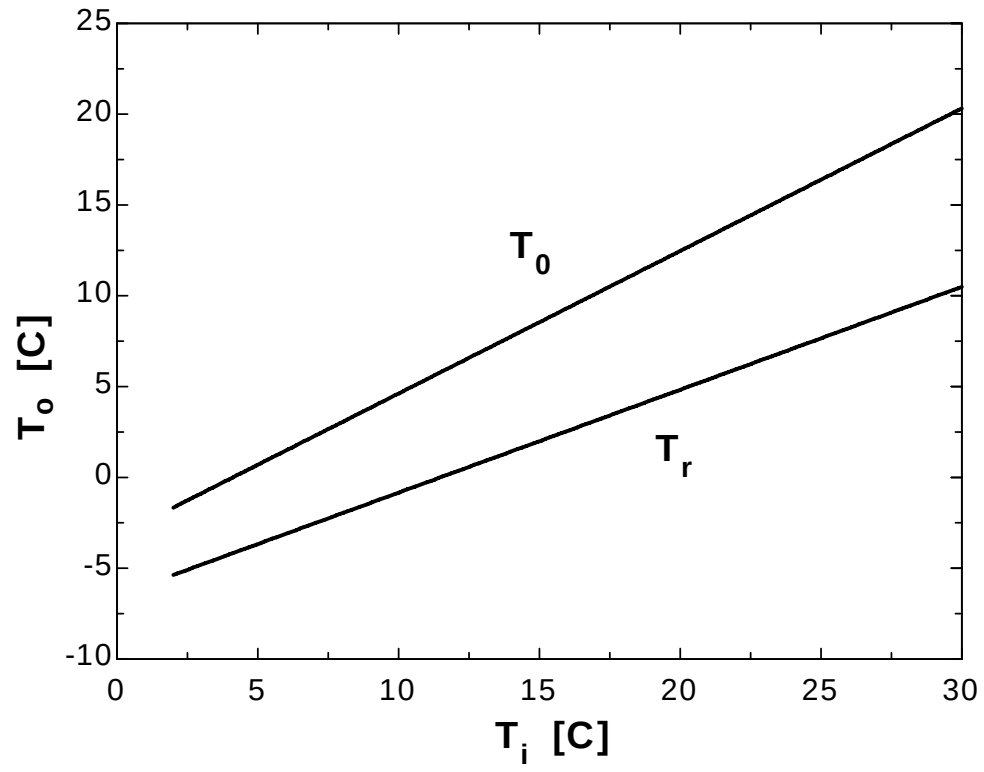
4-51

PROBLEM 4-51**GIVEN** $T_{\infty} = -15$ [C] $T_i = 20$ [C], parameter to be varied $h = 8$ [W/m²-C] $r_o = 0.09/2$ [m] $\text{time} = 1 \times 3600$ [s]**PROPERTIES** $k = 0.513$ [W/m-C] $\rho = 840$ [kg/m³] $C_p = 3.6$ [kJ/kg-C] $\alpha = 1.3 \times 10^{-7}$ [m²/s]**ANALYSIS** $Bi = (h \cdot r_o) / k$

From Table 4-1 corresponding to this Bi number, we read

 $\lambda_1 = 1.3525$ $A_1 = 1.1978$ $\tau = (\alpha \cdot \text{time}) / r_o^2$ $(T_o - T_{\infty}) / (T_i - T_{\infty}) = A_1 \exp(-\lambda_1^2 \tau)$ $(T_r - T_{\infty}) / (T_i - T_{\infty}) = A_1 \exp(-\lambda_1^2 \tau) \sin(\lambda_1 r_o / r_o) / (\lambda_1 r_o / r_o)$ $V = 4/3 \pi r_o^3$ $m = \rho \cdot V$ $Q_{\max} = m \cdot C_p (T_i - T_{\infty})$ $Q/Q_{\max} = 1 - 3 \cdot (T_o - T_{\infty}) / (T_i - T_{\infty}) \cdot (\sin(\lambda_1) - \lambda_1 \cos(\lambda_1)) / \lambda_1^3$

T_i [C]	T_o [C]	T_r [C]	Q [kJ]
2	-1.658	-5.369	6.861
4	-0.08803	-4.236	7.668
6	1.482	-3.103	8.476
8	3.051	-1.97	9.283
10	4.621	-0.8371	10.09
12	6.191	0.296	10.9
14	7.76	1.429	11.7
16	9.33	2.562	12.51
18	10.9	3.695	13.32
20	12.47	4.828	14.13
22	14.04	5.961	14.93
24	15.61	7.094	15.74
26	17.18	8.227	16.55
28	18.75	9.36	17.35
30	20.32	10.49	18.16



4-52 An orange is exposed to very cold ambient air. It is to be determined whether the orange will freeze in 4 h in subfreezing temperatures.

Assumptions 1 The orange is spherical in shape with a diameter of 8 cm. 2 Heat conduction in the orange is one-dimensional because of symmetry about the midpoint. 3 The thermal properties of the orange are constant, and are those of water. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of the orange are approximated by those of water at the average temperature of about 5°C, $k = 0.571 \text{ W/m} \cdot ^\circ\text{C}$ and $\alpha = k / \rho C_p = 0.571 / (1000 \times 4205) = 0.136 \times 10^{-6} \text{ m}^2 / \text{s}$ (Table A-9).

Analysis The Biot number is

$$Bi = \frac{hr_o}{k} = \frac{(15 \text{ W/m}^2 \cdot ^\circ\text{C})(0.04 \text{ m})}{(0.571 \text{ W/m} \cdot ^\circ\text{C})} = 1.051 \approx 1.0$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 1.5708 \quad \text{and} \quad A_1 = 1.2732$$

The Fourier number is

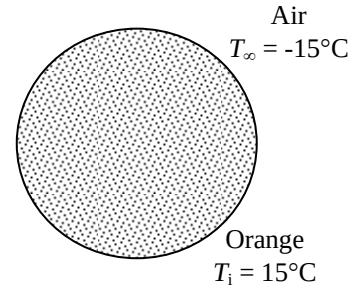
$$\tau = \frac{\alpha t}{L^2} = \frac{(0.136 \times 10^{-6} \text{ m}^2/\text{s})(4 \text{ h} \times 3600 \text{ s/h})}{(0.04 \text{ m})^2} = 1.224 > 0.2$$

Therefore, the one-term approximate solution (or the transient temperature charts) is applicable. Then the temperature at the surface of the oranges becomes

$$\theta(r_o, t)_{sph} = \frac{T(r_o, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r_o / r_o)}{\lambda_1 r_o / r_o} = (1.2732) e^{-(1.5708)^2 (1.224)} \frac{\sin(1.5708 \text{ rad})}{1.5708} = 0.0396$$

$$\frac{T(r_o, t) - (-6)}{15 - (-6)} = 0.0396 \longrightarrow T(r_o, t) = -5.2^\circ\text{C}$$

which is less than 0°C. Therefore, the oranges will freeze.



4-53 A hot baked potato is taken out of the oven and wrapped so that no heat is lost from it. The time the potato is baked in the oven and the final equilibrium temperature of the potato after it is wrapped are to be determined.

Assumptions 1 The potato is spherical in shape with a diameter of 8 cm. 2 Heat conduction in the potato is one-dimensional because of symmetry about the midpoint. 3 The thermal properties of the potato are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of the potato are given to be $k = 0.6 \text{ W/m}\cdot^\circ\text{C}$, $\rho = 1100 \text{ kg/m}^3$, $C_p = 3.9 \text{ kJ/kg}\cdot^\circ\text{C}$, and $\alpha = 1.4 \times 10^{-7} \text{ m}^2/\text{s}$.

Analysis (a) The Biot number is

$$Bi = \frac{hr_o}{k} = \frac{(25 \text{ W/m}^2\cdot^\circ\text{C})(0.04 \text{ m})}{(0.6 \text{ W/m}\cdot^\circ\text{C})} = 1.67$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 1.8777 \quad \text{and} \quad A_1 = 1.4113$$

Then the Fourier number and the time period become

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{70 - 170}{25 - 170} = 0.69 = (1.4113) e^{-(1.8777)^2 \tau} \longrightarrow \tau = 0.203 > 0.2$$

The baking time of the potatoes is determined to be

$$t = \frac{\tau_o^2}{\alpha} = \frac{(0.203)(0.04 \text{ m})^2}{(1.4 \times 10^{-7} \text{ m}^2/\text{s})} = 2320 \text{ s} = \mathbf{38.7 \text{ min}}$$

(b) The maximum amount of heat transfer is

$$m = \rho V = \rho \frac{4}{3} \pi r_o^3 = (1100 \text{ kg/m}^3) \left[\frac{4}{3} \pi (0.04 \text{ m})^3 \right] = 0.295 \text{ kg}$$

$$Q_{\max} = m C_p (T_\infty - T_i) = (0.295 \text{ kg})(3.900 \text{ kJ/kg}\cdot^\circ\text{C})(170 - 25)^\circ\text{C} = 166.76 \text{ kJ}$$

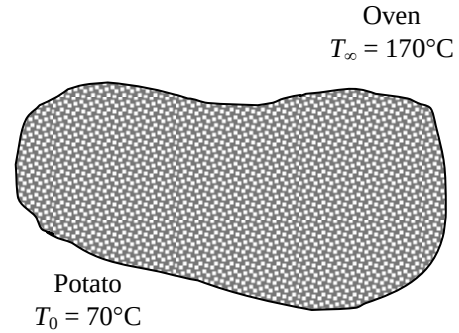
Then the actual amount of heat transfer becomes

$$\frac{Q}{Q_{\max}} = 1 - 3\theta_{0,sph} \frac{\sin(\lambda_1) - \lambda_1 \cos(\lambda_1)}{\lambda_1^3} = 1 - 3(0.69) \frac{\sin(1.8777) - (1.8777) \cos(1.8777)}{(1.8777)^3} = 0.525$$

$$Q = 0.525 Q_{\max} = (0.525)(166.76 \text{ kJ}) = \mathbf{87.5 \text{ kJ}}$$

The final equilibrium temperature of the potato after it is wrapped is

$$Q = m C_p (T_{eqv} - T_i) \longrightarrow T_{eqv} = T_i + \frac{Q}{m C_p} = 25^\circ\text{C} + \frac{87.5 \text{ kJ}}{(0.295 \text{ kg})(3.9 \text{ kJ/kg}\cdot^\circ\text{C})} = \mathbf{101^\circ\text{C}}$$



4-54 The center temperature of potatoes is to be lowered to 6°C during cooling. The cooling time and if any part of the potatoes will suffer chilling injury during this cooling process are to be determined.

Assumptions 1 The potatoes are spherical in shape with a radius of $r_0 = 3$ cm. 2 Heat conduction in the potato is one-dimensional in the radial direction because of the symmetry about the midpoint. 3 The thermal properties of the potato are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal conductivity and thermal diffusivity of potatoes are given to be $k = 0.50$ W/m·°C and $\alpha = 0.13 \times 10^{-6}$ m²/s.

Analysis First we find the Biot number:

$$Bi = \frac{hr_0}{k} = \frac{(19 \text{ W/m}^2 \cdot ^\circ\text{C})(0.03 \text{ m})}{0.5 \text{ W/m} \cdot ^\circ\text{C}} = 1.14$$

From Table 4-1 we read, for a sphere, $\lambda_1 = 1.635$ and $A_1 = 1.302$. Substituting these values into the one-term solution gives

$$\theta_0 = \frac{T_o - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \rightarrow \frac{6 - 2}{25 - 2} = 1.302 e^{-(1.635)^2 \tau} \rightarrow \tau = 0.753$$

which is greater than 0.2 and thus the one-term solution is applicable. Then the cooling time becomes

$$\tau = \frac{\alpha t}{r_0^2} \rightarrow t = \frac{\tau r_0^2}{\alpha} = \frac{(0.753)(0.03 \text{ m})^2}{0.13 \times 10^{-6} \text{ m}^2/\text{s}} = 5213 \text{ s} = \mathbf{1.45 \text{ h}}$$

The lowest temperature during cooling will occur on the surface ($r/r_0 = 1$), and is determined to be

$$\frac{T(r) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r / r_0)}{\lambda_1 r / r_0} \rightarrow \frac{T(r_0) - T_\infty}{T_i - T_\infty} = \theta_0 \frac{\sin(\lambda_1 r_0 / r_0)}{\lambda_1 r_0 / r_0} = \frac{T_o - T_\infty}{T_i - T_\infty} \frac{\sin(\lambda_1 r_0 / r_0)}{\lambda_1 r_0 / r_0}$$

$$\text{Substituting, } \frac{T(r_0) - 2}{25 - 2} = \left(\frac{6 - 2}{25 - 2} \right) \frac{\sin(1.635 \text{ rad})}{1.635} \rightarrow T(r_0) = 4.44^\circ\text{C}$$

which is above the temperature range of 3 to 4 °C for chilling injury for potatoes. Therefore, **no part** of the potatoes will experience chilling injury during this cooling process.

Alternative solution We could also solve this problem using transient temperature charts as follows:

$$\left. \begin{aligned} \frac{1}{Bi} &= \frac{k}{hr_0} = \frac{0.50 \text{ W/m} \cdot ^\circ\text{C}}{(19 \text{ W/m}^2 \cdot ^\circ\text{C})(0.03 \text{ m})} = 0.877 \\ \frac{T_o - T_\infty}{T_i - T_\infty} &= \frac{6 - 2}{25 - 2} = 0.174 \end{aligned} \right\} \tau = \frac{\alpha t}{r_0^2} = 0.75 \quad (\text{Fig. 4-15a})$$

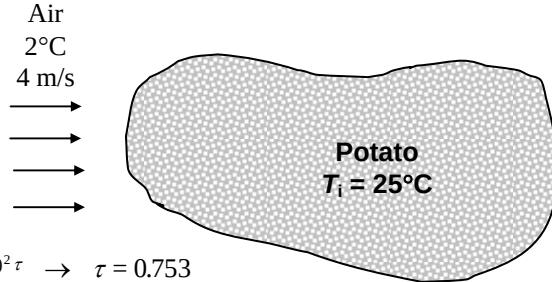
$$\text{Therefore, } t = \frac{\tau r_0^2}{\alpha} = \frac{(0.75)(0.03)^2}{0.13 \times 10^{-6} \text{ m}^2/\text{s}} = 5192 \text{ s} \cong \mathbf{1.44 \text{ h}}$$

The surface temperature is determined from

$$\left. \begin{aligned} \frac{1}{Bi} &= \frac{k}{hr_0} = 0.877 \\ \frac{r}{r_0} &= 1 \end{aligned} \right\} \frac{T(r) - T_\infty}{T_o - T_\infty} = 0.6 \quad (\text{Fig. 4-15b})$$

which gives $T_{\text{surface}} = T_\infty + 0.6(T_o - T_\infty) = 2 + 0.6(6 - 2) = 4.4^\circ\text{C}$

The slight difference between the two results is due to the reading error of the charts.



4-55E The center temperature of oranges is to be lowered to 40°F during cooling. The cooling time and if any part of the oranges will freeze during this cooling process are to be determined.

Assumptions 1 The oranges are spherical in shape with a radius of $r_0 = 1.25$ in = 0.1042 ft. 2 Heat conduction in the orange is one-dimensional in the radial direction because of the symmetry about the midpoint. 3 The thermal properties of the orange are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal conductivity and thermal diffusivity of oranges are given to be $k = 0.26$ Btu/h·ft·°F and $\alpha = 1.4 \times 10^{-6}$ ft²/s.

Analysis First we find the Biot number:

$$Bi = \frac{hr_0}{k} = \frac{(4.6 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(1.25/12 \text{ ft})}{0.26 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{C}} = 1.843$$

From Table 4-1 we read, for a sphere, $\lambda_1 = 1.9569$ and $A_1 = 1.447$. Substituting these values into the one-term solution gives

$$\theta_0 = \frac{T_o - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \rightarrow \frac{40 - 25}{78 - 25} = 1.447 e^{-(1.9569)^2 \tau} \rightarrow \tau = 0.426$$

which is greater than 0.2 and thus the one-term solution is applicable. Then the cooling time becomes

$$\tau = \frac{\alpha t}{r_0^2} \rightarrow t = \frac{\tau r_0^2}{\alpha} = \frac{(0.426)(1.25/12 \text{ ft})^2}{1.4 \times 10^{-6} \text{ ft}^2/\text{s}} = 3302 \text{ s} = \mathbf{55.0 \text{ min}}$$

The lowest temperature during cooling will occur on the surface ($r/r_0 = 1$), and is determined to be

$$\frac{T(r) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r / r_0)}{\lambda_1 r / r_0} \rightarrow \frac{T(r_0) - T_\infty}{T_i - T_\infty} = \theta_0 \frac{\sin(\lambda_1 r_0 / r_0)}{\lambda_1 r_0 / r_0} = \frac{T_o - T_\infty}{T_i - T_\infty} \frac{\sin(\lambda_1 r_0 / r_0)}{\lambda_1 r_0 / r_0}$$

Substituting,
$$\frac{T(r_0) - 25}{78 - 25} = \left(\frac{40 - 25}{78 - 25} \right) \frac{\sin(1.9569 \text{ rad})}{1.9569} \rightarrow T(r_0) = 32.1^\circ\text{F}$$

which is above the freezing temperature of 31 °C for oranges . Therefore, no part of the oranges will freeze during this cooling process.

Alternative solution We could also solve this problem using transient temperature charts as follows:

$$\left. \begin{aligned} \frac{1}{Bi} &= \frac{k}{hr_0} = \frac{0.26 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}}{(4.6 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(1.25/12 \text{ ft})} = 0.543 \\ \frac{T_o - T_\infty}{T_i - T_\infty} &= \frac{40 - 25}{78 - 25} = 0.283 \end{aligned} \right\} \tau = \frac{\alpha t}{r_0^2} = 0.43 \quad (\text{Fig.4-15a})$$

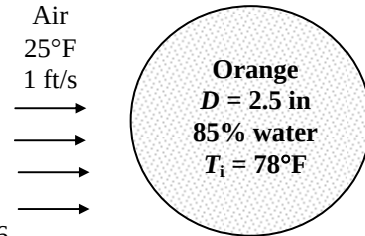
Therefore,
$$t = \frac{\tau r_0^2}{\alpha} = \frac{(0.43)(1.25/12 \text{ ft})^2}{1.4 \times 10^{-6} \text{ ft}^2/\text{s}} = 3333 \text{ s} = 55.5 \text{ min}$$

The lowest temperature during cooling will occur on the surface ($r/r_0 = 1$) of the oranges is determined to be

$$\left. \begin{aligned} \frac{1}{Bi} &= \frac{k}{hr_0} = 0.543 \\ \frac{r}{r_0} &= 1 \end{aligned} \right\} \frac{T(r) - T_\infty}{T_o - T_\infty} = 0.45 \quad (\text{Fig.4-15b})$$

which gives
$$T_{\text{surface}} = T_\infty + 0.45(T_o - T_\infty) = 25 + 0.45(40 - 25) = 31.8^\circ\text{F}$$

The slight difference between the two results is due to the reading error of the charts.



4-56 The center temperature of a beef carcass is to be lowered to 4°C during cooling. The cooling time and if any part of the carcass will suffer freezing injury during this cooling process are to be determined.

Assumptions 1 The beef carcass can be approximated as a cylinder with insulated top and base surfaces having a radius of $r_0 = 12$ cm and a height of $H = 1.4$ m. **2** Heat conduction in the carcass is one-dimensional in the radial direction because of the symmetry about the centerline. **3** The thermal properties of the carcass are constant. **4** The heat transfer coefficient is constant and uniform over the entire surface. **5** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal conductivity and thermal diffusivity of carcass are given to be $k = 0.47$ W/m·°C and $\alpha = 0.13 \times 10^{-6}$ m²/s.

Analysis First we find the Biot number:

$$Bi = \frac{hr_0}{k} = \frac{(22 \text{ W/m}^2 \cdot ^\circ\text{C})(0.12 \text{ m})}{0.47 \text{ W/m} \cdot ^\circ\text{C}} = 5.62$$

From Table 4-1 we read, for a cylinder, $\lambda_1 = 2.027$ and $A_1 = 1.517$.

Substituting these values into the one-term solution gives

$$\theta_0 = \frac{T_o - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \rightarrow \frac{4 - (-6)}{37 - (-6)} = 1.517 e^{-(2.027)^2 \tau} \rightarrow \tau = 0.456$$

which is greater than 0.2 and thus the one-term solution is applicable.

Then the cooling time becomes

$$\tau = \frac{\alpha t}{r_0^2} \rightarrow t = \frac{\tau r_0^2}{\alpha} = \frac{(0.456)(0.12 \text{ m})^2}{0.13 \times 10^{-6} \text{ m}^2/\text{s}} = 50,558 \text{ s} = \mathbf{14.0 \text{ h}}$$

The lowest temperature during cooling will occur on the surface ($r/r_0 = 1$), and is determined to be

$$\frac{T(r) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} J_0(\lambda_1 r / r_0) \rightarrow \frac{T(r_0) - T_\infty}{T_i - T_\infty} = \theta_0 J_0(\lambda_1 r / r_0) = \frac{T_o - T_\infty}{T_i - T_\infty} J_0(\lambda_1 r_0 / r_0)$$

$$\text{Substituting, } \frac{T(r_0) - (-6)}{37 - (-6)} = \left(\frac{4 - (-6)}{37 - (-6)} \right) J_0(\lambda_1) = 0.2326 \times 0.2084 = 0.0485 \rightarrow T(r_0) = -3.9^\circ\text{C}$$

which is below the freezing temperature of -1.7°C . Therefore, the outer part of the beef carcass will freeze during this cooling process.

Alternative solution We could also solve this problem using transient temperature charts as follows:

$$\left. \begin{aligned} \frac{1}{Bi} = \frac{k}{hr_0} &= \frac{0.47 \text{ W/m} \cdot ^\circ\text{C}}{(22 \text{ W/m}^2 \cdot ^\circ\text{C})(0.12 \text{ m})} = 0.178 \\ \frac{T_o - T_\infty}{T_i - T_\infty} &= \frac{4 - (-6)}{37 - (-6)} = 0.23 \end{aligned} \right\} \tau = \frac{\alpha t}{r_0^2} = 0.4 \quad (\text{Fig.4-14a})$$

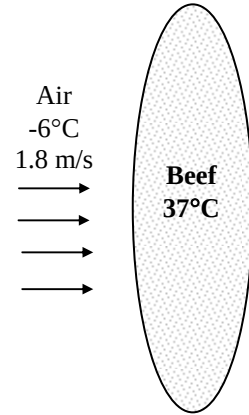
$$\text{Therefore, } t = \frac{\tau r_0^2}{\alpha} = \frac{(0.4)(0.12 \text{ m})^2}{0.13 \times 10^{-6} \text{ m}^2/\text{s}} = 44,308 \text{ s} \approx 12.3 \text{ h}$$

The surface temperature is determined from

$$\left. \begin{aligned} \frac{1}{Bi} = \frac{k}{hr_0} &= 0.178 \\ \frac{r}{r_0} &= 1 \end{aligned} \right\} \frac{T(r) - T_\infty}{T_o - T_\infty} = 0.17 \quad (\text{Fig.4-14b})$$

$$\text{which gives } T_{\text{surface}} = T_\infty + 0.17(T_o - T_\infty) = -6 + 0.17[4 - (-6)] = -4.3^\circ\text{C}$$

The difference between the two results is due to the reading error of the charts.



4-57 The center temperature of meat slabs is to be lowered to -18°C during cooling. The cooling time and the surface temperature of the slabs at the end of the cooling process are to be determined.

Assumptions 1 The meat slabs can be approximated as very large plane walls of half-thickness $L = 11.5$ cm. 2 Heat conduction in the meat slabs is one-dimensional because of the symmetry about the centerplane. 3 The thermal properties of the meat slabs are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified). 6 The phase change effects are not considered, and thus the actual cooling time will be much longer than the value determined.

Properties The thermal conductivity and thermal diffusivity of meat slabs are given to be $k = 0.47 \text{ W/m}\cdot^{\circ}\text{C}$ and $\alpha = 0.13 \times 10^{-6} \text{ m}^2/\text{s}$. These properties will be used for both fresh and frozen meat.

Analysis First we find the Biot number:

$$\text{Bi} = \frac{hr_0}{k} = \frac{(20 \text{ W/m}^2\cdot^{\circ}\text{C})(0.115 \text{ m})}{0.47 \text{ W/m}\cdot^{\circ}\text{C}} = 4.89$$

From Table 4-1 we read, for a plane wall, $\lambda_1 = 1.308$ and $A_1 = 1.239$. Substituting these values into the one-term solution gives

$$\theta_0 = \frac{T_o - T_{\infty}}{T_i - T_{\infty}} = A_1 e^{-\lambda_1^2 \tau} \rightarrow \frac{-18 - (-30)}{7 - (-30)} = 1.239 e^{-(1.308)^2 \tau} \rightarrow \tau = 0.783$$

which is greater than 0.2 and thus the one-term solution is applicable. Then the cooling time becomes

$$\tau = \frac{\alpha t}{L^2} \rightarrow t = \frac{\tau L^2}{\alpha} = \frac{(0.783)(0.115 \text{ m})^2}{0.13 \times 10^{-6} \text{ m}^2/\text{s}} = 79,650 \text{ s} = \mathbf{22.1 \text{ h}}$$

The lowest temperature during cooling will occur on the surface ($x/L = 1$), and is determined to be

$$\frac{T(x) - T_{\infty}}{T_i - T_{\infty}} = A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 x / L) \rightarrow \frac{T(L) - T_{\infty}}{T_i - T_{\infty}} = \theta_0 \cos(\lambda_1 L / L) = \frac{T_o - T_{\infty}}{T_i - T_{\infty}} \cos(\lambda_1)$$

Substituting,

$$\frac{T(L) - (-30)}{7 - (-30)} = \left(\frac{-18 - (-30)}{7 - (-30)} \right) \cos(\lambda_1) = 0.3243 \times 0.2598 = 0.08425 \rightarrow T(L) = \mathbf{-26.9^{\circ}\text{C}}$$

which is close the temperature of the refrigerated air.

Alternative solution We could also solve this problem using transient temperature charts as follows:

$$\left. \begin{aligned} \frac{1}{\text{Bi}} &= \frac{k}{hL} = \frac{0.47 \text{ W/m}\cdot^{\circ}\text{C}}{(20 \text{ W/m}^2\cdot^{\circ}\text{C})(0.115 \text{ m})} = 0.204 \\ \frac{T_o - T_{\infty}}{T_i - T_{\infty}} &= \frac{-18 - (-30)}{7 - (-30)} = 0.324 \end{aligned} \right\} \tau = \frac{\alpha t}{L^2} = 0.75 \quad (\text{Fig.4-13a})$$

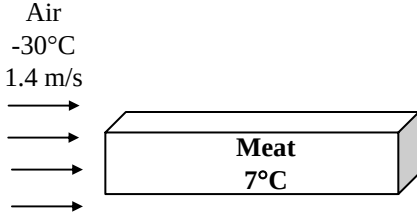
$$\text{Therefore, } t = \frac{\tau r_o^2}{\alpha} = \frac{(0.75)(0.115 \text{ m})^2}{0.13 \times 10^{-6} \text{ m}^2/\text{s}} = 76,300 \text{ s} \approx 21.2 \text{ h}$$

The surface temperature is determined from

$$\left. \begin{aligned} \frac{1}{\text{Bi}} &= \frac{k}{hL} = 0.204 \\ \frac{x}{L} &= 1 \end{aligned} \right\} \frac{T(x) - T_{\infty}}{T_o - T_{\infty}} = 0.22 \quad (\text{Fig.4-13b})$$

$$\text{which gives } T_{\text{surface}} = T_{\infty} + 0.22(T_o - T_{\infty}) = -30 + 0.22[-18 - (-30)] = -27.4^{\circ}\text{C}$$

The slight difference between the two results is due to the reading error of the charts.

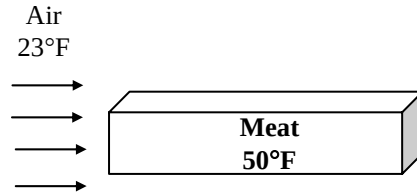


4-58E The center temperature of meat slabs is to be lowered to 36°F during 12-h of cooling. The average heat transfer coefficient during this cooling process is to be determined.

Assumptions **1** The meat slabs can be approximated as very large plane walls of half-thickness $L = 3$ -in. **2** Heat conduction in the meat slabs is one-dimensional because of symmetry about the centerplane. **3** The thermal properties of the meat slabs are constant. **4** The heat transfer coefficient is constant and uniform over the entire surface. **5** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal conductivity and thermal diffusivity of meat slabs are given to be $k = 0.26$ Btu/h·ft·°F and $\alpha = 1.4 \times 10^{-6}$ ft²/s.

Analysis The average heat transfer coefficient during this cooling process is determined from the transient temperature charts for a flat plate as follows:



$$\left. \begin{aligned} \tau &= \frac{\alpha t}{L^2} = \frac{(1.4 \times 10^{-6} \text{ ft}^2/\text{s})(12 \times 3600 \text{ s})}{(3/12 \text{ ft})^2} = 0.968 \\ \frac{T_o - T_\infty}{T_i - T_\infty} &= \frac{36 - 23}{50 - 23} = 0.481 \end{aligned} \right\} \frac{1}{Bi} = 0.7 \quad (\text{Fig. 4-13a})$$

Therefore,

$$h = \frac{kBi}{L} = \frac{(0.26 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F})(1/0.7)}{(3/12) \text{ ft}} = \mathbf{1.5 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}}$$

Discussion We could avoid the uncertainty associated with the reading of the charts and obtain a more accurate result by using the one-term solution relation for an infinite plane wall, but it would require a trial and error approach since the Bi number is not known.

4-59 Chickens are to be chilled by holding them in agitated brine for 2.5 h. The center and surface temperatures of the chickens are to be determined, and if any part of the chickens will freeze during this cooling process is to be assessed.

Assumptions 1 The chickens are spherical in shape. 2 Heat conduction in the chickens is one-dimensional in the radial direction because of symmetry about the midpoint. 3 The thermal properties of the chickens are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified). 6 The phase change effects are not considered, and thus the actual the temperatures will be much higher than the values determined since a considerable part of the cooling process will occur during phase change (freezing of chicken).

Properties The thermal conductivity, thermal diffusivity, and density of chickens are given to be $k = 0.45$ W/m·°C, $\alpha = 0.13 \times 10^{-6}$ m²/s, and $\rho = 950$ kg/m³. These properties will be used for both fresh and frozen chicken.

Analysis We first find the volume and equivalent radius of the chickens:

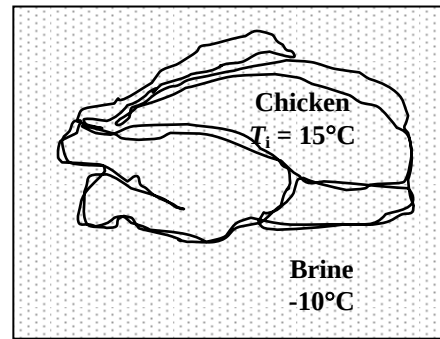
$$V = m / \rho = 1700 \text{ g} / (0.95 \text{ g/cm}^3) = 1789 \text{ cm}^3$$

$$r_o = \left(\frac{3}{4\pi} V \right)^{1/3} = \left(\frac{3}{4\pi} 1789 \text{ cm}^3 \right)^{1/3} = 7.53 \text{ cm} = 0.0753 \text{ m}$$

Then the Biot and Fourier numbers become

$$\text{Bi} = \frac{hr_o}{k} = \frac{(440 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0753 \text{ m})}{0.45 \text{ W/m} \cdot ^\circ\text{C}} = 73.6$$

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.13 \times 10^{-6} \text{ m}^2/\text{s})(2.5 \times 3600 \text{ s})}{(0.0753 \text{ m})^2} = 0.2063$$



Note that $\tau = 0.207 > 0.2$, and thus the one-term solution is applicable. From Table 4-1 we read, for a sphere, $\lambda_1 = 3.094$ and $A_1 = 1.998$. Substituting these values into the one-term solution gives

$$\theta_0 = \frac{T_o - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \rightarrow \frac{T_o - (-10)}{15 - (-10)} = 1.998 e^{-(3.094)^2 (0.2063)} = 0.277 \rightarrow T_o = -3.1^\circ\text{C}$$

The lowest temperature during cooling will occur on the surface ($r/r_o = 1$), and is determined to be

$$\frac{T(r) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r / r_o)}{\lambda_1 r / r_o} \rightarrow \frac{T(r_o) - T_\infty}{T_i - T_\infty} = \theta_0 \frac{\sin(\lambda_1 r_o / r_o)}{\lambda_1 r_o / r_o} = \frac{T_o - T_\infty}{T_i - T_\infty} \frac{\sin(\lambda_1 r_o / r_o)}{\lambda_1 r_o / r_o}$$

$$\text{Substituting, } \frac{T(r_o) - (-10)}{15 - (-10)} = 0.277 \frac{\sin(3.094 \text{ rad})}{3.094} \rightarrow T(r_o) = -9.9^\circ\text{C}$$

The entire chicken will freeze during this process since the freezing point of chicken is -2.8°C , and even the center temperature of chicken is below this value.

Discussion We could also solve this problem using transient temperature charts, but the data in this case falls at a point on the chart which is very difficult to read:

$$\left. \begin{aligned} \tau &= \frac{\alpha t}{r_o^2} = \frac{(0.13 \times 10^{-6} \text{ m}^2/\text{s})(2.5 \times 3600 \text{ s})}{(0.0753 \text{ m})^2} = 0.206 \\ \frac{1}{\text{Bi}} &= \frac{k}{hr_o} = \frac{0.45 \text{ W/m} \cdot ^\circ\text{C}}{(440 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0753 \text{ m})} = 0.0136 \end{aligned} \right\} \frac{T_o - T_\infty}{T_i - T_\infty} = 0.15 \dots 0.30 \text{ ?? (Fig.4-15)}$$

Transient Heat Conduction in Semi-Infinite Solids

4-60C A semi-infinite medium is an idealized body which has a single exposed plane surface and extends to infinity in all directions. The earth and thick walls can be considered to be semi-infinite media.

4-61C A thick plane wall can be treated as a semi-infinite medium if all we are interested in is the variation of temperature in a region near one of the surfaces for a time period during which the temperature in the mid section of the wall does not experience any change.

4-62C The total amount of heat transfer from a semi-infinite solid up to a specified time t_0 can be determined by integration from

$$Q = \int_0^{t_0} Ah[T(0, t) - T_\infty] dt$$

where the surface temperature $T(0, t)$ is obtained from Eq. 4-22 by substituting $x = 0$.

4-63 The water pipes are buried in the ground to prevent freezing. The minimum burial depth at a particular location is to be determined.

Assumptions 1 The temperature in the soil is affected by the thermal conditions at one surface only, and thus the soil can be considered to be a semi-infinite medium with a specified surface temperature. **2** The thermal properties of the soil are constant.

Properties The thermal properties of the soil are given to be $k = 0.35 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 0.15 \times 10^{-6} \text{ m}^2/\text{s}$.

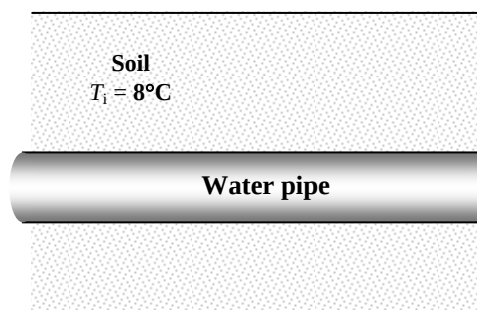
Analysis The length of time the snow pack stays on the ground is

$$t = (60 \text{ days})(24 \text{ hr / days})(3600 \text{ s / hr}) = 5.184 \times 10^6 \text{ s}$$

$$T_s = -8^\circ\text{C}$$

The surface is kept at -18°C at all times. The depth at which freezing at 0°C occurs can be determined from the analytical solution,

$$\begin{aligned} \frac{T(x, t) - T_i}{T_s - T_i} &= \text{erfc}\left(\frac{x}{\sqrt{\alpha t}}\right) \\ \frac{0 - 8}{-8 - 8} &= \text{erfc}\left(\frac{x}{2\sqrt{(0.15 \times 10^{-6} \text{ m}^2/\text{s})(5.184 \times 10^6 \text{ s})}}\right) \\ \longrightarrow 0.444 &= \text{erfc}\left(\frac{x}{1.7636}\right) \end{aligned}$$



Then from Table 4-3 we get $\frac{x}{1.7636} = 0.5297 \longrightarrow x = \mathbf{0.934 \text{ m}}$

Discussion The solution could also be determined using the chart, but it would be subject to reading error.

4-64 An area is subjected to cold air for a 10-h period. The soil temperatures at distances 0, 10, 20, and 50 cm from the earth's surface are to be determined.

Assumptions 1 The temperature in the soil is affected by the thermal conditions at one surface only, and thus the soil can be considered to be a semi-infinite medium with a specified surface temperature. **2** The thermal properties of the soil are constant.

Properties The thermal properties of the soil are given to be $k = 0.9 \text{ W/m} \cdot ^\circ\text{C}$ and $\alpha = 1.6 \times 10^{-5} \text{ m}^2/\text{s}$.

Analysis The one-dimensional transient temperature distribution in the ground can be determined from

$$\frac{T(x, t) - T_i}{T_\infty - T_i} = \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right) - \exp\left(\frac{hx}{k} + \frac{h^2 \alpha t}{k^2}\right) \left[\text{erfc}\left(\frac{x}{2\sqrt{\alpha t}} + \frac{h\sqrt{\alpha t}}{k}\right) \right]$$



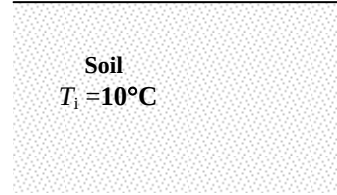
Winds

$T_\infty = -10^\circ\text{C}$

where

$$\frac{h\sqrt{\alpha t}}{k} = \frac{(40 \text{ W/m}^2 \cdot ^\circ\text{C})\sqrt{(1.6 \times 10^{-5} \text{ m}^2/\text{s})(10 \times 3600 \text{ s})}}{0.9 \text{ W/m} \cdot ^\circ\text{C}} = 33.7$$

$$\frac{h^2 \alpha t}{k^2} = \left(\frac{h\sqrt{\alpha t}}{k}\right)^2 = 33.7^2 = 1138$$



Then we conclude that the last term in the temperature distribution relation above must be zero regardless of x despite the exponential term tending to infinity since (1) $\text{erfc}(\xi) \rightarrow 0$ for $\xi > 4$ (see Table 4-3) and (2) the term has to remain less than 1 to have physically meaningful solutions. That is,

$$\exp\left(\frac{hx}{k} + \frac{h^2 \alpha t}{k^2}\right) \left[\text{erfc}\left(\frac{x}{2\sqrt{\alpha t}} + \frac{h\sqrt{\alpha t}}{k}\right) \right] = \exp\left(\frac{hx}{k} + 1138\right) \left[\text{erfc}\left(\frac{x}{2\sqrt{\alpha t}} + 33.3\right) \right] \cong 0$$

Therefore, the temperature distribution relation simplifies to

$$\frac{T(x, t) - T_i}{T_\infty - T_i} = \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right) \rightarrow T(x, t) = T_i + (T_\infty - T_i) \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

Then the temperatures at 0, 10, 20, and 50 cm depth from the ground surface become

$$x = 0: \quad T(0, 10 \text{ h}) = T_i + (T_\infty - T_i) \text{erfc}\left(\frac{0}{2\sqrt{\alpha t}}\right) = T_i + (T_\infty - T_i) \text{erfc}(0) = T_i + (T_\infty - T_i) \times 1 = T_\infty = -10^\circ\text{C}$$

$$x = 0.1 \text{ m}: \quad T(0.1 \text{ m}, 10 \text{ h}) = 10 + (-10 - 10) \text{erfc}\left(\frac{0.1 \text{ m}}{2\sqrt{(1.6 \times 10^{-5} \text{ m}^2/\text{s})(10 \text{ h} \times 3600 \text{ s/h})}}\right)$$

$$= 10 - 20 \text{erfc}(0.066) = 10 - 20 \times 0.9257 = -8.5^\circ\text{C}$$

$$x = 0.2 \text{ m}: \quad T(0.2 \text{ m}, 10 \text{ h}) = 10 + (-10 - 10) \text{erfc}\left(\frac{0.2 \text{ m}}{2\sqrt{(1.6 \times 10^{-5} \text{ m}^2/\text{s})(10 \text{ h} \times 3600 \text{ s/h})}}\right)$$

$$= 10 - 20 \text{erfc}(0.132) = 10 - 20 \times 0.8519 = -7.0^\circ\text{C}$$

$$x = 0.5 \text{ m}: \quad T(0.5 \text{ m}, 10 \text{ h}) = 10 + (-10 - 10) \text{erfc}\left(\frac{0.5 \text{ m}}{2\sqrt{(1.6 \times 10^{-5} \text{ m}^2/\text{s})(10 \text{ h} \times 3600 \text{ s/h})}}\right)$$

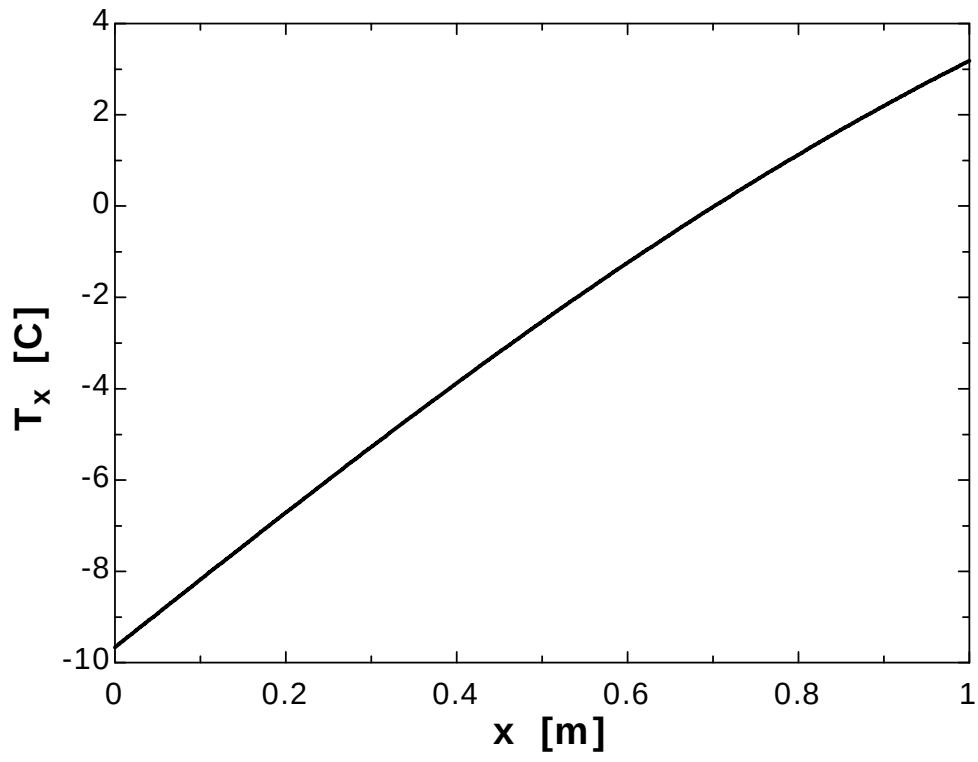
$$= 10 - 20 \text{erfc}(0.329) = 10 - 20 \times 0.6418 = -2.8^\circ\text{C}$$

4-65

"!PROBLEM 4-65"**"GIVEN"** $T_i = 10$ "[C]" $T_{\infty} = -10$ "[C]" $h = 40$ "[W/m²-C]" $\text{time} = 10 \times 3600$ "[s]" $x = 0.1$ [m], parameter to be varied**"PROPERTIES"** $k = 0.9$ "[W/m-C]" $\alpha = 1.6 \times 10^{-5}$ "[m²/s]"**"ANALYSIS"**

$$\frac{(T_x - T_i)}{(T_{\infty} - T_i)} = \text{erfc}\left(\frac{x}{2\sqrt{\alpha \text{time}}}\right) - \exp\left(\frac{h^2 x}{k^2 \alpha \text{time}}\right) \text{erfc}\left(\frac{x}{2\sqrt{\alpha \text{time}}}\right) + \frac{h\sqrt{\alpha \text{time}}}{k}$$

x [m]	T_x [C]
0	-9.666
0.05	-8.923
0.1	-8.183
0.15	-7.447
0.2	-6.716
0.25	-5.993
0.3	-5.277
0.35	-4.572
0.4	-3.878
0.45	-3.197
0.5	-2.529
0.55	-1.877
0.6	-1.24
0.65	-0.6207
0.7	-0.01894
0.75	0.5643
0.8	1.128
0.85	1.672
0.9	2.196
0.95	2.7
1	3.183



4-66 The walls of a furnace made of concrete are exposed to hot gases at the inner surfaces. The time it will take for the temperature of the outer surface of the furnace to change is to be determined.

Assumptions 1 The temperature in the wall is affected by the thermal conditions at inner surfaces only and the convection heat transfer coefficient inside is given to be very large. Therefore, the wall can be considered to be a semi-infinite medium with a specified surface temperature of 1800°F. 2 The thermal properties of the concrete wall are constant.

Properties The thermal properties of the concrete are given to be $k = 0.64 \text{ Btu/h.ft.}^\circ\text{F}$ and $\alpha = 0.023 \text{ ft}^2/\text{h}$.

Analysis The one-dimensional transient temperature distribution in the wall for that time period can be determined from

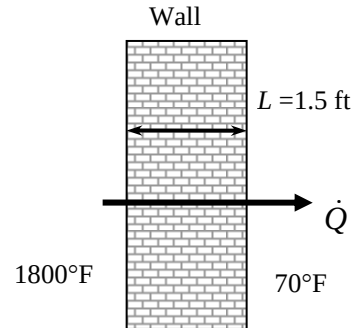
$$\frac{T(x,t) - T_i}{T_s - T_i} = \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

But,

$$\frac{T(x,t) - T_i}{T_s - T_i} = \frac{70.1 - 70}{1800 - 70} = 0.00006 \rightarrow 0.00006 = \text{erfc}(2.85) \quad (\text{Table 4-3})$$

Therefore,

$$\frac{x}{2\sqrt{\alpha t}} = 2.85 \rightarrow t = \frac{x^2}{4 \times (2.85)^2 \alpha} = \frac{(1.5 \text{ ft})^2}{4 \times (2.85)^2 (0.023 \text{ ft}^2/\text{h})} = 3.01 \text{ h} = \mathbf{181 \text{ min}}$$



4-67 A thick wood slab is exposed to hot gases for a period of 5 minutes. It is to be determined whether the wood will ignite.

Assumptions **1** The wood slab is treated as a semi-infinite medium subjected to convection at the exposed surface. **2** The thermal properties of the wood slab are constant. **3** The heat transfer coefficient is constant and uniform over the entire surface.

Properties The thermal properties of the wood are $k = 0.17 \text{ W/m} \cdot ^\circ\text{C}$ and $\alpha = 1.28 \times 10^{-7} \text{ m}^2/\text{s}$.

Analysis The one-dimensional transient temperature distribution in the wood can be determined from

$$\frac{T(x, t) - T_i}{T_\infty - T_i} = \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right) - \exp\left(\frac{hx}{k} + \frac{h^2 \alpha t}{k^2}\right) \left[\text{erfc}\left(\frac{x}{2\sqrt{\alpha t}} + \frac{h\sqrt{\alpha t}}{k}\right) \right]$$

where

$$\frac{h\sqrt{\alpha t}}{k} = \frac{(35 \text{ W/m}^2 \cdot ^\circ\text{C})\sqrt{(1.28 \times 10^{-7} \text{ m}^2/\text{s})(5 \times 60 \text{ s})}}{0.17 \text{ W/m} \cdot ^\circ\text{C}} = 1.276$$

$$\frac{h^2 \alpha t}{k^2} = \left(\frac{h\sqrt{\alpha t}}{k}\right)^2 = 1.276^2 = 1.628$$

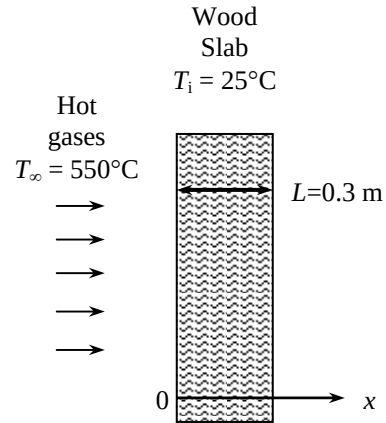
Noting that $x = 0$ at the surface and using Table 4-3 for *erfc* values,

$$\begin{aligned} \frac{T(x, t) - 25}{550 - 25} &= \text{erfc}(0) - \exp(0 + 1.628) \text{erfc}(0 + 1.276) \\ &= 1 - (5.0937)(0.0727) \\ &= 0.630 \end{aligned}$$

Solving for $T(x, t)$ gives

$$T(x, t) = 356^\circ\text{C}$$

which is less than the ignition temperature of 450°C . Therefore, the wood will not ignite.



4-68 The outer surfaces of a large cast iron container filled with ice are exposed to hot water. The time before the ice starts melting and the rate of heat transfer to the ice are to be determined.

Assumptions 1 The temperature in the container walls is affected by the thermal conditions at outer surfaces only and the convection heat transfer coefficient outside inside is given to be very large. Therefore, the wall can be considered to be a semi-infinite medium with a specified surface temperature. **2** The thermal properties of the wall are constant.

Properties The thermal properties of the cast iron are given to be $k = 52 \text{ W/m} \cdot ^\circ\text{C}$ and $\alpha = 1.70 \times 10^{-5} \text{ m}^2/\text{s}$.

Analysis The one-dimensional transient temperature distribution in the wall for that time period can be determined from

$$\frac{T(x,t) - T_i}{T_s - T_i} = \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

But,

$$\frac{T(x,t) - T_i}{T_s - T_i} = \frac{0.1 - 0}{60 - 0} = 0.00167 \rightarrow 0.00167 = \text{erfc}(2.225) \quad (\text{Table 4-3})$$

Therefore,

$$\frac{x}{2\sqrt{\alpha t}} = 2.225 \rightarrow t = \frac{x^2}{4 \times (2.225)^2 \alpha} = \frac{(0.05 \text{ m})^2}{4(2.225)^2 (1.7 \times 10^{-5} \text{ m}^2/\text{s})} = 7.4 \text{ s}$$

The rate of heat transfer to the ice when steady operation conditions are reached can be determined by applying the thermal resistance network concept as

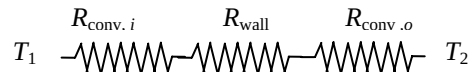
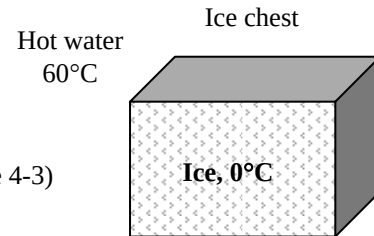
$$R_{\text{conv},i} = \frac{1}{h_i A} = \frac{1}{(250 \text{ W/m}^2 \cdot ^\circ\text{C})(1.2 \times 2 \text{ m}^2)} = 0.00167^\circ\text{C/W}$$

$$R_{\text{wall}} = \frac{L}{kA} = \frac{0.05 \text{ m}}{(52 \text{ W/m} \cdot ^\circ\text{C})(1.2 \times 2 \text{ m}^2)} = 0.00040^\circ\text{C/W}$$

$$R_{\text{conv},o} = \frac{1}{h_o A} = \frac{1}{(\infty)(1.2 \times 2 \text{ m}^2)} \cong 0^\circ\text{C/W}$$

$$R_{\text{total}} = R_{\text{conv},1} + R_{\text{wall}} + R_{\text{conv},2} = 0.00167 + 0.00040 + 0 = 0.00207^\circ\text{C/W}$$

$$\dot{Q} = \frac{T_2 - T_1}{R_{\text{total}}} = \frac{(60 - 0)^\circ\text{C}}{0.00207^\circ\text{C/W}} = 28,990 \text{ W}$$



Transient Heat Conduction in Multidimensional Systems

4-69C The product solution enables us to determine the dimensionless temperature of two- or three-dimensional heat transfer problems as the product of dimensionless temperatures of one-dimensional heat transfer problems. The dimensionless temperature for a two-dimensional problem is determined by determining the dimensionless temperatures in both directions, and taking their product.

4-70C The dimensionless temperature for a three-dimensional heat transfer is determined by determining the dimensionless temperatures of one-dimensional geometries whose intersection is the three dimensional geometry, and taking their product.

4-71C This short cylinder is physically formed by the intersection of a long cylinder and a plane wall. The dimensionless temperatures at the center of plane wall and at the center of the cylinder are determined first. Their product yields the dimensionless temperature at the center of the short cylinder.

4-72C The heat transfer in this short cylinder is one-dimensional since there is no heat transfer in the axial direction. The temperature will vary in the radial direction only.

4-73 A short cylinder is allowed to cool in atmospheric air. The temperatures at the centers of the cylinder and the top surface as well as the total heat transfer from the cylinder for 15 min of cooling are to be determined.

Assumptions 1 Heat conduction in the short cylinder is two-dimensional, and thus the temperature varies in both the axial x - and the radial r - directions. 2 The thermal properties of the cylinder are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of brass are given to be $\rho = 8530 \text{ kg/m}^3$, $C_p = 0.389 \text{ kJ/kg} \cdot ^\circ\text{C}$, $k = 110 \text{ W/m} \cdot ^\circ\text{C}$, and $\alpha = 3.39 \times 10^{-5} \text{ m}^2/\text{s}$.

Analysis This short cylinder can physically be formed by the intersection of a long cylinder of radius $D/2 = 4 \text{ cm}$ and a plane wall of thickness $2L = 15 \text{ cm}$. We measure x from the midplane.

(a) The Biot number is calculated for the plane wall to be

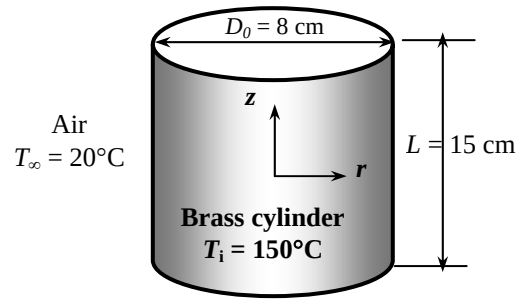
$$Bi = \frac{hL}{k} = \frac{(40 \text{ W/m}^2 \cdot ^\circ\text{C})(0.075 \text{ m})}{(110 \text{ W/m} \cdot ^\circ\text{C})} = 0.02727$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 0.164 \quad \text{and} \quad A_1 = 1.0050$$

The Fourier number is

$$\tau = \frac{\alpha t}{L^2} = \frac{(3.39 \times 10^{-5} \text{ m}^2/\text{s})(15 \text{ min} \times 60 \text{ s/min})}{(0.075 \text{ m})^2} = 5.424 > 0.2$$



Therefore, the one-term approximate solution (or the transient temperature charts) is applicable. Then the dimensionless temperature at the center of the plane wall is determined from

$$\theta_{o,wall} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.0050) e^{-(0.164)^2 (5.424)} = 0.869$$

We repeat the same calculations for the long cylinder,

$$Bi = \frac{hr_0}{k} = \frac{(40 \text{ W/m}^2 \cdot ^\circ\text{C})(0.04 \text{ m})}{(110 \text{ W/m} \cdot ^\circ\text{C})} = 0.01455$$

$$\lambda_1 = 0.1704 \quad \text{and} \quad A_1 = 1.0038$$

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(3.39 \times 10^{-5} \text{ m}^2/\text{s})(15 \times 60 \text{ s})}{(0.04 \text{ m})^2} = 19.069 > 0.2$$

$$\theta_{o,cyl} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.0038) e^{-(0.1704)^2 (19.069)} = 0.577$$

Then the center temperature of the short cylinder becomes

$$\left[\frac{T(0,0,t) - T_\infty}{T_i - T_\infty} \right]_{\text{short cylinder}} = \theta_{o,wall} \times \theta_{o,cyl} = 0.869 \times 0.577 = 0.501$$

$$\frac{T(0,0,t) - 20}{150 - 20} = 0.501 \longrightarrow T(0,0,t) = 85.1^\circ\text{C}$$

(b) The center of the top surface of the cylinder is still at the center of the long cylinder ($r = 0$), but at the outer surface of the plane wall ($x = L$). Therefore, we first need to determine the dimensionless temperature at the surface of the wall.

$$\theta(L, t)_{wall} = \frac{T(x, t) - T_{\infty}}{T_i - T_{\infty}} = A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 L / L) = (1.0050) e^{-(0.164)^2 (5.424)} \cos(0.164) = 0.857$$

Then the center temperature of the top surface of the cylinder becomes

$$\left[\frac{T(L, 0, t) - T_{\infty}}{T_i - T_{\infty}} \right]_{short\ cylinder} = \theta(L, t)_{wall} \times \theta_{o, cyl} = 0.857 \times 0.577 = 0.494$$

$$\frac{T(L, 0, t) - 20}{150 - 20} = 0.494 \longrightarrow T(L, 0, t) = \mathbf{84.2^{\circ}C}$$

(c) We first need to determine the maximum heat can be transferred from the cylinder

$$m = \rho V = \rho \pi r_o^2 L = (8530 \text{ kg/m}^3) [\pi (0.04 \text{ m})^2 (0.15 \text{ m})] = 6.43 \text{ kg}$$

$$Q_{max} = m C_p (T_i - T_{\infty}) = (6.43 \text{ kg})(0.389 \text{ kJ/kg} \cdot ^{\circ}\text{C})(150 - 20)^{\circ}\text{C} = 325 \text{ kJ}$$

Then we determine the dimensionless heat transfer ratios for both geometries as

$$\left(\frac{Q}{Q_{max}} \right)_{wall} = 1 - \theta_{o, wall} \frac{\sin(\lambda_1)}{\lambda_1} = 1 - (0.869) \frac{\sin(0.164)}{0.164} = 0.135$$

$$\left(\frac{Q}{Q_{max}} \right)_{cyl} = 1 - 2\theta_{o, cyl} \frac{J_1(\lambda_1)}{\lambda_1} = 1 - 2(0.577) \frac{0.0846}{0.1704} = 0.427$$

The heat transfer ratio for the short cylinder is

$$\left(\frac{Q}{Q_{max}} \right)_{short\ cylinder} = \left(\frac{Q}{Q_{max}} \right)_{plane\ wall} + \left(\frac{Q}{Q_{max}} \right)_{long\ cylinder} \left[1 - \left(\frac{Q}{Q_{max}} \right)_{plane\ wall} \right] = 0.135 + (0.427)(1 - 0.135) = 0.504$$

Then the total heat transfer from the short cylinder during the first 15 minutes of cooling becomes

$$Q = 0.503 Q_{max} = (0.504)(325 \text{ kJ}) = \mathbf{164 \text{ kJ}}$$

4-74

PROBLEM 4-74**GIVEN** $D = 0.08 \text{ [m]}$ $r_o = D/2$ $\text{height} = 0.15 \text{ [m]}$ $L = \text{height}/2$ $T_i = 150 \text{ [}^\circ\text{C]}$ $T_\infty = 20 \text{ [}^\circ\text{C]}$ $h = 40 \text{ [W/m}^2\text{-}^\circ\text{C]}$ $\text{time} = 15 \text{ [min]}$, parameter to be varied**PROPERTIES** $k = 110 \text{ [W/m-}^\circ\text{C]}$ $\rho = 8530 \text{ [kg/m}^3\text{]}$ $C_p = 0.389 \text{ [kJ/kg-}^\circ\text{C]}$ $\alpha = 3.39 \times 10^{-5} \text{ [m}^2\text{/s]}$ **ANALYSIS****(a)**

This short cylinder can physically be formed by the intersection of a long cylinder of radius r_o and a plane wall of thickness $2L$.

For plane wall

 $B_{i_w} = (hL)/k$

From Table 4-1 corresponding to this B_i number, we read

 $\lambda_{1_w} = 0.2282$ "w stands for wall" $A_{1_w} = 1.0060$ $\tau_w = (\alpha \cdot \text{time} \cdot \text{Convert}(\text{min}, \text{s}))/L^2$ $\theta_{o_w} = A_{1_w} \exp(-\lambda_{1_w}^2 \tau_w)$ " $\theta_{o_w} = (T_{o_w} - T_\infty)/(T_i - T_\infty)$ "

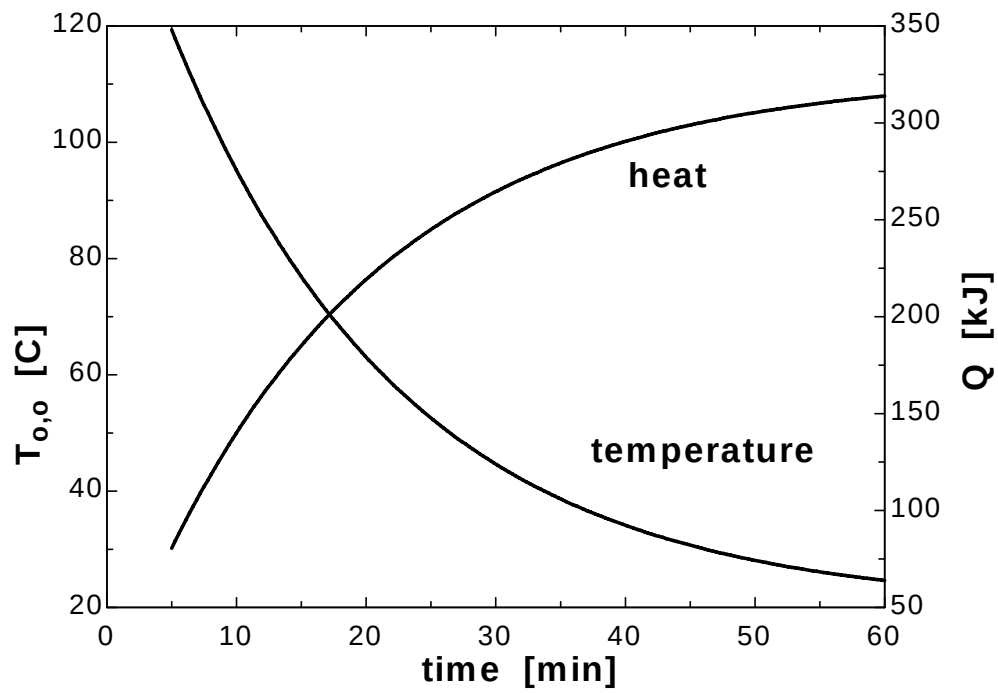
For long cylinder

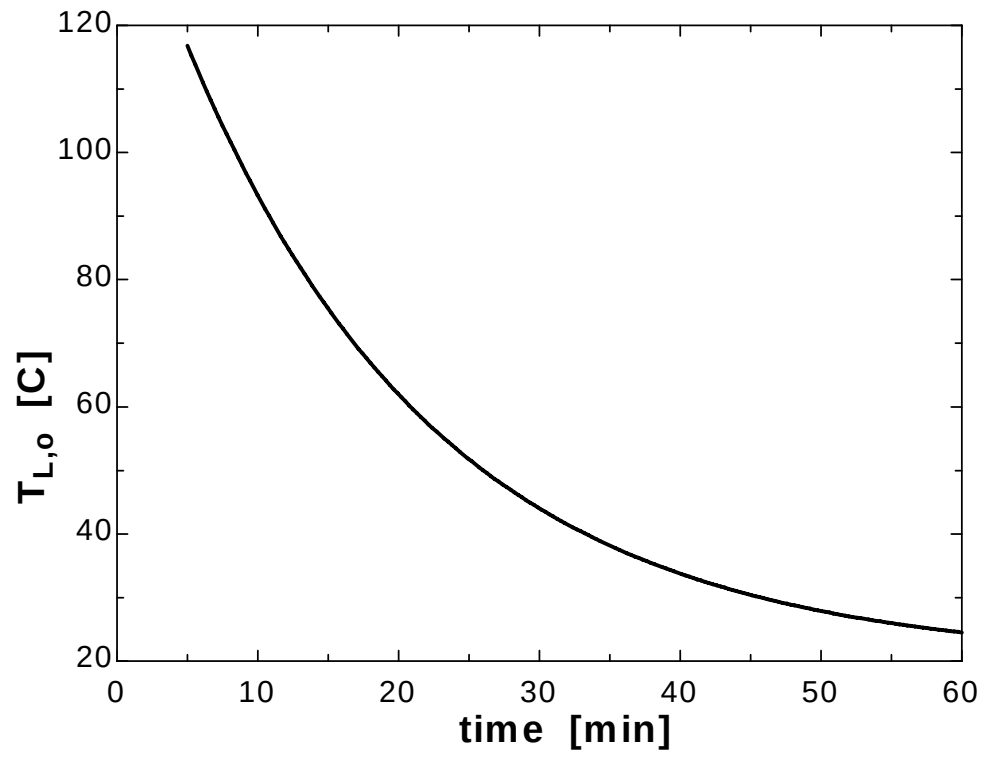
 $B_{i_c} = (hr_o)/k$ "c stands for cylinder"

From Table 4-1 corresponding to this B_i number, we read

 $\lambda_{1_c} = 0.1704$ $A_{1_c} = 1.0038$ $\tau_c = (\alpha \cdot \text{time} \cdot \text{Convert}(\text{min}, \text{s}))/r_o^2$ $\theta_{o_c} = A_{1_c} \exp(-\lambda_{1_c}^2 \tau_c)$ " $\theta_{o_c} = (T_{o_c} - T_\infty)/(T_i - T_\infty)$ " $(T_{o_o} - T_\infty)/(T_i - T_\infty) = \theta_{o_w} \theta_{o_c}$ "center temperature of short cylinder"**(b)** $\theta_{L_w} = A_{1_w} \exp(-\lambda_{1_w}^2 \tau_w) \cos(\lambda_{1_w} L/L)$ " $\theta_{L_w} = (T_{L_w} - T_\infty)/(T_i - T_\infty)$ " $(T_{L_o} - T_\infty)/(T_i - T_\infty) = \theta_{L_w} \theta_{o_c}$ "center temperature of the top surface"**(c)** $V = \pi r_o^2 (2L)$ $m = \rho V$ $Q_{\max} = m C_p (T_i - T_\infty)$ $Q_w = 1 - \theta_{o_w} \sin(\lambda_{1_w})/\lambda_{1_w}$ " $Q_w = (Q/Q_{\max})_w$ " $Q_c = 1 - 2\theta_{o_c} J_1/\lambda_{1_c}$ " $Q_c = (Q/Q_{\max})_c$ " $J_1 = 0.0846$ "From Table 4-2, at λ_{1_c} " $Q/Q_{\max} = Q_w + Q_c (1 - Q_w)$ "total heat transfer"

time [min]	$T_{o,o}$ [C]	$T_{L,o}$ [C]	Q [kJ]
5	119.3	116.8	80.58
10	95.18	93.23	140.1
15	76.89	75.42	185.1
20	63.05	61.94	219.2
25	52.58	51.74	245
30	44.66	44.02	264.5
35	38.66	38.18	279.3
40	34.12	33.75	290.5
45	30.69	30.41	298.9
50	28.09	27.88	305.3
55	26.12	25.96	310.2
60	24.63	24.51	313.8





4-75 A semi-infinite aluminum cylinder is cooled by water. The temperature at the center of the cylinder 10 cm from the end surface is to be determined.

Assumptions 1 Heat conduction in the semi-infinite cylinder is two-dimensional, and thus the temperature varies in both the axial x - and the radial r - directions. 2 The thermal properties of the cylinder are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of aluminum are given to be $k = 237 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 9.71 \times 10^{-5} \text{ m}^2/\text{s}$.

Analysis This semi-infinite cylinder can physically be formed by the intersection of a long cylinder of radius $r_o = D/2 = 7.5 \text{ cm}$ and a semi-infinite medium. The dimensionless temperature 5 cm from the surface of a semi-infinite medium is first determined from

$$\begin{aligned} \frac{T(x,t)-T_i}{T_\infty-T_i} &= \text{erfc}\left(\frac{x}{\sqrt{\alpha t}}\right) - \exp\left(\frac{hx}{k} + \frac{h^2 \alpha t}{k^2}\right) \left[\text{erfc}\left(\frac{x}{2\sqrt{\alpha t}} + \frac{h\sqrt{\alpha t}}{k}\right) \right] \\ &= \text{erfc}\left(\frac{0.05}{2\sqrt{(9.71 \times 10^{-5})(8 \times 60)}}\right) - \exp\left(\frac{(140)(0.05)}{237} + \frac{(140)^2 (9.71 \times 10^{-5})(8 \times 60)}{(237)^2}\right) \\ &\quad \times \left[\text{erfc}\left(\frac{0.05}{2\sqrt{(9.71 \times 10^{-5})(8 \times 60)}} + \frac{(140)\sqrt{(9.71 \times 10^{-5})(8 \times 60)}}{237}\right) \right] \\ &= \text{erfc}(0.1158) - \exp(0.0458) \text{erfc}(0.2433) = 0.8699 - (1.0468)(0.7308) = 0.1049 \end{aligned}$$

$$\theta_{\text{semi-inf}} = \frac{T(x,t)-T_\infty}{T_i-T_\infty} = 1 - 0.1049 = 0.8951$$

The Biot number is calculated for the long cylinder to be

$$Bi = \frac{hr_o}{k} = \frac{(140 \text{ W/m}^2 \cdot ^\circ\text{C})(0.075 \text{ m})}{(237 \text{ W/m}\cdot^\circ\text{C})} = 0.0443$$

The constants λ_1 and A_1 corresponding to this

Biot number are, from Table 4-1,

$$\lambda_1 = 0.2948 \quad \text{and} \quad A_1 = 1.0110$$

The Fourier number is

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(9.71 \times 10^{-5} \text{ m}^2/\text{s})(8 \times 60 \text{ s})}{(0.075 \text{ m})^2} = 8.286 > 0.2$$

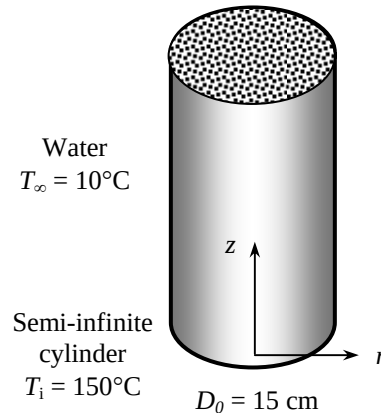
Therefore, the one-term approximate solution (or the transient temperature charts) is applicable.

Then the dimensionless temperature at the center of the plane wall is determined from

$$\theta_{o,\text{cyl}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.0110) e^{-(0.2948)^2 (8.286)} = 0.4921$$

The center temperature of the semi-infinite cylinder then becomes

$$\begin{aligned} \left[\frac{T(x,0,t)-T_\infty}{T_i-T_\infty} \right]_{\text{semi-infinite cylinder}} &= \theta_{\text{semi-inf}}(x,t) \times \theta_{o,\text{cyl}} = 0.8951 \times 0.4921 = 0.4405 \\ \left[\frac{T(x,0,t)-10}{150-10} \right]_{\text{semi-infinite cylinder}} &= 0.4405 \longrightarrow T(x,0,t) = 71.7^\circ\text{C} \end{aligned}$$



4-76E A hot dog is dropped into boiling water. The center temperature of the hot dog is to be determined by treating hot dog as a finite cylinder and also as an infinitely long cylinder.

Assumptions 1 When treating hot dog as a finite cylinder, heat conduction in the hot dog is two-dimensional, and thus the temperature varies in both the axial x - and the radial r - directions. When treating hot dog as an infinitely long cylinder, heat conduction is one-dimensional in the radial r - direction. 2 The thermal properties of the hot dog are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of the hot dog are given to be $k = 0.44 \text{ Btu/h.ft.}^\circ\text{F}$, $\rho = 61.2 \text{ lbm/ft}^3$, $C_p = 0.93 \text{ Btu/lbm.}^\circ\text{F}$, and $\alpha = 0.0077 \text{ ft}^2/\text{h}$.

Analysis (a) This hot dog can physically be formed by the intersection of a long cylinder of radius $r_o = D/2 = (0.4/12) \text{ ft}$ and a plane wall of thickness $2L = (5/12) \text{ ft}$. The distance x is measured from the midplane.

After 5 minutes

First the Biot number is calculated for the plane wall to be

$$Bi = \frac{hL}{k} = \frac{(120 \text{ Btu/h.ft.}^\circ\text{F})(2.5/12 \text{ ft})}{(0.44 \text{ Btu/h.ft.}^\circ\text{F})} = 56.8$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 1.5421 \quad \text{and} \quad A_1 = 1.2728$$

The Fourier number is

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(5/60 \text{ h})}{(2.5/12 \text{ ft})^2} = 0.015 < 0.2 \quad (\text{Be cautious!})$$

Then the dimensionless temperature at the center of the plane wall is determined from

$$\theta_{o,\text{wall}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.2728)e^{-(1.5421)^2(0.015)} \cong 1$$

We repeat the same calculations for the long cylinder,

$$Bi = \frac{hr_o}{k} = \frac{(120 \text{ Btu/h.ft.}^\circ\text{F})(0.4/12 \text{ ft})}{(0.44 \text{ Btu/h.ft.}^\circ\text{F})} = 9.1$$

$$\lambda_1 = 2.1589 \quad \text{and} \quad A_1 = 1.5618$$

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(5/60 \text{ h})}{(0.4/12 \text{ ft})^2} = 0.578 > 0.2$$

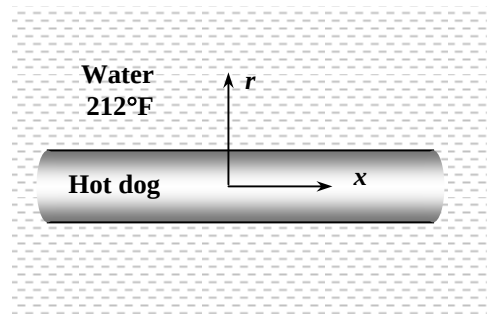
$$\theta_{o,\text{cyl}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.5618)e^{-(2.1589)^2(0.578)} = 0.106$$

Then the center temperature of the short cylinder becomes

$$\left[\frac{T(0,0,t) - T_\infty}{T_i - T_\infty} \right]_{\text{short cylinder}} = \theta_{o,\text{wall}} \times \theta_{o,\text{cyl}} = 1 \times 0.106 = 0.106$$

$$\frac{T(0,0,t) - 212}{40 - 212} = 0.106 \longrightarrow T(0,0,t) = \mathbf{194^\circ\text{F}}$$

After 10 minutes



$$\tau = \frac{\alpha t}{L^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(10/60 \text{ h})}{(2.5/12 \text{ ft})^2} = 0.03 < 0.2 \quad (\text{Be cautious!})$$

$$\theta_{o,wall} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.2728)e^{-(1.5421)^2 (0.03)} \cong 1$$

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(10/60 \text{ h})}{(0.4/12 \text{ ft})^2} = 1.156 > 0.2$$

$$\theta_{o,cyl} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.5618)e^{-(2.1589)^2 (1.156)} = 0.007$$

$$\left[\frac{T(0,0,t) - T_\infty}{T_i - T_\infty} \right]_{\text{short cylinder}} = \theta_{o,wall} \times \theta_{o,cyl} = 1 \times 0.007 = 0.007$$

$$\frac{T(0,0,t) - 212}{40 - 212} = 0.007 \longrightarrow T(0,0,t) = 211^\circ\text{F}$$

After 15 minutes

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(15/60 \text{ h})}{(2.5/12 \text{ ft})^2} = 0.045 < 0.2 \quad (\text{Be cautious!})$$

$$\theta_{o,wall} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.2728)e^{-(1.5421)^2 (0.045)} \cong 1$$

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(15/60 \text{ h})}{(0.4/12 \text{ ft})^2} = 1.734 > 0.2$$

$$\theta_{o,cyl} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.5618)e^{-(2.1589)^2 (1.734)} = 0.0005$$

$$\left[\frac{T(0,0,t) - T_\infty}{T_i - T_\infty} \right]_{\text{short cylinder}} = \theta_{o,wall} \times \theta_{o,cyl} = 1 \times 0.0005 = 0.0005$$

$$\frac{T(0,0,t) - 212}{40 - 212} = 0.0005 \longrightarrow T(0,0,t) = 212^\circ\text{F}$$

(b) Treating the hot dog as an infinitely long cylinder will not change the results obtained in the part (a) since dimensionless temperatures for the plane wall is 1 for all cases.

4-77E A hot dog is dropped into boiling water. The center temperature of the hot dog is to be determined by treating hot dog as a finite cylinder and an infinitely long cylinder.

Assumptions **1** When treating hot dog as a finite cylinder, heat conduction in the hot dog is two-dimensional, and thus the temperature varies in both the axial x - and the radial r - directions. When treating hot dog as an infinitely long cylinder, heat conduction is one-dimensional in the radial r - direction. **2** The thermal properties of the hot dog are constant. **3** The heat transfer coefficient is constant and uniform over the entire surface. **4** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of the hot dog are given to be $k = 0.44 \text{ Btu/h.ft.}^\circ\text{F}$, $\rho = 61.2 \text{ lbm/ft}^3$, $C_p = 0.93 \text{ Btu/lbm.}^\circ\text{F}$, and $\alpha = 0.0077 \text{ ft}^2/\text{h}$.

Analysis (a) This hot dog can physically be formed by the intersection of a long cylinder of radius $r_o = D/2 = (0.4/12) \text{ ft}$ and a plane wall of thickness $2L = (5/12) \text{ ft}$. The distance x is measured from the midplane.

After 5 minutes

First the Biot number is calculated for the plane wall to be

$$Bi = \frac{hL}{k} = \frac{(120 \text{ Btu/h.ft.}^\circ\text{F})(2.5/12 \text{ ft})}{(0.44 \text{ Btu/h.ft.}^\circ\text{F})} = 56.8$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 1.5421 \quad \text{and} \quad A_1 = 1.2728$$

The Fourier number is

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(5/60 \text{ h})}{(2.5/12 \text{ ft})^2} = 0.015 < 0.2 \quad (\text{Be cautious!})$$

Then the dimensionless temperature at the center of the plane wall is determined from

$$\theta_{o,\text{wall}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.2728)e^{-(1.5421)^2 (0.015)} \cong 1$$

We repeat the same calculations for the long cylinder,

$$Bi = \frac{hr_o}{k} = \frac{(120 \text{ Btu/h.ft.}^\circ\text{F})(0.4/12 \text{ ft})}{(0.44 \text{ Btu/h.ft.}^\circ\text{F})} = 9.1$$

$$\lambda_1 = 2.1589 \quad \text{and} \quad A_1 = 1.5618$$

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(5/60 \text{ h})}{(0.4/12 \text{ ft})^2} = 0.578 > 0.2$$

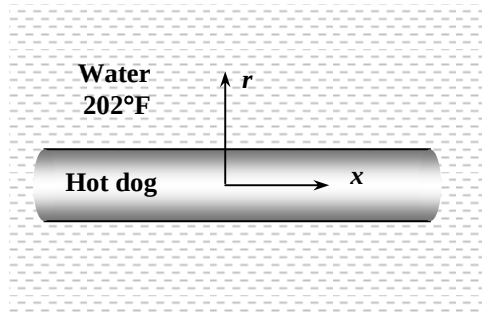
$$\theta_{o,\text{cyl}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.5618)e^{-(2.1589)^2 (0.578)} = 0.106$$

Then the center temperature of the short cylinder becomes

$$\left[\frac{T(0,0,t) - T_\infty}{T_i - T_\infty} \right]_{\text{short cylinder}} = \theta_{o,\text{wall}} \times \theta_{o,\text{cyl}} = 1 \times 0.106 = 0.106$$

$$\frac{T(0,0,t) - 202}{40 - 202} = 0.106 \longrightarrow T(0,0,t) = \mathbf{185^\circ\text{F}}$$

After 10 minutes



$$\tau = \frac{\alpha t}{L^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(10/60 \text{ h})}{(2.5/12 \text{ ft})^2} = 0.03 < 0.2 \quad (\text{Be cautious!})$$

$$\theta_{o,wall} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.2728)e^{-(1.5421)^2 (0.03)} \cong 1$$

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(10/60 \text{ h})}{(0.4/12 \text{ ft})^2} = 1.156 > 0.2$$

$$\theta_{o,cyl} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.5618)e^{-(2.1589)^2 (1.156)} = 0.007$$

$$\left[\frac{T(0,0,t) - T_\infty}{T_i - T_\infty} \right]_{\text{short cylinder}} = \theta_{o,wall} \times \theta_{o,cyl} = 1 \times 0.007 = 0.007$$

$$\frac{T(0,0,t) - 202}{40 - 202} = 0.007 \longrightarrow T(0,0,t) = \mathbf{201^\circ\text{F}}$$

After 15 minutes

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(15/60 \text{ h})}{(2.5/12 \text{ ft})^2} = 0.045 < 0.2 \quad (\text{Be cautious!})$$

$$\theta_{o,wall} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.2728)e^{-(1.5421)^2 (0.045)} \cong 1$$

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.0077 \text{ ft}^2/\text{h})(15/60 \text{ h})}{(0.4/12 \text{ ft})^2} = 1.734 > 0.2$$

$$\theta_{o,cyl} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.5618)e^{-(2.1589)^2 (1.734)} = 0.0005$$

$$\left[\frac{T(0,0,t) - T_\infty}{T_i - T_\infty} \right]_{\text{short cylinder}} = \theta_{o,wall} \times \theta_{o,cyl} = 1 \times 0.0005 = 0.0005$$

$$\frac{T(0,0,t) - 202}{40 - 202} = 0.0005 \longrightarrow T(0,0,t) = \mathbf{202^\circ\text{F}}$$

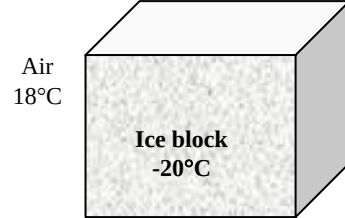
(b) Treating the hot dog as an infinitely long cylinder will not change the results obtained in the part (a) since dimensionless temperatures for the plane wall is 1 for all cases.

4-78 A rectangular ice block is placed on a table. The time the ice block starts melting is to be determined.

Assumptions 1 Heat conduction in the ice block is two-dimensional, and thus the temperature varies in both x - and y - directions. 2 The thermal properties of the ice block are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of the ice are given to be $k = 2.22 \text{ W/m} \cdot ^\circ\text{C}$ and $\alpha = 0.124 \times 10^{-7} \text{ m}^2/\text{s}$.

Analysis This rectangular ice block can be treated as a short rectangular block that can physically be formed by the intersection of two infinite plane wall of thickness $2L = 4 \text{ cm}$ and an infinite plane wall of thickness $2L = 10 \text{ cm}$. We measure x from the bottom surface of the block since this surface represents the adiabatic center surface of the plane wall of thickness $2L = 10 \text{ cm}$. Since the melting starts at the corner of the top surface, we need to determine the time required to melt ice block which will happen when the temperature drops below 0°C at this location. The Biot numbers and the corresponding constants are first determined to be



$$Bi_{\text{wall},1} = \frac{hL_1}{k} = \frac{(12 \text{ W/m}^2 \cdot ^\circ\text{C})(0.02 \text{ m})}{(2.22 \text{ W/m} \cdot ^\circ\text{C})} = 0.1081 \longrightarrow \lambda_1 = 0.3208 \quad \text{and} \quad A_1 = 1.0173$$

$$Bi_{\text{wall},3} = \frac{hL_3}{k} = \frac{(12 \text{ W/m}^2 \cdot ^\circ\text{C})(0.05 \text{ m})}{(2.22 \text{ W/m} \cdot ^\circ\text{C})} = 0.2703 \longrightarrow \lambda_1 = 0.4951 \quad \text{and} \quad A_1 = 1.0408$$

The ice will start melting at the corners because of the maximum exposed surface area there. Noting that $\tau = \alpha t / L^2$ and assuming that $\tau > 0.2$ in all dimensions so that the one-term approximate solution for transient heat conduction is applicable, the product solution method can be written for this problem as

$$\begin{aligned} \theta(L_1, L_2, L_3, t)_{\text{block}} &= \theta(L_1, t)_{\text{wall},1}^2 \theta(L_3, t)_{\text{wall},2} \\ \frac{0-18}{-20-18} &= \left[A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 L_1 / L_1) \right]^2 \left[A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 L_3 / L_3) \right] \\ 0.4737 &= \left\{ (1.0173) \exp \left[-(0.3208)^2 \frac{(0.124 \times 10^{-7})t}{(0.02)^2} \right] \cos(0.3208) \right\}^2 \\ &\quad \times \left\{ (1.0408) \exp \left[-(0.4951)^2 \frac{(0.124 \times 10^{-7})t}{(0.05)^2} \right] \cos(0.4951) \right\} \\ &\longrightarrow t = 108,135 \text{ s} = \mathbf{30.04 \text{ hours}} \end{aligned}$$

Therefore, the ice will start melting in about 30 hours.

Discussion Note that

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.124 \times 10^{-7} \text{ m}^2/\text{s})(1108,135 \text{ s/h})}{(0.05 \text{ m})^2} = 0.536 > 0.2$$

and thus the assumption of $\tau > 0.2$ for the applicability of the one-term approximate solution is verified.

4-79

"!PROBLEM 4-79"**"GIVEN"**

$$2L_1 = 0.04 \text{ [m]}$$

$$L_2 = L_1$$

$$2L_3 = 0.10 \text{ [m]}$$

$T_i = -20 \text{ [C]}$, parameter to be varied

$$T_{\infty} = 18 \text{ [C]}$$

$$h = 12 \text{ [W/m}^2\text{-C]}$$

$$T_{L1_L2_L3} = 0 \text{ [C]}$$

"PROPERTIES"

$$k = 2.22 \text{ [W/m-C]}$$

$$\alpha = 0.124 \times 10^{-7} \text{ [m}^2\text{/s]}$$

"ANALYSIS"

"This block can physically be formed by the intersection of two infinite plane wall of thickness $2L = 4 \text{ cm}$ and an infinite plane wall of thickness $2L = 10 \text{ cm}$ "

"For the two plane walls"

$$Bi_{w1} = (hL_1)/k$$

"From Table 4-1 corresponding to this Bi number, we read"

$$\lambda_{1_w1} = 0.3208 \text{ "w stands for wall"}$$

$$A_{1_w1} = 1.0173$$

$$\text{time} \times \text{Convert}(\text{min}, \text{s}) = \tau_{w1} L_1^2 / \alpha$$

"For the third plane wall"

$$Bi_{w3} = (hL_3)/k$$

"From Table 4-1 corresponding to this Bi number, we read"

$$\lambda_{1_w3} = 0.4951$$

$$A_{1_w3} = 1.0408$$

$$\text{time} \times \text{Convert}(\text{min}, \text{s}) = \tau_{w3} L_3^2 / \alpha$$

$$\theta_{L_w1} = A_{1_w1} \exp(-\lambda_{1_w1}^2 \tau_{w1}) \cos(\lambda_{1_w1} L_1 / L_1)$$

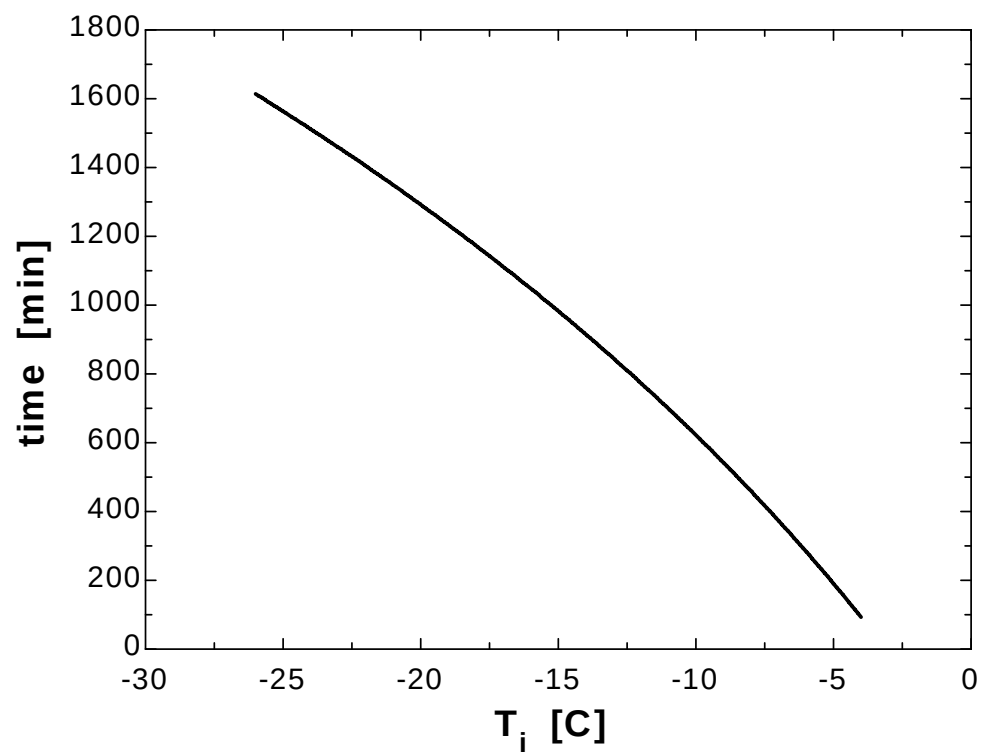
$$\theta_{L_w1} = (T_{L_w1} - T_{\infty}) / (T_i - T_{\infty})$$

$$\theta_{L_w3} = A_{1_w3} \exp(-\lambda_{1_w3}^2 \tau_{w3}) \cos(\lambda_{1_w3} L_3 / L_3)$$

$$\theta_{L_w3} = (T_{L_w3} - T_{\infty}) / (T_i - T_{\infty})$$

$$(T_{L1_L2_L3} - T_{\infty}) / (T_i - T_{\infty}) = \theta_{L_w1}^2 \theta_{L_w3} \text{ "corner temperature"}$$

$T_i \text{ [C]}$	time [min]
-26	1614
-24	1512
-22	1405
-20	1292
-18	1173
-16	1048
-14	914.9
-12	773.3
-10	621.9
-8	459.4
-6	283.7
-4	92.84

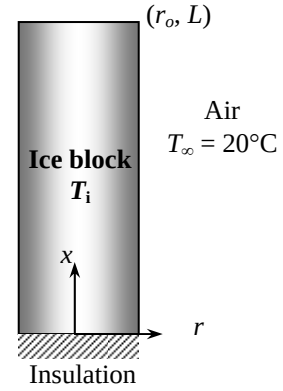


4-80 A cylindrical ice block is placed on a table. The initial temperature of the ice block to avoid melting for 2 h is to be determined.

Assumptions 1 Heat conduction in the ice block is two-dimensional, and thus the temperature varies in both x - and r - directions. 2 Heat transfer from the base of the ice block to the table is negligible. 3 The thermal properties of the ice block are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of the ice are given to be $k = 2.22 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 0.124 \times 10^{-7} \text{ m}^2/\text{s}$.

Analysis This cylindrical ice block can be treated as a short cylinder that can physically be formed by the intersection of a long cylinder of diameter $D = 2 \text{ cm}$ and an infinite plane wall of thickness $2L = 4 \text{ cm}$. We measure x from the bottom surface of the block since this surface represents the adiabatic center surface of the plane wall of thickness $2L = 4 \text{ cm}$. The melting starts at the outer surfaces of the top surface when the temperature drops below 0°C at this location. The Biot numbers, the corresponding constants, and the Fourier numbers are



$$Bi_{\text{wall}} = \frac{hL}{k} = \frac{(13 \text{ W/m}^2 \cdot ^\circ\text{C})(0.02 \text{ m})}{(2.22 \text{ W/m}\cdot^\circ\text{C})} = 0.1171 \longrightarrow \lambda_1 = 0.3318 \quad \text{and} \quad A_1 = 1.0187$$

$$Bi_{\text{cyl}} = \frac{hr_o}{k} = \frac{(13 \text{ W/m}^2 \cdot ^\circ\text{C})(0.01 \text{ m})}{(2.22 \text{ W/m}\cdot^\circ\text{C})} = 0.05856 \longrightarrow \lambda_1 = 0.3407 \quad \text{and} \quad A_1 = 1.0146$$

$$\tau_{\text{wall}} = \frac{\alpha t}{L^2} = \frac{(0.124 \times 10^{-7} \text{ m}^2/\text{s})(2 \text{ h} \times 3600 \text{ s/h})}{(0.02 \text{ m})^2} = 0.2232 > 0.2$$

$$\tau_{\text{cyl}} = \frac{\alpha t}{r_o^2} = \frac{(0.124 \times 10^{-7} \text{ m}^2/\text{s})(2 \text{ h} \times 3600 \text{ s/h})}{(0.01 \text{ m})^2} = 0.8928 > 0.2$$

Note that $\tau > 0.2$ in all dimensions and thus the one-term approximate solution for transient heat conduction is applicable. The product solution for this problem can be written as

$$\begin{aligned} \theta(L, r_o, t)_{\text{block}} &= \theta(L, t)_{\text{wall}} \theta(r_o, t)_{\text{cyl}} \\ \frac{0 - 20}{T_i - 20} &= \left[A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 L / L) \right] \left[A_1 e^{-\lambda_1^2 \tau} J_0(\lambda_1 r_o / r_o) \right] \\ \frac{0 - 20}{T_i - 20} &= \left[(1.0187) e^{-(0.3318)^2 (0.2232)} \cos(0.3318) \right] \left[(1.0146) e^{-(0.3407)^2 (0.8928)} (0.9707) \right] \end{aligned}$$

which gives $T_i = -4^\circ\text{C}$

Therefore, the ice will not start melting for at least 2 hours if its initial temperature is -4°C or below.

4-81 A cubic block and a cylindrical block are exposed to hot gases on all of their surfaces. The center temperatures of each geometry in 10, 20, and 60 min are to be determined.

Assumptions 1 Heat conduction in the cubic block is three-dimensional, and thus the temperature varies in all x -, y -, and z - directions. 2 Heat conduction in the cylindrical block is two-dimensional, and thus the temperature varies in both axial x - and radial r - directions. 3 The thermal properties of the granite are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of the granite are given to be $k = 2.5 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 1.15 \times 10^{-6} \text{ m}^2/\text{s}$.

Analysis:

Cubic block: This cubic block can physically be formed by the intersection of three infinite plane walls of thickness $2L = 5 \text{ cm}$.

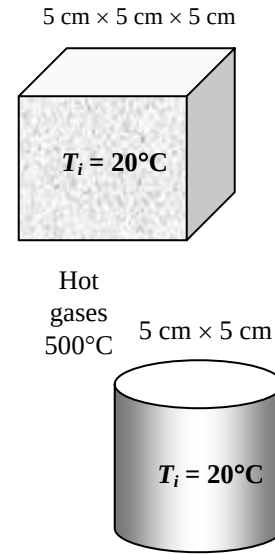
After 10 minutes: The Biot number, the corresponding constants, and the Fourier number are

$$Bi = \frac{hL}{k} = \frac{(40 \text{ W/m}^2 \cdot ^\circ\text{C})(0.025 \text{ m})}{(2.5 \text{ W/m}\cdot^\circ\text{C})} = 0.400 \longrightarrow \lambda_1 = 0.5932 \quad \text{and} \quad A_1 = 1.0580$$

$$\tau = \frac{\alpha t}{L^2} = \frac{(1.15 \times 10^{-6} \text{ m}^2/\text{s})(10 \text{ min} \times 60 \text{ s/min})}{(0.025 \text{ m})^2} = 1.104 > 0.2$$

To determine the center temperature, the product solution can be written as

$$\begin{aligned} \theta(0,0,0,t)_{\text{block}} &= [\theta(0,t)_{\text{wall}}]^3 \\ \frac{T(0,0,0,t) - T_\infty}{T_i - T_\infty} &= \left(A_1 e^{-\lambda_1^2 \tau} \right)^3 \\ \frac{T(0,0,0,t) - 500}{20 - 500} &= \left\{ (1.0580) e^{-(0.5932)^2 (1.104)} \right\}^3 = 0.369 \\ T(0,0,0,t) &= 323^\circ\text{C} \end{aligned}$$



After 20 minutes

$$\tau = \frac{\alpha t}{L^2} = \frac{(1.15 \times 10^{-6} \text{ m}^2/\text{s})(20 \text{ min} \times 60 \text{ s/min})}{(0.025 \text{ m})^2} = 2.208 > 0.2$$

$$\frac{T(0,0,0,t) - 500}{20 - 500} = \left\{ (1.0580) e^{-(0.5932)^2 (2.208)} \right\}^3 = 0.115 \longrightarrow T(0,0,0,t) = 445^\circ\text{C}$$

After 60 minutes

$$\tau = \frac{\alpha t}{L^2} = \frac{(1.15 \times 10^{-6} \text{ m}^2/\text{s})(60 \text{ min} \times 60 \text{ s/min})}{(0.025 \text{ m})^2} = 6.624 > 0.2$$

$$\frac{T(0,0,0,t) - 500}{20 - 500} = \left\{ (1.0580) e^{-(0.5932)^2 (6.624)} \right\}^3 = 0.00109 \longrightarrow T(0,0,0,t) = 500^\circ\text{C}$$

Note that $\tau > 0.2$ in all dimensions and thus the one-term approximate solution for transient heat conduction is applicable.

Cylinder: This cylindrical block can physically be formed by the intersection of a long cylinder of radius $r_o = D/2 = 2.5$ cm and a plane wall of thickness $2L = 5$ cm.

After 10 minutes: The Biot number and the corresponding constants for the long cylinder are

$$Bi = \frac{hr_o}{k} = \frac{(40 \text{ W/m}^2 \cdot ^\circ\text{C})(0.025 \text{ m})}{(2.5 \text{ W/m} \cdot ^\circ\text{C})} = 0.400 \longrightarrow \lambda_1 = 0.8516 \quad \text{and} \quad A_1 = 1.0931$$

To determine the center temperature, the product solution can be written as

$$\begin{aligned} \theta(0,0,t)_{block} &= [\theta(0,t)_{wall}] [\theta(0,t)_{cyl}] \\ \frac{T(0,0,t) - T_\infty}{T_i - T_\infty} &= \left(A_1 e^{-\lambda_1^2 \tau} \right)_{wall} \left(A_1 e^{-\lambda_1^2 \tau} \right)_{cyl} \\ \frac{T(0,0,t) - 500}{20 - 500} &= \left\{ (1.0580) e^{-(0.5932)^2 (1.104)} \right\} \left\{ (1.0931) e^{-(0.8516)^2 (1.104)} \right\} = 0.352 \longrightarrow T(0,0,t) = 331^\circ\text{C} \end{aligned}$$

After 20 minutes

$$\frac{T(0,0,t) - 500}{20 - 500} = \left\{ (1.0580) e^{-(0.5932)^2 (2.208)} \right\} \left\{ (1.0931) e^{-(0.8516)^2 (2.208)} \right\} = 0.107 \longrightarrow T(0,0,t) = 449^\circ\text{C}$$

After 60 minutes

$$\frac{T(0,0,t) - 500}{20 - 500} = \left\{ (1.0580) e^{-(0.5932)^2 (6.624)} \right\} \left\{ (1.0931) e^{-(0.8516)^2 (6.624)} \right\} = 0.00092 \longrightarrow T(0,0,t) = 500^\circ\text{C}$$

Note that $\tau > 0.2$ in all dimensions and thus the one-term approximate solution for transient heat conduction is applicable.

4-82 A cubic block and a cylindrical block are exposed to hot gases on all of their surfaces. The center temperatures of each geometry in 10, 20, and 60 min are to be determined.

Assumptions 1 Heat conduction in the cubic block is three-dimensional, and thus the temperature varies in all x -, y , and z - directions. 2 Heat conduction in the cylindrical block is two-dimensional, and thus the temperature varies in both axial x - and radial r - directions. 3 The thermal properties of the granite are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of the granite are $k = 2.5 \text{ W/m} \cdot ^\circ\text{C}$ and $\alpha = 1.15 \times 10^{-6} \text{ m}^2/\text{s}$.

Analysis:

Cubic block: This cubic block can physically be formed by the intersection of three infinite plane wall of thickness $2L = 5 \text{ cm}$. Two infinite plane walls are exposed to the hot gases with a heat transfer coefficient of $h = 40 \text{ W/m}^2 \cdot ^\circ\text{C}$ and one with $h = 80 \text{ W/m}^2 \cdot ^\circ\text{C}$.

After 10 minutes: The Biot number and the corresponding constants for $h = 40 \text{ W/m}^2 \cdot ^\circ\text{C}$ are

$$Bi = \frac{hL}{k} = \frac{(40 \text{ W/m}^2 \cdot ^\circ\text{C})(0.025 \text{ m})}{(2.5 \text{ W/m} \cdot ^\circ\text{C})} = 0.400 \longrightarrow \lambda_1 = 0.5932 \quad \text{and} \quad A_1 = 1.0580$$

The Biot number and the corresponding constants for $h = 80 \text{ W/m}^2 \cdot ^\circ\text{C}$ are

$$Bi = \frac{hL}{k} = \frac{(80 \text{ W/m}^2 \cdot ^\circ\text{C})(0.025 \text{ m})}{(2.5 \text{ W/m} \cdot ^\circ\text{C})} = 0.800$$

$$\longrightarrow \lambda_1 = 0.7910 \quad \text{and} \quad A_1 = 1.1016$$

The Fourier number is

$$\tau = \frac{\alpha t}{L^2} = \frac{(1.15 \times 10^{-6} \text{ m}^2/\text{s})(10 \text{ min} \times 60 \text{ s/min})}{(0.025 \text{ m})^2} = 1.104 > 0.2$$

To determine the center temperature, the product solution method can be written as

$$\begin{aligned} \theta(0,0,0,t)_{\text{block}} &= [\theta(0,t)_{\text{wall}}]^2 [\theta(0,t)_{\text{wall}}] \\ \frac{T(0,0,0,t) - T_\infty}{T_i - T_\infty} &= \left(A_1 e^{-\lambda_1^2 \tau} \right)^2 \left(A_1 e^{-\lambda_1^2 \tau} \right) \\ \frac{T(0,0,0,t) - 500}{20 - 500} &= \left\{ (1.0580) e^{-(0.5932)^2 (1.104)} \right\}^2 \left\{ (1.1016) e^{-(0.7910)^2 (1.104)} \right\} = 0.284 \end{aligned}$$

$$T(0,0,0,t) = 364^\circ\text{C}$$

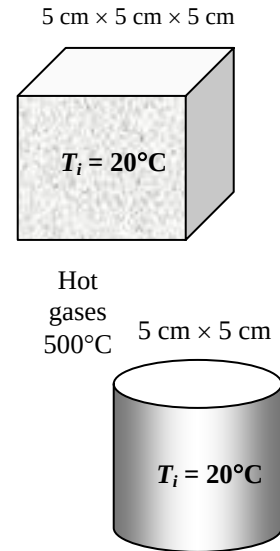
After 20 minutes

$$\tau = \frac{\alpha t}{L^2} = \frac{(1.15 \times 10^{-6} \text{ m}^2/\text{s})(20 \text{ min} \times 60 \text{ s/min})}{(0.025 \text{ m})^2} = 2.208 > 0.2$$

$$\frac{T(0,0,0,t) - 500}{20 - 500} = \left\{ (1.0580) e^{-(0.5932)^2 (2.208)} \right\}^2 \left\{ (1.1016) e^{-(0.7910)^2 (2.208)} \right\} = 0.0654$$

$$\longrightarrow T(0,0,0,t) = 469^\circ\text{C}$$

After 60 minutes



$$\tau = \frac{\alpha t}{L^2} = \frac{(1.15 \times 10^{-6} \text{ m}^2/\text{s})(60 \text{ min} \times 60 \text{ s/min})}{(0.025 \text{ m})^2} = 6.624 > 0.2$$

$$\frac{T(0,0,0,t) - 500}{20 - 500} = \left\{ (1.0580)e^{-(0.5932)^2(6.624)} \right\}^2 \left\{ (1.1016)e^{-(0.7910)^2(6.624)} \right\} = 0.000186$$

$$\longrightarrow T(0,0,0,t) = 500^\circ\text{C}$$

Note that $\tau > 0.2$ in all dimensions and thus the one-term approximate solution for transient heat conduction is applicable.

Cylinder: This cylindrical block can physically be formed by the intersection of a long cylinder of radius $r_o = D/2 = 2.5 \text{ cm}$ exposed to the hot gases with a heat transfer coefficient of $h = 40 \text{ W/m}^2 \cdot ^\circ\text{C}$ and a plane wall of thickness $2L = 5 \text{ cm}$ exposed to the hot gases with $h = 80 \text{ W/m}^2 \cdot ^\circ\text{C}$.

After 10 minutes: The Biot number and the corresponding constants for the long cylinder are

$$Bi = \frac{hr_o}{k} = \frac{(40 \text{ W/m}^2 \cdot ^\circ\text{C})(0.025 \text{ m})}{(2.5 \text{ W/m} \cdot ^\circ\text{C})} = 0.400 \longrightarrow \lambda_1 = 0.8516 \text{ and } A_1 = 1.0931$$

To determine the center temperature, the product solution method can be written as

$$\begin{aligned} \theta(0,0,t)_{\text{block}} &= [\theta(0,t)_{\text{wall}}][\theta(0,t)_{\text{cyl}}] \\ \frac{T(0,0,t) - T_\infty}{T_i - T_\infty} &= \left(A_1 e^{-\lambda_1^2 \tau} \right)_{\text{wall}} \left(A_1 e^{-\lambda_1^2 \tau} \right)_{\text{cyl}} \\ \frac{T(0,0,t) - 500}{20 - 500} &= \left\{ (1.1016)e^{-(0.7910)^2(1.104)} \right\} \left\{ (1.0931)e^{-(0.8516)^2(1.104)} \right\} = 0.271 \end{aligned}$$

$$T(0,0,t) = 370^\circ\text{C}$$

After 20 minutes

$$\frac{T(0,0,t) - 500}{20 - 500} = \left\{ (1.1016)e^{-(0.7910)^2(2.208)} \right\} \left\{ (1.0931)e^{-(0.8516)^2(2.208)} \right\} = 0.06094 \longrightarrow T(0,0,t) = 471^\circ\text{C}$$

After 60 minutes

$$\frac{T(0,0,t) - 500}{20 - 500} = \left\{ (1.1016)e^{-(0.7910)^2(6.624)} \right\} \left\{ (1.0931)e^{-(0.8516)^2(6.624)} \right\} = 0.0001568 \longrightarrow T(0,0,t) = 500^\circ\text{C}$$

Note that $\tau > 0.2$ in all dimensions and thus the one-term approximate solution for transient heat conduction is applicable.

4-83 A cylindrical aluminum block is heated in a furnace. The length of time the block should be kept in the furnace and the amount of heat transfer to the block are to be determined.

Assumptions 1 Heat conduction in the cylindrical block is two-dimensional, and thus the temperature varies in both axial x - and radial r - directions. 2 The thermal properties of the aluminum are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (it will be verified).

Properties The thermal properties of the aluminum block are given to be $k = 236 \text{ W/m} \cdot ^\circ\text{C}$, $\rho = 2702 \text{ kg/m}^3$, $C_p = 0.896 \text{ kJ/kg} \cdot ^\circ\text{C}$, and $\alpha = 9.75 \times 10^{-5} \text{ m}^2/\text{s}$.

Analysis This cylindrical aluminum block can physically be formed by the intersection of an infinite plane wall of thickness $2L = 20 \text{ cm}$, and a long cylinder of radius $r_0 = D/2 = 7.5 \text{ cm}$. The Biot numbers and the corresponding constants are first determined to be

$$Bi = \frac{hL}{k} = \frac{(80 \text{ W/m}^2 \cdot ^\circ\text{C})(0.1 \text{ m})}{(236 \text{ W/m} \cdot ^\circ\text{C})} = 0.0339 \quad \longrightarrow \lambda_1 = 0.1811 \quad \text{and} \quad A_1 = 1.0056$$

$$Bi = \frac{hr_0}{k} = \frac{(80 \text{ W/m}^2 \cdot ^\circ\text{C})(0.075 \text{ m})}{236 \text{ W/m} \cdot ^\circ\text{C}} = 0.0254 \quad \longrightarrow \lambda_1 = 0.2217 \quad \text{and} \quad A_1 = 1.0063$$

Noting that $\tau = \alpha t / L^2$ and assuming $\tau > 0.2$ in all dimensions and thus the one-term approximate solution for transient heat conduction is applicable, the product solution for this problem can be written as

$$\theta(0,0,t)_{\text{block}} = \theta(0,t)_{\text{wall}} \theta(0,t)_{\text{cyl}} = \left(A_1 e^{-\lambda_1^2 \tau} \right)_{\text{wall}} \left(A_1 e^{-\lambda_1^2 \tau} \right)_{\text{cyl}}$$

$$\frac{300 - 1200}{20 - 1200} = \left\{ (1.0056) \exp \left[- (0.1811)^2 \frac{(9.75 \times 10^{-5}) t}{(0.1)^2} \right] \right\} \left\{ (1.0063) \exp \left[- (0.2217)^2 \frac{(9.75 \times 10^{-5}) t}{(0.075)^2} \right] \right\} = 0.7627$$

Solving for the time t gives $t = 241 \text{ s} = \mathbf{4.0 \text{ min}}$. We note that

$$\tau = \frac{\alpha t}{L^2} = \frac{(9.75 \times 10^{-5} \text{ m}^2/\text{s})(241 \text{ s})}{(0.1 \text{ m})^2} = 2.35 > 0.2$$

and thus the assumption of $\tau > 0.2$ for the applicability of the one-term approximate solution is verified.

The maximum amount of heat transfer is

$$m = \rho V = \rho \pi r_0^2 L = (2702 \text{ kg/m}^3) [\pi (0.075 \text{ m})^2 (0.2 \text{ m})] = 9.550 \text{ kg}$$

$$Q_{\max} = m C_p (T_i - T_\infty) = (9.550 \text{ kg})(0.896 \text{ kJ/kg} \cdot ^\circ\text{C})(20 - 1200)^\circ\text{C} = 10,100 \text{ kJ}$$

Then we determine the dimensionless heat transfer ratios for both geometries as

$$\left(\frac{Q}{Q_{\max}} \right)_{\text{wall}} = 1 - \theta_{o,\text{wall}} \frac{\sin(\lambda_1)}{\lambda_1} = 1 - (0.7627) \frac{\sin(0.1811)}{0.1811} = 0.2415$$

$$\left(\frac{Q}{Q_{\max}} \right)_{\text{cyl}} = 1 - 2\theta_{o,\text{cyl}} \frac{J_1(\lambda_1)}{\lambda_1} = 1 - 2(0.7627) \frac{0.1101}{0.2217} = 0.2425$$

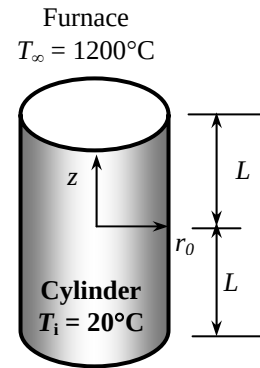
The heat transfer ratio for the short cylinder is

$$\left(\frac{Q}{Q_{\max}} \right)_{\text{short cylinder}} = \left(\frac{Q}{Q_{\max}} \right)_{\text{plane wall}} + \left(\frac{Q}{Q_{\max}} \right)_{\text{long cylinder}} \left[1 - \left(\frac{Q}{Q_{\max}} \right)_{\text{plane wall}} \right] = 0.2415 + (0.2425)(1 - 0.2415) = 0.4254$$

Then the total heat transfer from the short cylinder as it is cooled from 300°C at the center to 20°C becomes

$$Q = 0.4236 Q_{\max} = (0.4254)(10,100 \text{ kJ}) = \mathbf{4297 \text{ kJ}}$$

which is identical to the heat transfer to the cylinder as the cylinder at 20°C is heated to 300°C at the center.



4-84 A cylindrical aluminum block is heated in a furnace. The length of time the block should be kept in the furnace and the amount of heat transferred to the block are to be determined.

Assumptions 1 Heat conduction in the cylindrical block is two-dimensional, and thus the temperature varies in both axial x - and radial r - directions. 2 Heat transfer from the bottom surface of the block is negligible. 3 The thermal properties of the aluminum are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of the aluminum block are given to be $k = 236 \text{ W/m}\cdot^\circ\text{C}$, $\rho = 2702 \text{ kg/m}^3$, $C_p = 0.896 \text{ kJ/kg}\cdot^\circ\text{C}$, and $\alpha = 9.75 \times 10^{-5} \text{ m}^2/\text{s}$.

Analysis This cylindrical aluminum block can physically be formed by the intersection of an infinite plane wall of thickness $2L = 40 \text{ cm}$ and a long cylinder of radius $r_0 = D/2 = 7.5 \text{ cm}$. Note that the height of the short cylinder represents the half thickness of the infinite plane wall where the bottom surface of the short cylinder is adiabatic. The Biot numbers and corresponding constants are first determined to be

$$Bi = \frac{hL}{k} = \frac{(80 \text{ W/m}^2\cdot^\circ\text{C})(0.2 \text{ m})}{(236 \text{ W/m}\cdot^\circ\text{C})} = 0.0678 \quad \longrightarrow \lambda_1 = 0.2568 \quad \text{and} \quad A_1 = 1.0110$$

$$Bi = \frac{hr_0}{k} = \frac{(80 \text{ W/m}^2\cdot^\circ\text{C})(0.075 \text{ m})}{(236 \text{ W/m}\cdot^\circ\text{C})} = 0.0254 \quad \longrightarrow \lambda_1 = 0.2217 \quad \text{and} \quad A_1 = 1.0063$$

Noting that $\tau = \alpha t / L^2$ and assuming $\tau > 0.2$ in all dimensions and thus the one-term approximate solution for transient heat conduction is applicable, the product solution for this problem can be written as

$$\theta(0,0,t)_{\text{block}} = \theta(0,t)_{\text{wall}} \theta(0,t)_{\text{cyl}} = \left(A_1 e^{-\lambda_1^2 \tau} \right)_{\text{wall}} \left(A_1 e^{-\lambda_1^2 \tau} \right)_{\text{cyl}}$$

$$\frac{300 - 1200}{20 - 1200} = \left\{ (1.0110) \exp \left[- (0.2568)^2 \frac{(9.75 \times 10^{-5}) t}{(0.2)^2} \right] \right\} \left\{ (1.0063) \exp \left[- (0.2217)^2 \frac{(9.75 \times 10^{-5}) t}{(0.075)^2} \right] \right\} = 0.7627$$

Solving for the time t gives $t = 285 \text{ s} = \mathbf{4.7 \text{ min}}$. We note that

$$\tau = \frac{\alpha t}{L^2} = \frac{(9.75 \times 10^{-5} \text{ m}^2/\text{s})(285 \text{ s})}{(0.2 \text{ m})^2} = 0.69 > 0.2$$

and thus the assumption of $\tau > 0.2$ for the applicability of the one-term approximate solution is verified. The maximum amount of heat transfer is

$$m = \rho V = \rho \pi r_0 L = (2702 \text{ kg/m}^3) [\pi (0.075 \text{ m})^2 (0.2 \text{ m})] = 9.55 \text{ kg}$$

$$Q_{\max} = m C_p (T_i - T_\infty) = (9.55 \text{ kg})(0.896 \text{ kJ/kg}\cdot^\circ\text{C})(20 - 1200)^\circ\text{C} = 10,100 \text{ kJ}$$

Then we determine the dimensionless heat transfer ratios for both geometries as

$$\left(\frac{Q}{Q_{\max}} \right)_{\text{wall}} = 1 - \theta_{o,\text{wall}} \frac{\sin(\lambda_1)}{\lambda_1} = 1 - (0.7627) \frac{\sin(0.2568)}{0.2568} = 0.2457$$

$$\left(\frac{Q}{Q_{\max}} \right)_{\text{cyl}} = 1 - 2\theta_{o,\text{cyl}} \frac{J_1(\lambda_1)}{\lambda_1} = 1 - 2(0.7627) \frac{0.1101}{0.2217} = 0.2425$$

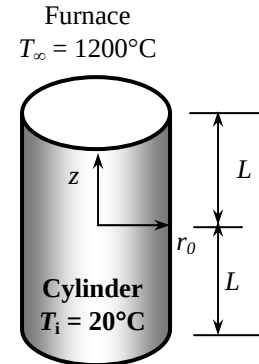
The heat transfer ratio for the short cylinder is

$$\left(\frac{Q}{Q_{\max}} \right)_{\text{short cylinder}} = \left(\frac{Q}{Q_{\max}} \right)_{\text{plane wall}} + \left(\frac{Q}{Q_{\max}} \right)_{\text{long cylinder}} \left[1 - \left(\frac{Q}{Q_{\max}} \right)_{\text{plane wall}} \right] = 0.2457 + (0.2425)(1 - 0.2457) = 0.4286$$

Then the total heat transfer from the short cylinder as it is cooled from 300°C at the center to 20°C becomes

$$Q = 0.4286 Q_{\max} = (0.4286)(10,100 \text{ kJ}) = \mathbf{4239 \text{ kJ}}$$

which is identical to the heat transfer to the cylinder as the cylinder at 20°C is heated to 300°C at the center.



4-85

PROBLEM 4-85**GIVEN**

$$2L = 0.20 \text{ [m]}$$

$$r_o = 0.15 \text{ [m]}$$

$$T_i = 20 \text{ [C]}$$

$$T_{\infty} = 1200 \text{ [C]}$$

$$T_{o_o} = 300 \text{ [C], parameter to be varied}$$

$$h = 80 \text{ [W/m}^2\text{-C]}$$

PROPERTIES

$$k = 236 \text{ [W/m-C]}$$

$$\rho = 2702 \text{ [kg/m}^3\text{]}$$

$$C_p = 0.896 \text{ [kJ/kg-C]}$$

$$\alpha = 9.75 \times 10^{-5} \text{ [m}^2\text{/s]}$$

ANALYSIS

This short cylinder can physically be formed by the intersection of a long cylinder of radius r_o and a plane wall of thickness $2L$

For plane wall

$$Bi_w = (hL)/k$$

From Table 4-1 corresponding to this Bi number, we read

$$\lambda_{1w} = 0.1439 \text{ "w stands for wall"}$$

$$A_{1w} = 1.0035$$

$$\tau_w = (\alpha \text{time})/L^2$$

$$\theta_{ow} = A_{1w} \exp(-\lambda_{1w}^2 \tau_w) \text{ "}\theta_{ow} = (T_{ow} - T_{\infty})/(T_i - T_{\infty})\text{"}$$

For long cylinder

$$Bi_c = (hr_o)/k \text{ "c stands for cylinder"}$$

From Table 4-1 corresponding to this Bi number, we read

$$\lambda_{1c} = 0.1762$$

$$A_{1c} = 1.0040$$

$$\tau_c = (\alpha \text{time})/r_o^2$$

$$\theta_{oc} = A_{1c} \exp(-\lambda_{1c}^2 \tau_c) \text{ "}\theta_{oc} = (T_{oc} - T_{\infty})/(T_i - T_{\infty})\text{"}$$

$$(T_{oo} - T_{\infty})/(T_i - T_{\infty}) = \theta_{ow} \theta_{oc} \text{ "center temperature of cylinder"}$$

$$V = \pi r_o^2 (2L)$$

$$m = \rho V$$

$$Q_{\max} = m C_p (T_{\infty} - T_i)$$

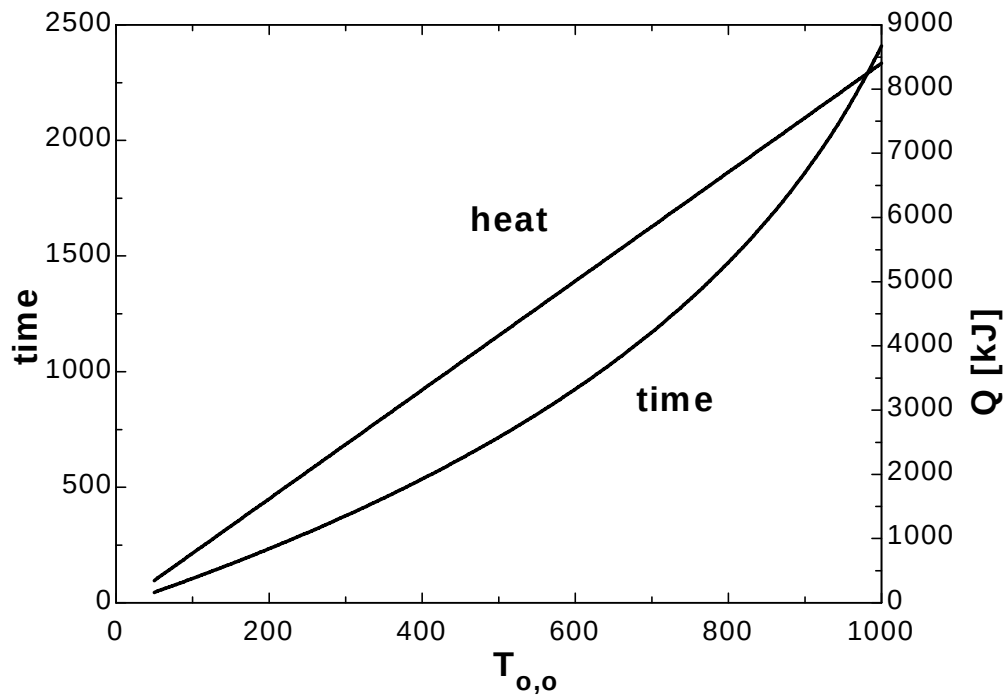
$$Q_w = 1 - \theta_{ow} \sin(\lambda_{1w})/\lambda_{1w} \text{ "Q}_w = (Q/Q_{\max})_w\text{"}$$

$$Q_c = 1 - 2\theta_{oc} J_1/\lambda_{1c} \text{ "Q}_c = (Q/Q_{\max})_c\text{"}$$

$$J_1 = 0.0876 \text{ "From Table 4-2, at } \lambda_{1c}\text{"}$$

$$Q/Q_{\max} = Q_w + Q_c (1 - Q_w) \text{ "total heat transfer"}$$

$T_{o,o}$ [C]	time [s]	Q [kJ]
50	44.91	346.3
100	105	770.2
150	167.8	1194
200	233.8	1618
250	303.1	2042
300	376.1	2466
350	453.4	2890
400	535.3	3314
450	622.5	3738
500	715.7	4162
550	815.9	4586
600	924	5010
650	1042	5433
700	1170	5857
750	1313	6281
800	1472	6705
850	1652	7129
900	1861	7553
950	2107	7977
1000	2409	8401



Special Topic: Refrigeration and Freezing of Foods

4-86C The common kinds of microorganisms are bacteria, yeasts, molds, and viruses. The undesirable changes caused by microorganisms are off-flavors and colors, slime production, changes in the texture and appearances, and the spoilage of foods.

4-87C Microorganisms are the prime cause for the spoilage of foods. Refrigeration prevents or delays the spoilage of foods by reducing the rate of growth of microorganisms. Freezing extends the storage life of foods for months by preventing the growths of microorganisms.

4-88C The environmental factors that affect the growth rate of microorganisms are the temperature, the relative humidity, the oxygen level of the environment, and air motion.

4-89C Cooking kills the microorganisms in foods, and thus prevents spoilage of foods. It is important to raise the internal temperature of a roast in an oven above 70°C since most microorganisms, including some that cause diseases, may survive temperatures below 70°C.

4-90C The contamination of foods with microorganisms can be prevented or minimized by (1) preventing contamination by following strict sanitation practices such as washing hands and using fine filters in ventilation systems, (2) inhibiting growth by altering the environmental conditions, and (3) destroying the organisms by heat treatment or chemicals.

The growth of microorganisms in foods can be retarded by keeping the temperature below 4°C and relative humidity below 60 percent. Microorganisms can be destroyed by heat treatment, chemicals, ultraviolet light, and solar radiation.

4-91C (a) High air motion retards the growth of microorganisms in foods by keeping the food surfaces dry, and creating an undesirable environment for the microorganisms. (b) Low relative humidity (dry) environments also retard the growth of microorganisms by depriving them of water that they need to grow. Moist air supplies the microorganisms with the water they need, and thus encourages their growth. Relative humidities below 60 percent prevent the growth rate of most microorganisms on food surfaces.

4-92C Cooling the carcass with refrigerated air is at -10°C would certainly reduce the cooling time, but this proposal should be rejected since it will cause the outer parts of the carcasses to freeze, which is undesirable. Also, the refrigeration unit will consume more power to reduce the temperature to -10°C, and thus it will have a lower efficiency.

4-93C The freezing time could be decreased by (a) lowering the temperature of the refrigerated air, (b) increasing the velocity of air, (c) increasing the capacity of the refrigeration system, and (d) decreasing the size of the meat boxes.

4-94C The rate of freezing can affect color, tenderness, and drip. Rapid freezing increases tenderness and reduces the tissue damage and the amount of drip after thawing.

4-95C This claim is reasonable since the lower the storage temperature, the longer the storage life of beef. This is because some water remains unfrozen even at subfreezing temperatures, and the lower the temperature, the smaller the unfrozen water content of the beef.

4-96C A refrigerated shipping dock is a refrigerated space where the orders are assembled and shipped out. Such docks save valuable storage space from being used for shipping purpose, and provide a more acceptable working environment for the employees. The refrigerated shipping docks are usually maintained at 1.5°C, and therefore the air that flows into the freezer during shipping is already cooled to about 1.5°C. This reduces the refrigeration load of the cold storage rooms.

4-97C (a) The heat transfer coefficient during immersion cooling is much higher, and thus the cooling time during immersion chilling is much lower than that during forced air chilling. (b) The cool air chilling can cause a moisture loss of 1 to 2 percent while water immersion chilling can actually cause moisture absorption of 4 to 15 percent. (c) The chilled water circulated during immersion cooling encourages microbial growth, and thus immersion chilling is associated with more microbial growth. The problem can be minimized by adding chloride to the water.

4-98C The proper storage temperature of frozen poultry is about -18°C or below. The primary freezing methods of poultry are the air blast tunnel freezing, cold plates, immersion freezing, and cryogenic cooling.

4-99C The factors, which affect the quality of frozen, fish are the condition of the fish before freezing, the freezing method, and the temperature and humidity during storage and transportation, and the length of storage time.

4-100 The chilling room of a meat plant with a capacity of 350 beef carcasses is considered. The cooling load, the air flow rate, and the heat transfer area of the evaporator are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Specific heats of beef carcass and air are constant.

Properties The density and specific heat of air at 0°C are given to be 1.28 kg/m³ and 1.0 kJ/kg·°C. The specific heat of beef carcass is given to be 3.14 kJ/kg·°C.

Analysis (a) The amount of beef mass that needs to be cooled per unit time is

$$\begin{aligned}\dot{m}_{beef} &= (\text{Total beef mass cooled})/(\text{cooling time}) \\ &= (350 \times 280 \text{ kg/carcass})/(12 \text{ h} \times 3600 \text{ s}) = 2.27 \text{ kg/s}\end{aligned}$$

The product refrigeration load can be viewed as the energy that needs to be removed from the beef carcass as it is cooled from 35 to 16°C at a rate of 2.27 kg/s, and is determined to be

$$\begin{aligned}\dot{Q}_{beef} &= (\dot{m}C_p\Delta T)_{beef} \\ &= (2.27 \text{ kg/s})(3.14 \text{ kJ/kg}\cdot^\circ\text{C})(35 - 16)^\circ\text{C} = 135 \text{ kW}\end{aligned}$$

Then the total refrigeration load of the chilling room becomes

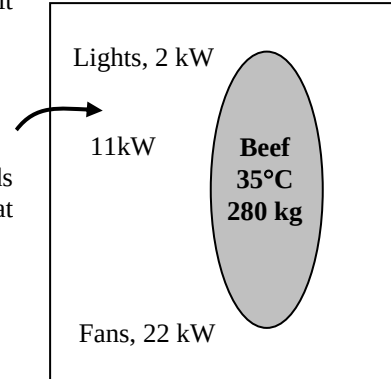
$$\dot{Q}_{\text{total, chilling room}} = \dot{Q}_{beef} + \dot{Q}_{fan} + \dot{Q}_{lights} + \dot{Q}_{\text{heat gain}} = 135 + 22 + 2 + 11 = \mathbf{170 \text{ kW}}$$

(b) Heat is transferred to air at the rate determined above, and the temperature of air rises from -2.2°C to 0.5°C as a result. Therefore, the mass flow rate of air is

$$\dot{m}_{air} = \frac{\dot{Q}_{air}}{(C_p\Delta T)_{air}} = \frac{170 \text{ kW}}{(1.0 \text{ kJ/kg}\cdot^\circ\text{C})[0.5 - (-2.2)^\circ\text{C}]} = 63.0 \text{ kg/s}$$

Then the volume flow rate of air becomes

$$\dot{V}_{air} = \frac{\dot{m}_{air}}{\rho_{air}} = \frac{63.0 \text{ kg/s}}{1.28 \text{ kg/m}^3} = \mathbf{49.2 \text{ m}^3/\text{s}}$$



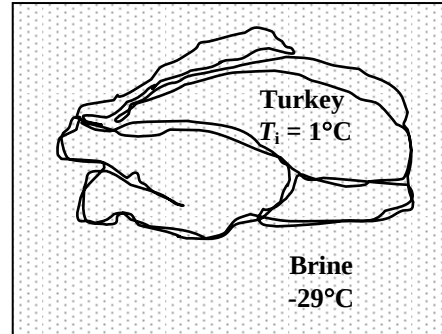
4-101 Turkeys are to be frozen by submerging them into brine at -29°C . The time it will take to reduce the temperature of turkey breast at a depth of 3.8 cm to -18°C and the amount of heat transfer per turkey are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The thermal properties of turkeys are constant.

Properties It is given that the specific heats of turkey are 2.98 and 1.65 kJ/kg $\cdot^{\circ}\text{C}$ above and below the freezing point of -2.8°C , respectively, and the latent heat of fusion of turkey is 214 kJ/kg.

Analysis The time required to freeze the turkeys from 1°C to -18°C with brine at -29°C can be determined directly from Fig. 4-45 to be

$$t \cong 180 \text{ min.} \cong \mathbf{3 \text{ hours}}$$



(a) Assuming the entire water content of turkey is frozen, the amount of heat that needs to be removed from the turkey as it is cooled from 1°C to -18°C is

$$\text{Cooling to } -2.8^{\circ}\text{C}: Q_{\text{cooling, fresh}} = (mC \Delta T)_{\text{fresh}} = (7 \text{ kg})(2.98 \text{ kJ/kg} \cdot ^{\circ}\text{C})[1 - (-2.8)^{\circ}\text{C}] = 79.3 \text{ kJ}$$

$$\text{Freezing at } -2.8^{\circ}\text{C}: Q_{\text{freezing}} = mh_{\text{latent}} = (7 \text{ kg})(214 \text{ kJ/kg}) = 1498 \text{ kJ}$$

$$\text{Cooling } -18^{\circ}\text{C}: Q_{\text{cooling, frozen}} = (mC \Delta T)_{\text{frozen}} = (7 \text{ kg})(1.65 \text{ kJ/kg} \cdot ^{\circ}\text{C})[-2.8 - (-18)]^{\circ}\text{C} = 175.6 \text{ kJ}$$

Therefore, the total amount of heat removal per turkey is

$$Q_{\text{total}} = Q_{\text{cooling, fresh}} + Q_{\text{freezing}} + Q_{\text{cooling, frozen}} = 79.3 + 1498 + 175.6 \cong \mathbf{1753 \text{ kJ}}$$

(b) Assuming only 90 percent of the water content of turkey is frozen, the amount of heat that needs to be removed from the turkey as it is cooled from 1°C to -18°C is

$$\text{Cooling to } -2.8^{\circ}\text{C}: Q_{\text{cooling, fresh}} = (mC \Delta T)_{\text{fresh}} = (7 \text{ kg})(2.98 \text{ kJ/kg} \cdot ^{\circ}\text{C})[1 - (-2.98)^{\circ}\text{C}] = 79.3 \text{ kJ}$$

$$\text{Freezing at } -2.8^{\circ}\text{C}: Q_{\text{freezing}} = mh_{\text{latent}} = (7 \times 0.9 \text{ kg})(214 \text{ kJ / kg}) = 1,348 \text{ kJ}$$

$$\text{Cooling } -18^{\circ}\text{C}: Q_{\text{cooling, frozen}} = (mC \Delta T)_{\text{frozen}} = (7 \times 0.9 \text{ kg})(1.65 \text{ kJ / kg} \cdot ^{\circ}\text{C})[-2.8 - (-18)]^{\circ}\text{C} = 158 \text{ kJ}$$

$$Q_{\text{cooling, unfrozen}} = (mC \Delta T)_{\text{fresh}} = (7 \times 0.1 \text{ kg})(2.98 \text{ kJ/kg} \cdot ^{\circ}\text{C})[-2.8 - (-18)^{\circ}\text{C}] = 31.7 \text{ kJ}$$

Therefore, the total amount of heat removal per turkey is

$$Q_{\text{total}} = Q_{\text{cooling, fresh}} + Q_{\text{freezing}} + Q_{\text{cooling, frozen \& unfrozen}} = 79.3 + 1348 + 158 + 31.7 = \mathbf{1617 \text{ kJ}}$$

4-102E Chickens are to be frozen by refrigerated air. The cooling time of the chicken is to be determined for the cases of cooling air being at -40°F and -80°F .

Assumptions 1 Steady operating conditions exist. 2 The thermal properties of chickens are constant.

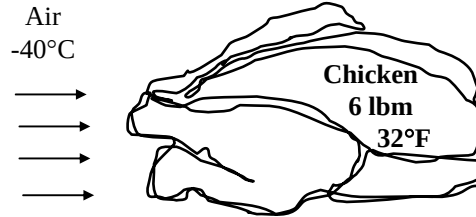
Analysis The time required to reduce the inner surface temperature of the chickens from 32°F to 25°F with refrigerated air at -40°F is determined from Fig. 4-44 to be

$$t \cong 2.3 \text{ hours}$$

If the air temperature were -80°F , the freezing time would be

$$t \cong 1.4 \text{ hours}$$

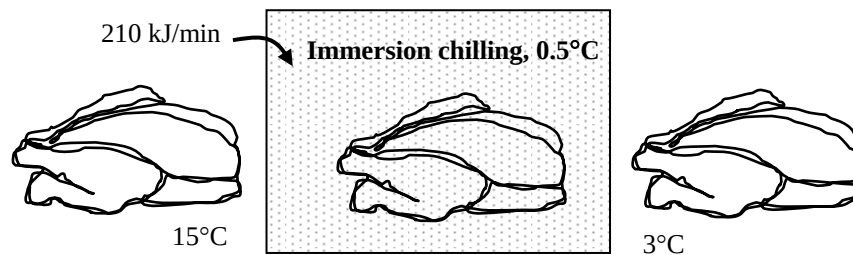
Therefore, the time required to cool the chickens to 25°F is reduced considerably when the refrigerated air temperature is decreased.



4-103 Chickens are to be cooled by chilled water in an immersion chiller. The rate of heat removal from the chicken and the mass flow rate of water are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The thermal properties of chickens are constant.

Properties The specific heat of chicken are given to be $3.54 \text{ kJ/kg}\cdot^{\circ}\text{C}$. The specific heat of water is $4.18 \text{ kJ/kg}\cdot^{\circ}\text{C}$ (Table A-9).



Analysis (a) Chickens are dropped into the chiller at a rate of 500 per hour. Therefore, chickens can be considered to flow steadily through the chiller at a mass flow rate of

$$\dot{m}_{\text{chicken}} = (500 \text{ chicken} / \text{h})(2.2 \text{ kg} / \text{chicken}) = 1100 \text{ kg} / \text{h} = 0.3056 \text{ kg} / \text{s}$$

Then the rate of heat removal from the chickens as they are cooled from 15°C to 3°C at this rate becomes

$$\dot{Q}_{\text{chicken}} = (\dot{m}C_p\Delta T)_{\text{chicken}} = (0.3056 \text{ kg/s})(3.54 \text{ kJ/kg}\cdot^{\circ}\text{C})(15 - 3)^{\circ}\text{C} = \mathbf{13.0 \text{ kW}}$$

(b) The chiller gains heat from the surroundings as a rate of $210 \text{ kJ/min} = 3.5 \text{ kJ/s}$. Then the total rate of heat gain by the water is

$$\dot{Q}_{\text{water}} = \dot{Q}_{\text{chicken}} + \dot{Q}_{\text{heat gain}} = 13.0 + 3.5 = 16.5 \text{ kW}$$

Noting that the temperature rise of water is not to exceed 2°C as it flows through the chiller, the mass flow rate of water must be at least

$$\dot{m}_{\text{water}} = \frac{\dot{Q}_{\text{water}}}{(C_p\Delta T)_{\text{water}}} = \frac{16.5 \text{ kW}}{(4.18 \text{ kJ/kg}\cdot^{\circ}\text{C})(2^{\circ}\text{C})} = \mathbf{1.97 \text{ kg/s}}$$

If the mass flow rate of water is less than this value, then the temperature rise of water will have to be more than 2°C .

4-104 The center temperature of meat slabs is to be lowered by chilled air to below 5°C while the surface temperature remains above -1°C to avoid freezing. The average heat transfer coefficient during this cooling process is to be determined.

Assumptions **1** The meat slabs can be approximated as very large plane walls of half-thickness $L = 5\text{-cm}$. **2** Heat conduction in the meat slabs is one-dimensional because of symmetry about the centerplane. **3** The thermal properties of the meat slab are constant. **4** The heat transfer coefficient is constant and uniform over the entire surface. **5** The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

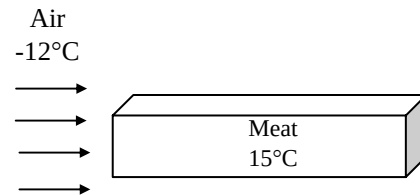
Properties The thermal properties of the beef slabs are given to be $\rho = 1090\text{ kg/m}^3$, $C_p = 3.54\text{ kJ/kg}\cdot^\circ\text{C}$, $k = 0.47\text{ W/m}\cdot^\circ\text{C}$, and $\alpha = 0.13 \times 10^{-6}\text{ m}^2/\text{s}$.

Analysis The lowest temperature in the steak will occur at the surfaces and the highest temperature at the center at a given time since the inner part of the steak will be last place to be cooled. In the limiting case, the surface temperature at $x = L = 5\text{ cm}$ from the center will be -1°C while the mid plane temperature is 5°C in an environment at -12°C . Then from Fig. 4-13b we obtain

$$\left. \begin{aligned} \frac{x}{L} = \frac{5\text{ cm}}{5\text{ cm}} = 1 \\ \frac{T(L, t) - T_\infty}{T_o - T_\infty} = \frac{-1 - (-12)}{5 - (-12)} = 0.65 \end{aligned} \right\} \frac{1}{\text{Bi}} = \frac{k}{hL} = 0.95$$

which gives

$$h = \frac{k}{L} \text{Bi} = \frac{0.47\text{ W/m}\cdot^\circ\text{C}}{0.05\text{ m}} \left(\frac{1}{0.95} \right) = \mathbf{9.9\text{ W/m}^2\cdot^\circ\text{C}}$$



Therefore, the convection heat transfer coefficient should be kept below this value to satisfy the constraints on the temperature of the steak during refrigeration. We can also meet the constraints by using a lower heat transfer coefficient, but doing so would extend the refrigeration time unnecessarily.

Discussion We could avoid the uncertainty associated with the reading of the charts and obtain a more accurate result by using the one-term solution relation for an infinite plane wall, but it would require a trial and error approach since the Bi number is not known.

Review Problems

4-105 Two large steel plates are stuck together because of the freezing of the water between the two plates. Hot air is blown over the exposed surface of the plate on the top to melt the ice. The length of time the hot air should be blown is to be determined.

Assumptions 1 Heat conduction in the plates is one-dimensional since the plate is large relative to its thickness and there is thermal symmetry about the center plane. 3 The thermal properties of the steel plates are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of steel plates are given to be $k = 43 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 1.17 \times 10^{-5} \text{ m}^2/\text{s}$

Analysis The characteristic length of the plates and the Biot number are

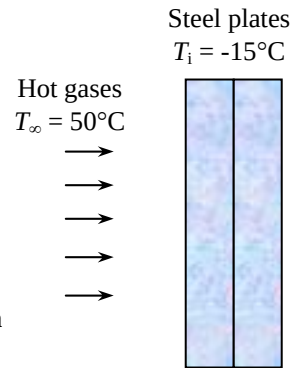
$$L_c = \frac{V}{A_s} = L = 0.02 \text{ m}$$

$$Bi = \frac{hL_c}{k} = \frac{(40 \text{ W/m}^2\cdot^\circ\text{C})(0.02 \text{ m})}{(43 \text{ W/m}\cdot^\circ\text{C})} = 0.019 < 0.1$$

Since $Bi < 0.1$, the lumped system analysis is applicable. Therefore,

$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{40 \text{ W/m}^2\cdot^\circ\text{C}}{(3.675 \times 10^6 \text{ J/m}^3\cdot^\circ\text{C})(0.02 \text{ m})} = 0.000544 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{0 - 50}{-15 - 50} = e^{-(0.000544 \text{ s}^{-1})t} \longrightarrow t = \mathbf{482 \text{ s} = 8.0 \text{ min}}$$



where $\rho C_p = \frac{k}{\alpha} = \frac{43 \text{ W/m}\cdot^\circ\text{C}}{1.17 \times 10^{-5} \text{ m}^2/\text{s}} = 3.675 \times 10^6 \text{ J/m}^3\cdot^\circ\text{C}$

Alternative solution: This problem can also be solved using the transient chart Fig. 4-13a,

$$\left. \begin{aligned} \frac{1}{Bi} &= \frac{1}{0.019} = 52.6 \\ \frac{T_o - T_\infty}{T_i - T_\infty} &= \frac{0 - 50}{-15 - 50} = 0.769 \end{aligned} \right\} \tau = \frac{\alpha t}{r_o^2} = 15 > 0.2$$

Then,

$$t = \frac{\tau r_o^2}{\alpha} = \frac{(15)(0.02 \text{ m})^2}{(1.17 \times 10^{-5} \text{ m}^2/\text{s})} = \mathbf{513 \text{ s}}$$

The difference is due to the reading error of the chart.

4-106 A curing kiln is heated by injecting steam into it and raising its inner surface temperature to a specified value. It is to be determined whether the temperature at the outer surfaces of the kiln changes during the curing period.

Assumptions 1 The temperature in the wall is affected by the thermal conditions at inner surfaces only and the convection heat transfer coefficient inside is very large. Therefore, the wall can be considered to be a semi-infinite medium with a specified surface temperature of 45°C. 2 The thermal properties of the concrete wall are constant.

Properties The thermal properties of the concrete wall are given to be $k = 0.9 \text{ W/m} \cdot ^\circ\text{C}$ and $\alpha = 0.23 \times 10^{-5} \text{ m}^2/\text{s}$.

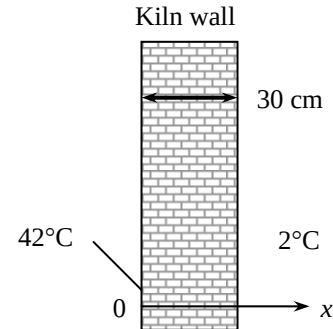
Analysis We determine the temperature at a depth of $x = 0.3 \text{ m}$ in 3 h using the analytical solution,

$$\frac{T(x, t) - T_i}{T_s - T_i} = \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

Substituting,

$$\begin{aligned} \frac{T(x, t) - 2}{42 - 2} &= \text{erfc}\left(\frac{0.3 \text{ m}}{2\sqrt{(0.23 \times 10^{-5} \text{ m}^2/\text{s})(3 \text{ h} \times 3600 \text{ s/h})}}\right) \\ &= \text{erfc}(0.952) = 0.1782 \end{aligned}$$

$$T(x, t) = \mathbf{9.1^\circ\text{C}}$$



which is greater than the initial temperature of 2°C. Therefore, heat will propagate through the 0.3 m thick wall in 3 h, and thus it may be desirable to insulate the outer surface of the wall to save energy.

4-107 The water pipes are buried in the ground to prevent freezing. The minimum burial depth at a particular location is to be determined.

Assumptions 1 The temperature in the soil is affected by the thermal conditions at one surface only, and thus the soil can be considered to be a semi-infinite medium with a specified surface temperature of -10°C. 2 The thermal properties of the soil are constant.

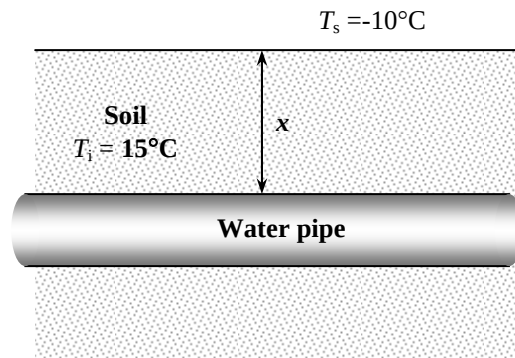
Properties The thermal properties of the soil are given to be $k = 0.7 \text{ W/m} \cdot ^\circ\text{C}$ and $\alpha = 1.4 \times 10^{-5} \text{ m}^2/\text{s}$.

Analysis The depth at which the temperature drops to 0°C in 75 days is determined using the analytical solution,

$$\frac{T(x, t) - T_i}{T_s - T_i} = \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

Substituting,

$$\begin{aligned} \frac{0 - 15}{-10 - 15} &= \text{erfc}\left(\frac{x}{2\sqrt{(1.4 \times 10^{-5} \text{ m}^2/\text{s})(75 \text{ day} \times 24 \text{ h/day} \times 3600 \text{ s/h})}}\right) \\ &\longrightarrow \mathbf{x = 7.05 \text{ m}} \end{aligned}$$



Therefore, the pipes must be buried at a depth of at least 7.05 m.

4-108 A hot dog is to be cooked by dropping it into boiling water. The time of cooking is to be determined.

Assumptions 1 Heat conduction in the hot dog is two-dimensional, and thus the temperature varies in both the axial x - and the radial r - directions. 2 The thermal properties of the hot dog are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of the hot dog are given to be $k = 0.76 \text{ W/m}\cdot^\circ\text{C}$, $\rho = 980 \text{ kg/m}^3$, $C_p = 3.9 \text{ kJ/kg}\cdot^\circ\text{C}$, and $\alpha = 2 \times 10^{-7} \text{ m}^2/\text{s}$.

Analysis This hot dog can physically be formed by the intersection of an infinite plane wall of thickness $2L = 12 \text{ cm}$, and a long cylinder of radius $r_o = D/2 = 1 \text{ cm}$. The Biot numbers and corresponding constants are first determined to be

$$Bi = \frac{hL}{k} = \frac{(600 \text{ W/m}^2\cdot^\circ\text{C})(0.06 \text{ m})}{(0.76 \text{ W/m}\cdot^\circ\text{C})} = 47.37 \longrightarrow \lambda_1 = 1.5381 \text{ and } A_1 = 1.2726$$

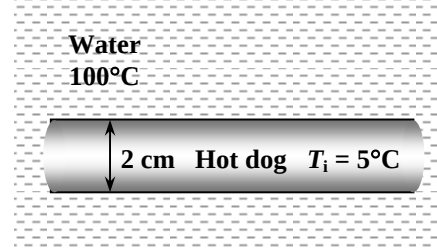
$$Bi = \frac{hr_o}{k} = \frac{(600 \text{ W/m}^2\cdot^\circ\text{C})(0.01 \text{ m})}{(0.76 \text{ W/m}\cdot^\circ\text{C})} = 7.895 \longrightarrow \lambda_1 = 2.1251 \text{ and } A_1 = 1.5515$$

Noting that $\tau = \alpha t / L^2$ and assuming $\tau > 0.2$ in all dimensions and thus the one-term approximate solution for transient heat conduction is applicable, the product solution for this problem can be written as

$$\theta(0,0,t)_{block} = \theta(0,t)_{wall} \theta(0,t)_{cyl} = \left(A_1 e^{-\lambda_1^2 \tau} \right) \left(A_1 e^{-\lambda_1^2 \tau} \right)$$

$$\frac{80-100}{5-100} = \left\{ (1.2726) \exp \left[- (1.5381)^2 \frac{(2 \times 10^{-7})t}{(0.06)^2} \right] \right\}$$

$$\times \left\{ (1.5515) \exp \left[- (2.1251)^2 \frac{(2 \times 10^{-7})t}{(0.01)^2} \right] \right\} = 0.2105$$



which gives

$$t = 244 \text{ s} = 4.1 \text{ min}$$

Therefore, it will take about 4.1 min for the hot dog to cook. Note that

$$\tau_{cyl} = \frac{\alpha t}{r_o^2} = \frac{(2 \times 10^{-7} \text{ m}^2/\text{s})(244 \text{ s})}{(0.01 \text{ m})^2} = 0.49 > 0.2$$

and thus the assumption $\tau > 0.2$ for the applicability of the one-term approximate solution is verified.

Discussion This problem could also be solved by treating the hot dog as an infinite cylinder since heat transfer through the end surfaces will have little effect on the mid section temperature because of the large distance.

4-109 A long roll of large 1-Mn manganese steel plate is to be quenched in an oil bath at a specified rate. The temperature of the sheet metal after quenching and the rate at which heat needs to be removed from the oil in order to keep its temperature constant are to be determined.

Assumptions 1 The thermal properties of the balls are constant. 2 The heat transfer coefficient is constant and uniform over the entire surface. 3 The Biot number is $Bi < 0.1$ so that the lumped system analysis is applicable (this assumption will be checked).

Properties The properties of the steel plate are given to be $k = 60.5 \text{ W/m} \cdot ^\circ\text{C}$, $\rho = 7854 \text{ kg/m}^3$, and $C_p = 434 \text{ J/kg} \cdot ^\circ\text{C}$ (Table A-3).

Analysis The characteristic length of the steel plate and the Biot number are

$$L_c = \frac{V}{A_s} = L = 0.0025 \text{ m}$$

$$Bi = \frac{hL_c}{k} = \frac{(860 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0025 \text{ m})}{60.5 \text{ W/m} \cdot ^\circ\text{C}} = 0.036 < 0.1$$

Since $Bi < 0.1$, the lumped system analysis is applicable. Therefore,

$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{860 \text{ W/m}^2 \cdot ^\circ\text{C}}{(7854 \text{ kg/m}^3)(434 \text{ J/kg} \cdot ^\circ\text{C})(0.0025 \text{ m})} = 0.10092 \text{ s}^{-1}$$

$$\text{time} = \frac{\text{length}}{\text{velocity}} = \frac{5 \text{ m}}{10 \text{ m/min}} = 0.5 \text{ min} = 30 \text{ s}$$

Then the temperature of the sheet metal when it leaves the oil bath is determined to be

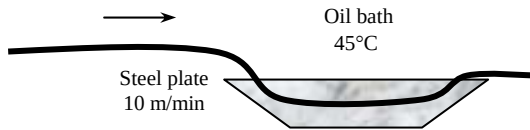
$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{T(t) - 45}{820 - 45} = e^{-(0.10092 \text{ s}^{-1})(30 \text{ s})} \longrightarrow T(t) = \mathbf{82.53^\circ\text{C}}$$

The mass flow rate of the sheet metal through the oil bath is

$$\dot{m} = \rho \dot{V} = \rho w t V = (7854 \text{ kg/m}^3)(2 \text{ m})(0.005 \text{ m})(10 \text{ m/min}) = 785.4 \text{ kg/min}$$

Then the rate of heat transfer from the sheet metal to the oil bath and thus the rate at which heat needs to be removed from the oil in order to keep its temperature constant at 45°C becomes

$$\dot{Q} = \dot{m} C_p [T(t) - T_\infty] = (785.4 \text{ kg/min})(434 \text{ J/kg} \cdot ^\circ\text{C})(82.53 - 45)^\circ\text{C} = 1.279 \times 10^7 \text{ J/min} = \mathbf{213.2 \text{ kW}}$$



4-110E A stuffed turkey is cooked in an oven. The average heat transfer coefficient at the surface of the turkey, the temperature of the skin of the turkey in the oven and the total amount of heat transferred to the turkey in the oven are to be determined.

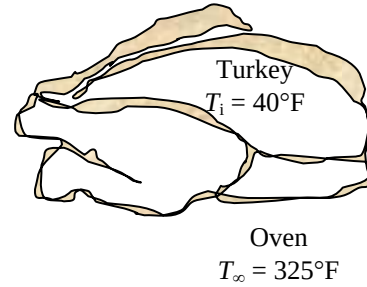
Assumptions 1 The turkey is a homogeneous spherical object. 2 Heat conduction in the turkey is one-dimensional because of symmetry about the midpoint. 3 The thermal properties of the turkey are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions are applicable (this assumption will be verified).

Properties The properties of the turkey are given to be $k = 0.26 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$, $\rho = 75 \text{ lbm/ft}^3$, $C_p = 0.98 \text{ Btu/lbm}\cdot^\circ\text{F}$, and $\alpha = 0.0035 \text{ ft}^2/\text{h}$.

Analysis (a) Assuming the turkey to be spherical in shape, its radius is determined to be

$$m = \rho V \longrightarrow V = \frac{m}{\rho} = \frac{14 \text{ lbm}}{75 \text{ lbm/ft}^3} = 0.1867 \text{ ft}^3$$

$$V = \frac{4}{3}\pi r_o^3 \longrightarrow r_o = \sqrt[3]{\frac{3V}{4\pi}} = \sqrt[3]{\frac{3(0.1867 \text{ ft}^3)}{4\pi}} = 0.3545 \text{ ft}$$



The Fourier number is $\tau = \frac{\alpha t}{r_o^2} = \frac{(3.5 \times 10^{-3} \text{ ft}^2/\text{h})(5 \text{ h})}{(0.3545 \text{ ft})^2} = 0.1392$

which is close to 0.2 but a little below it. Therefore, assuming the one-term approximate solution for transient heat conduction to be applicable, the one-term solution formulation at one-third the radius from the center of the turkey can be expressed as

$$\theta(x, t)_{sph} = \frac{T(x, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r / r_o)}{\lambda_1 r / r_o}$$

$$\frac{185 - 325}{40 - 325} = 0.491 = A_1 e^{-\lambda_1^2 (0.14)} \frac{\sin(0.333 \lambda_1)}{0.333 \lambda_1}$$

By trial and error, it is determined from Table 4-1 that the equation above is satisfied when $Bi = 20$ corresponding to $\lambda_1 = 2.9857$ and $A_1 = 1.9781$. Then the heat transfer coefficient can be determined from

$$Bi = \frac{hr_o}{k} \longrightarrow h = \frac{kBi}{r_o} = \frac{(0.26 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F})(20)}{(0.3545 \text{ ft})} = \mathbf{14.7 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}}$$

(b) The temperature at the surface of the turkey is

$$\frac{T(r_o, t) - 325}{40 - 325} = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r_o / r_o)}{\lambda_1 r_o / r_o} = (1.9781) e^{-(2.9857)^2 (0.14)} \frac{\sin(2.9857)}{2.9857} = 0.02953$$

$$\longrightarrow T(r_o, t) = \mathbf{317^\circ\text{F}}$$

(c) The maximum possible heat transfer is

$$Q_{\max} = mC_p (T_\infty - T_i) = (14 \text{ lbm})(0.98 \text{ Btu/lbm}\cdot^\circ\text{F})(325 - 40)^\circ\text{F} = 3910 \text{ Btu}$$

Then the actual amount of heat transfer becomes

$$\frac{Q}{Q_{\max}} = 1 - 3\theta_{o, sph} \frac{\sin(\lambda_1) - \lambda_1 \cos(\lambda_1)}{\lambda_1^3} = 1 - 3(0.491) \frac{\sin(2.9857) - (2.9857) \cos(2.9857)}{(2.9857)^3} = 0.828$$

$$Q = 0.828 Q_{\max} = (0.828)(3910 \text{ Btu}) = \mathbf{3238 \text{ Btu}}$$

Discussion The temperature of the outer parts of the turkey will be greater than that of the inner parts when the turkey is taken out of the oven. Then heat will continue to be transferred from the outer parts of the turkey to the inner as a result of temperature difference. Therefore, after 5 minutes, the thermometer reading will probably be more than 185°F .

4-111 The trunks of some dry oak trees are exposed to hot gases. The time for the ignition of the trunks is to be determined.

Assumptions 1 Heat conduction in the trunks is one-dimensional since it is long and it has thermal symmetry about the center line. 2 The thermal properties of the trunks are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of the trunks are given to be $k = 0.17 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 1.28 \times 10^{-7} \text{ m}^2/\text{s}$.

Analysis We treat the trunks of the trees as an infinite cylinder since heat transfer is primarily in the radial direction. Then the Biot number becomes

$$Bi = \frac{hr_o}{k} = \frac{(65 \text{ W/m}^2\cdot^\circ\text{C})(0.1 \text{ m})}{(0.17 \text{ W/m}\cdot^\circ\text{C})} = 38.24$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 2.3420 \quad \text{and} \quad A_1 = 1.5989$$

The Fourier number is

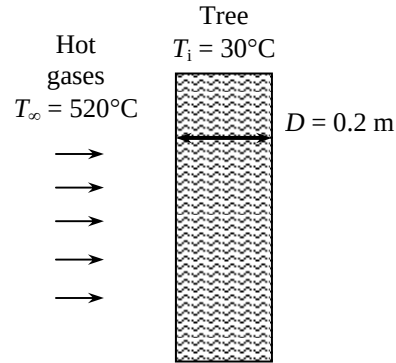
$$\tau = \frac{\alpha t}{r_o^2} = \frac{(1.28 \times 10^{-7} \text{ m}^2/\text{s})(4 \text{ h} \times 3600 \text{ s/h})}{(0.1 \text{ m})^2} = 0.184$$

which is slightly below 0.2 but close to it. Therefore, assuming the one-term approximate solution for transient heat conduction to be applicable, the temperature at the surface of the trees in 4 h becomes

$$\theta(r_o, t)_{\text{cyl}} = \frac{T(r_o, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} J_0(\lambda_1 r / r_o)$$

$$\frac{T(r_o, t) - 520}{30 - 520} = (1.5989) e^{-(2.3420)^2 (0.184)} (0.0332) = 0.01935 \longrightarrow T(r_o, t) = 511^\circ\text{C} > 410^\circ\text{C}$$

Therefore, the trees will ignite. (Note: J_0 is read from Table 4-2).



4-112 A spherical watermelon that is cut into two equal parts is put into a freezer. The time it will take for the center of the exposed cut surface to cool from 25 to 3°C is to be determined.

Assumptions 1 The temperature of the exposed surfaces of the watermelon is affected by the convection heat transfer at those surfaces only. Therefore, the watermelon can be considered to be a semi-infinite medium 2 The thermal properties of the watermelon are constant.

Properties The thermal properties of the water is closely approximated by those of water at room temperature, $k = 0.607 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = k / \rho C_p = 0.146 \times 10^{-6} \text{ m}^2/\text{s}$ (Table A-9).

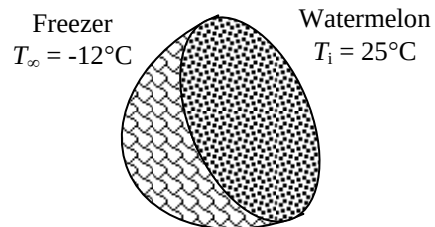
Analysis We use the transient chart in Fig. 4-23 in this case for convenience (instead of the analytic solution),

$$1 - \frac{T(x, t) - T_\infty}{T_i - T_\infty} = 1 - \frac{3 - (-12)}{25 - (-12)} = 0.595$$

$$\left. \begin{aligned} \xi = \frac{x}{2\sqrt{\alpha t}} = 0 \\ \frac{h\sqrt{\alpha t}}{k} = 1 \end{aligned} \right\}$$

Therefore,

$$t = \frac{(1)^2 k^2}{h^2 \alpha} = \frac{(0.607 \text{ W/m}\cdot^\circ\text{C})^2}{(30 \text{ W/m}^2\cdot^\circ\text{C})^2 (0.146 \times 10^{-6} \text{ m}^2/\text{s})} = 2804 \text{ s} = 46.7 \text{ min}$$



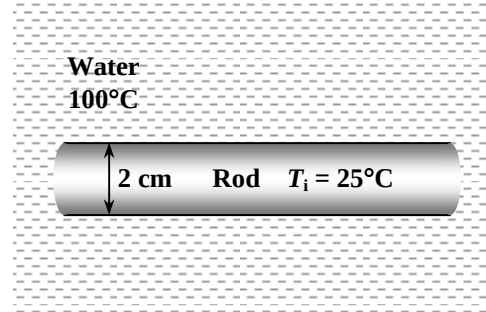
4-113 A cylindrical rod is dropped into boiling water. The thermal diffusivity and the thermal conductivity of the rod are to be determined.

Assumptions 1 Heat conduction in the rod is one-dimensional since the rod is sufficiently long, and thus temperature varies in the radial direction only. 2 The thermal properties of the rod are constant.

Properties The thermal properties of the rod available are given to be $\rho = 3700 \text{ kg/m}^3$ and $C_p = 920 \text{ J/kg} \cdot ^\circ\text{C}$.

Analysis From Fig. 4-14b we have

$$\left. \begin{aligned} \frac{T - T_\infty}{T_o - T_\infty} &= \frac{93 - 100}{75 - 100} = 0.28 \\ \frac{x}{r_o} &= \frac{r_o}{r_o} = 1 \end{aligned} \right\} \frac{1}{Bi} = \frac{k}{hr_o} = 0.25$$



From Fig. 4-14a we have

$$\left. \begin{aligned} \frac{1}{Bi} &= \frac{k}{hr_o} = 0.25 \\ \frac{T_o - T_\infty}{T_i - T_\infty} &= \frac{75 - 100}{25 - 100} = 0.33 \end{aligned} \right\} \tau = \frac{\alpha t}{r_o^2} = 0.40$$

Then the thermal diffusivity and the thermal conductivity of the material become

$$\alpha = \frac{0.40 r_o^2}{t} = \frac{(0.40)(0.01 \text{ m})^2}{3 \text{ min} \times 60 \text{ s/min}} = 2.22 \times 10^{-7} \text{ m}^2/\text{s}$$

$$\alpha = \frac{k}{\rho C_p} \longrightarrow k = \alpha \rho C_p = (2.22 \times 10^{-7} \text{ m}^2/\text{s})(3700 \text{ kg/m}^3)(920 \text{ J/kg} \cdot ^\circ\text{C}) = 0.756 \text{ W/m} \cdot ^\circ\text{C}$$

4-114 The time it will take for the diameter of a raindrop to reduce to a certain value as it falls through ambient air is to be determined.

Assumptions 1 The water temperature remains constant. 2 The thermal properties of the water are constant.

Properties The density and heat of vaporization of the water are $\rho = 1000 \text{ kg/m}^3$ and $h_{fg} = 2490 \text{ kJ/kg}$ (Table A-9).

Analysis The initial and final masses of the raindrop are

$$m_i = \rho V_i = \rho \frac{4}{3} \pi r_i^3 = (1000 \text{ kg/m}^3) \frac{4}{3} \pi (0.0025 \text{ m})^3 = 0.0000654 \text{ kg}$$

$$m_f = \rho V_f = \rho \frac{4}{3} \pi r_f^3 = (1000 \text{ kg/m}^3) \frac{4}{3} \pi (0.0015 \text{ m})^3 = 0.0000141 \text{ kg}$$

whose difference is

$$m = m_i - m_f = 0.0000654 - 0.0000141 = 0.0000513 \text{ kg}$$

The amount of heat transfer required to cause this much evaporation is

$$Q = (0.0000513 \text{ kg})(2490 \text{ kJ/kg}) = 0.1278 \text{ kJ}$$

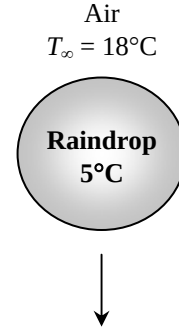
The average heat transfer surface area and the rate of heat transfer are

$$A_s = \frac{4\pi(r_i^2 + r_f^2)}{2} = \frac{4\pi[(0.0025 \text{ m})^2 + (0.0015 \text{ m})^2]}{2} = 5.341 \times 10^{-5} \text{ m}^2$$

$$\dot{Q} = hA_s(T_i - T_\infty) = (400 \text{ W/m}^2 \cdot ^\circ\text{C})(5.341 \times 10^{-5} \text{ m}^2)(18 - 5)^\circ\text{C} = 0.2777 \text{ J/s}$$

Then the time required for the raindrop to experience this reduction in size becomes

$$\dot{Q} = \frac{Q}{\Delta t} \longrightarrow \Delta t = \frac{Q}{\dot{Q}} = \frac{127.8 \text{ J}}{0.2777 \text{ J/s}} = 460 \text{ s} = \mathbf{7.7 \text{ min}}$$



4-115E A plate, a long cylinder, and a sphere are exposed to cool air. The center temperature of each geometry is to be determined.

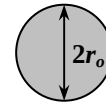
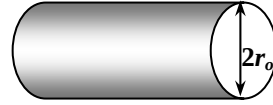
Assumptions 1 Heat conduction in each geometry is one-dimensional. 2 The thermal properties of the bodies are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of bronze are given to be $k = 15 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$ and $\alpha = 0.333 \text{ ft}^2/\text{h}$.

Analysis After 5 minutes

Plate: First the Biot number is calculated to be

$$Bi = \frac{hL}{k} = \frac{(7 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F})(0.5/12 \text{ ft})}{(15 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F})} = 0.01944$$

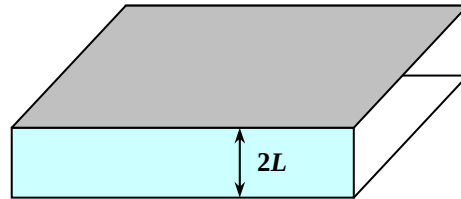


The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 0.1410 \quad \text{and} \quad A_1 = 1.0033$$

The Fourier number is

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.333 \text{ ft}^2/\text{h})(5 \text{ min}/60 \text{ min/h})}{(0.5/12 \text{ ft})^2} = 15.98 > 0.2$$



Then the center temperature of the plate becomes

$$\theta_{0,\text{wall}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0033) e^{-(0.1410)^2 (15.98)} = 0.730 \longrightarrow T_0 = 312^\circ\text{F}$$

Cylinder:

$$Bi = 0.02 \xrightarrow{\text{Table 9-1}} \lambda_1 = 0.1995 \quad \text{and} \quad A_1 = 1.0050$$

$$\theta_{0,\text{cyl}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0050) e^{-(0.1995)^2 (15.98)} = 0.532 \longrightarrow T_0 = 248^\circ\text{F}$$

Sphere:

$$Bi = 0.02 \xrightarrow{\text{Table 9-1}} \lambda_1 = 0.2445 \quad \text{and} \quad A_1 = 1.0060$$

$$\theta_{0,\text{sph}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0060) e^{-(0.2445)^2 (15.98)} = 0.387 \longrightarrow T_0 = 201^\circ\text{F}$$

After 10 minutes

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.333 \text{ ft}^2/\text{h})(10 \text{ min}/60 \text{ min/h})}{(0.5/12 \text{ ft})^2} = 31.97 > 0.2$$

Plate:

$$\theta_{0,\text{wall}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0033) e^{-(0.1410)^2 (31.97)} = 0.531 \longrightarrow T_0 = 248^\circ\text{F}$$

Cylinder:

$$\theta_{0,cyl} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0050)e^{-(0.1995)^2(31.97)} = 0.282 \longrightarrow T_0 = \mathbf{167^\circ F}$$

Sphere:

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0060)e^{-(0.2445)^2(31.97)} = 0.149 \longrightarrow T_0 = \mathbf{123^\circ F}$$

After 30 minutes

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.333 \text{ ft}^2 / \text{h})(30 \text{ min} / 60 \text{ min} / \text{h})}{(0.5 / 12 \text{ ft})^2} = 95.9 > 0.2$$

Plate:

$$\theta_{0,wall} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0033)e^{-(0.1410)^2(95.9)} = 0.149 \longrightarrow T_0 = \mathbf{123^\circ F}$$

Cylinder:

$$\theta_{0,cyl} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0050)e^{-(0.1995)^2(95.9)} = 0.0221 \longrightarrow T_0 = \mathbf{82^\circ F}$$

Sphere:

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0060)e^{-(0.2445)^2(95.9)} = 0.00326 \longrightarrow T_0 = \mathbf{76^\circ F}$$

The sphere has the largest surface area through which heat is transferred per unit volume, and thus the highest rate of heat transfer. Consequently, the center temperature of the sphere is always the lowest.

4-116E A plate, a long cylinder, and a sphere are exposed to cool air. The center temperature of each geometry is to be determined. ✓

Assumptions 1 Heat conduction in each geometry is one-dimensional. 2 The thermal properties of the geometries are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of cast iron are given to be $k = 29 \text{ Btu/h.ft.}^\circ\text{F}$ and $\alpha = 0.61 \text{ ft}^2/\text{h}$.

Analysis After 5 minutes

Plate: First the Biot number is calculated to be

$$Bi = \frac{hL}{k} = \frac{(7 \text{ Btu/h.ft}^2.\text{°F})(0.5/12 \text{ ft})}{(29 \text{ Btu/h.ft.}^\circ\text{F})} = 0.01006$$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

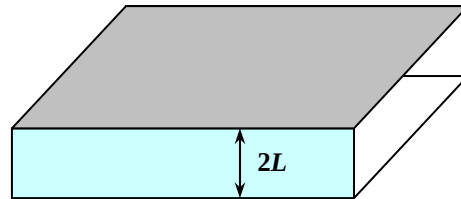
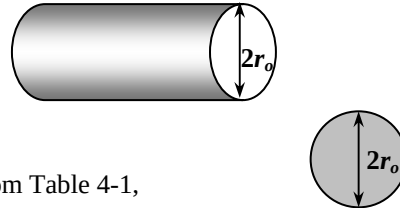
$$\lambda_1 = 0.0998 \quad \text{and} \quad A_1 = 1.0017$$

The Fourier number is

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.61 \text{ ft}^2/\text{h})(5 \text{ min}/60 \text{ min/h})}{(0.5/12 \text{ ft})^2} = 29.28 > 0.2$$

Then the center temperature of the plate becomes

$$\theta_{0,\text{wall}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0017) e^{-(0.0998)^2 (29.28)} = 0.748 \longrightarrow T_0 = 318^\circ\text{F}$$



Cylinder:

$$Bi = 0.01 \xrightarrow{\text{Table 4-1}} \lambda_1 = 0.1412 \quad \text{and} \quad A_1 = 1.0025$$

$$\theta_{0,\text{cyl}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0025) e^{-(0.1412)^2 (29.28)} = 0.559 \longrightarrow T_0 = 257^\circ\text{F}$$

Sphere:

$$Bi = 0.01 \xrightarrow{\text{Table 4-1}} \lambda_1 = 0.1730 \quad \text{and} \quad A_1 = 1.0030$$

$$\theta_{0,\text{sph}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0030) e^{-(0.1730)^2 (29.28)} = 0.418 \longrightarrow T_0 = 211^\circ\text{F}$$

After 10 minutes

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.61 \text{ ft}^2/\text{h})(10 \text{ min}/60 \text{ min/h})}{(0.5/12 \text{ ft})^2} = 58.56 > 0.2$$

Plate:

$$\theta_{0,\text{wall}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0017) e^{-(0.0998)^2 (58.56)} = 0.559 \longrightarrow T_0 = 257^\circ\text{F}$$

Cylinder:

$$\theta_{0,cyl} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0025)e^{-(0.1412)^2 (58.56)} = 0.312 \longrightarrow T_0 = \mathbf{176^\circ F}$$

Sphere:

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0030)e^{-(0.1730)^2 (58.56)} = 0.174 \longrightarrow T_0 = \mathbf{132^\circ F}$$

After 30 minutes

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.61 \text{ ft}^2/\text{h})(30 \text{ min}/60 \text{ min/h})}{(0.5/12 \text{ ft})^2} = 175.68 > 0.2$$

Plate:

$$\theta_{0,wall} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0017)e^{-(0.0998)^2 (175.68)} = 0.174 \longrightarrow T_0 = \mathbf{132^\circ F}$$

Cylinder:

$$\theta_{0,cyl} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0025)e^{-(0.1412)^2 (175.68)} = 0.030 \longrightarrow T_0 = \mathbf{84.8^\circ F}$$

Sphere:

$$\theta_{0,sph} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{T_0 - 75}{400 - 75} = (1.0030)e^{-(0.1730)^2 (175.68)} = 0.0052 \longrightarrow T_0 = \mathbf{76.7^\circ F}$$

The sphere has the largest surface area through which heat is transferred per unit volume, and thus the highest rate of heat transfer. Consequently, the center temperature of the sphere is always the lowest.

4-117E "PROBLEM 4-117E"

"GIVEN"

$$2L = 1/12 \text{ [ft]}$$

$$2r_{o,c} = 1/12 \text{ [ft], c stands for cylinder}$$

$$2r_{o,s} = 1/12 \text{ [ft], s stands for sphere}$$

$$T_i = 400 \text{ [F]}$$

$$T_{\infty} = 75 \text{ [F]}$$

$$h = 7 \text{ [Btu/h-ft}^2\text{-F]}$$

"time = 5 [min], parameter to be varied"

"PROPERTIES"

$$k = 15 \text{ [Btu/h-ft-F]}$$

$$\alpha = 0.333 \text{ Convert(ft}^2\text{/h, ft}^2\text{/min) [ft}^2\text{/min]}$$

"ANALYSIS"

"For plane wall"

$$Bi_w = (hL)/k$$

"From Table 4-1 corresponding to this Bi number, we read"

$$\lambda_{1,w} = 0.1410$$

$$A_{1,w} = 1.0033$$

$$\tau_w = (\alpha \text{time})/L^2$$

$$(T_{o,w} - T_{\infty})/(T_i - T_{\infty}) = A_{1,w} \exp(-\lambda_{1,w}^2 \tau_w)$$

"For long cylinder"

$$Bi_c = (hr_{o,c})/k$$

"From Table 4-1 corresponding to this Bi number, we read"

$$\lambda_{1,c} = 0.1995$$

$$A_{1,c} = 1.0050$$

$$\tau_c = (\alpha \text{time})/r_{o,c}^2$$

$$(T_{o,c} - T_{\infty})/(T_i - T_{\infty}) = A_{1,c} \exp(-\lambda_{1,c}^2 \tau_c)$$

"For sphere"

$$Bi_s = (hr_{o,s})/k$$

"From Table 4-1 corresponding to this Bi number, we read"

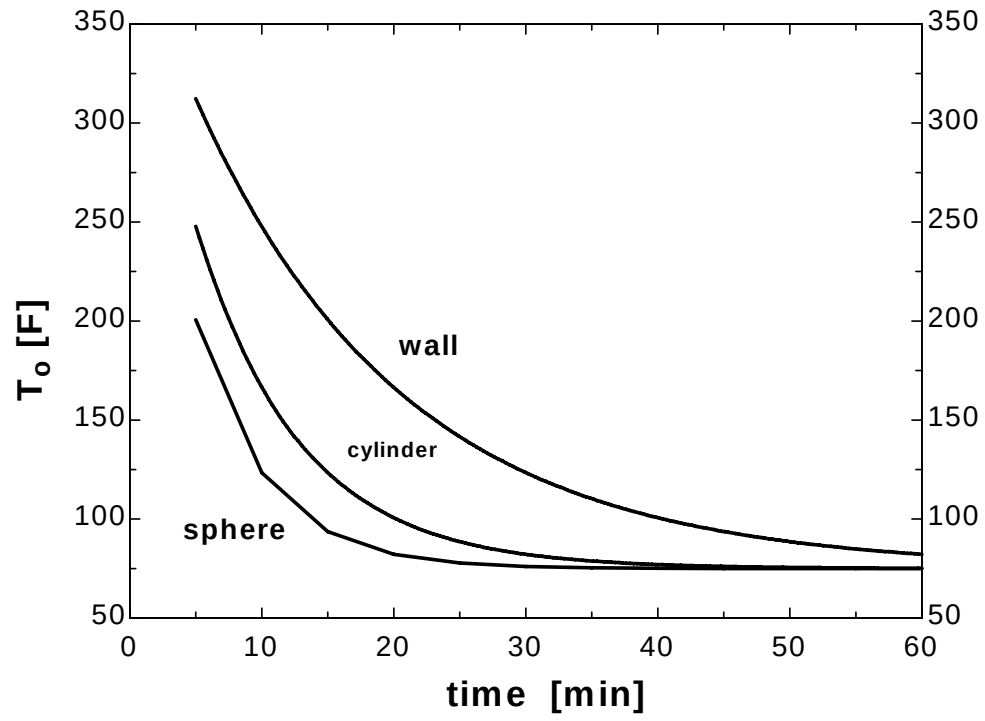
$$\lambda_{1,s} = 0.2445$$

$$A_{1,s} = 1.0060$$

$$\tau_s = (\alpha \text{time})/r_{o,s}^2$$

$$(T_{o,s} - T_{\infty})/(T_i - T_{\infty}) = A_{1,s} \exp(-\lambda_{1,s}^2 \tau_s)$$

time [min]	$T_{o,w}$ [F]	$T_{o,c}$ [F]	$T_{o,s}$ [F]
5	312.3	247.9	200.7
10	247.7	166.5	123.4
15	200.7	123.4	93.6
20	166.5	100.6	82.15
25	141.6	88.57	77.75
30	123.4	82.18	76.06
35	110.3	78.8	75.41
40	100.7	77.01	75.16
45	93.67	76.07	75.06
50	88.59	75.56	75.02
55	84.89	75.3	75.01
60	82.2	75.16	75



4-118 Internal combustion engine valves are quenched in a large oil bath. The time it takes for the valve temperature to drop to specified temperatures and the maximum heat transfer are to be determined.

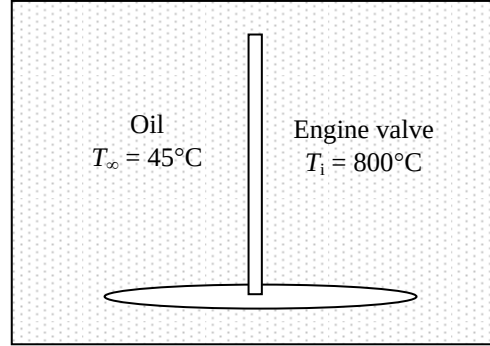
Assumptions **1** The thermal properties of the valves are constant. **2** The heat transfer coefficient is constant and uniform over the entire surface. **3** Depending on the size of the oil bath, the oil bath temperature will increase during quenching. However, an average constant temperature as specified in the problem will be used. **4** The Biot number is $Bi < 0.1$ so that the lumped system analysis is applicable (this assumption will be verified).

Properties The thermal conductivity, density, and specific heat of the balls are given to be $k = 48 \text{ W/m} \cdot ^\circ\text{C}$, $\rho = 7840 \text{ kg/m}^3$, and $C_p = 440 \text{ J/kg} \cdot ^\circ\text{C}$.

Analysis (a) The characteristic length of the balls and the Biot number are

$$L_c = \frac{V}{A_s} = \frac{1.8(\pi D^2 L / 4)}{2\pi DL} = \frac{1.8D}{8} = \frac{1.8(0.008 \text{ m})}{8} = 0.0018 \text{ m}$$

$$Bi = \frac{hL_c}{k} = \frac{(650 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0018 \text{ m})}{(48 \text{ W/m} \cdot ^\circ\text{C})} = 0.024 < 0.1$$



Therefore, we can use lumped system analysis. Then the time for a final valve temperature of 400°C becomes

$$b = \frac{hA_s}{\rho C_p V} = \frac{8h}{1.8\rho C_p D} = \frac{8(650 \text{ W/m}^2 \cdot ^\circ\text{C})}{1.8(7840 \text{ kg/m}^3)(440 \text{ J/kg} \cdot ^\circ\text{C})(0.008 \text{ m})} = 0.10468 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{400 - 45}{800 - 45} = e^{-(0.10468 \text{ s}^{-1})t} \longrightarrow t = \mathbf{7.2 \text{ s}}$$

(b) The time for a final valve temperature of 200°C is

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{200 - 45}{800 - 45} = e^{-(0.10468 \text{ s}^{-1})t} \longrightarrow t = \mathbf{15.1 \text{ s}}$$

(c) The time for a final valve temperature of 46°C is

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{46 - 45}{800 - 45} = e^{-(0.10468 \text{ s}^{-1})t} \longrightarrow t = \mathbf{63.3 \text{ s}}$$

(d) The maximum amount of heat transfer from a single valve is determined from

$$m = \rho V = \rho \frac{1.8\pi D^2 L}{4} = (7840 \text{ kg/m}^3) \frac{1.8\pi(0.008 \text{ m})^2(0.10 \text{ m})}{4} = 0.0709 \text{ kg}$$

$$Q = mC_p[T_f - T_i] = (0.0709 \text{ kg})(440 \text{ J/kg} \cdot ^\circ\text{C})(800 - 45)^\circ\text{C} = 23,564 \text{ J} = \mathbf{23.56 \text{ kJ}} \text{ (per valve)}$$

4-119 A watermelon is placed into a lake to cool it. The heat transfer coefficient at the surface of the watermelon and the temperature of the outer surface of the watermelon are to be determined.

Assumptions 1 The watermelon is a homogeneous spherical object. 2 Heat conduction in the watermelon is one-dimensional because of symmetry about the midpoint. 3 The thermal properties of the watermelon are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

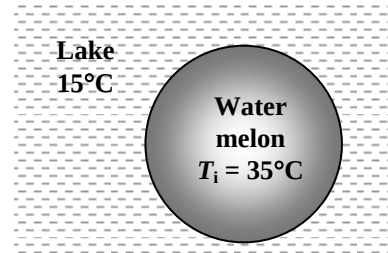
Properties The properties of the watermelon are given to be $k = 0.618 \text{ W/m}\cdot^\circ\text{C}$, $\alpha = 0.15 \times 10^{-6} \text{ m}^2/\text{s}$, $\rho = 995 \text{ kg/m}^3$ and $C_p = 4.18 \text{ kJ/kg}\cdot^\circ\text{C}$.

Analysis The Fourier number is

$$\tau = \frac{\alpha t}{r_o^2} = \frac{(0.15 \times 10^{-6} \text{ m}^2/\text{s})[(4 \times 60 + 40 \text{ min}) \times 60 \text{ s/min}]}{(0.10 \text{ m})^2} = 0.252$$

which is greater than 0.2. Then the one-term solution can be written in the form

$$\theta_{0,\text{sph}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \longrightarrow \frac{20 - 15}{35 - 15} = 0.25 = A_1 e^{-\lambda_1^2 (0.252)}$$



It is determined from Table 4-1 by trial and error that this equation is satisfied when $Bi = 10$, which corresponds to $\lambda_1 = 2.8363$ and $A_1 = 1.9249$. Then the heat transfer coefficient can be determined from

$$Bi = \frac{hr_o}{k} \longrightarrow h = \frac{kBi}{r_o} = \frac{(0.618 \text{ W/m}\cdot^\circ\text{C})(10)}{(0.10 \text{ m})} = \mathbf{61.8 \text{ W/m}^2\cdot^\circ\text{C}}$$

The temperature at the surface of the watermelon is

$$\theta(r_o, t)_{\text{sph}} = \frac{T(r_o, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r_o / r_o)}{\lambda_1 r_o / r_o} = (1.9249) e^{-(2.8363)^2 (0.252)} \frac{\sin(2.8363 \text{ rad})}{2.8363}$$

$$\frac{T(r_o, t) - 15}{35 - 15} = 0.0269 \longrightarrow T(r_o, t) = \mathbf{15.5^\circ\text{C}}$$

4-120 Large food slabs are cooled in a refrigeration room. Center temperatures are to be determined for different foods.

Assumptions 1 Heat conduction in the slabs is one-dimensional since the slab is large relative to its thickness and there is thermal symmetry about the center plane. 3 The thermal properties of the slabs are constant. 4 The heat transfer coefficient is constant and uniform over the entire surface. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of foods are given to be $k = 0.233 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 0.11 \times 10^{-6} \text{ m}^2/\text{s}$ for margarine, $k = 0.082 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 0.10 \times 10^{-6} \text{ m}^2/\text{s}$ for white cake, and $k = 0.106 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 0.12 \times 10^{-6} \text{ m}^2/\text{s}$ for chocolate cake.

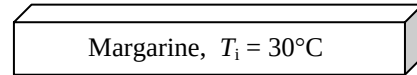
Analysis (a) In the case of margarine, the Biot number is

$$Bi = \frac{hL}{k} = \frac{(25 \text{ W/m}^2\cdot^\circ\text{C})(0.05 \text{ m})}{(0.233 \text{ W/m}\cdot^\circ\text{C})} = 5.365$$

Air
 $T_\infty = 0^\circ\text{C}$

The constants λ_1 and A_1 corresponding to this Biot number are, from Table 4-1,

$$\lambda_1 = 1.3269 \quad \text{and} \quad A_1 = 1.2431$$



The Fourier number is $\tau = \frac{\alpha t}{L^2} = \frac{(0.11 \times 10^{-6} \text{ m}^2/\text{s})(6 \text{ h} \times 3600 \text{ s/h})}{(0.05 \text{ m})^2} = 0.9504 > 0.2$

Therefore, the one-term approximate solution (or the transient temperature charts) is applicable. Then the temperature at the center of the box if the box contains margarine becomes

$$\theta(0, t)_{\text{wall}} = \frac{T(0, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.2431)e^{-(1.3269)^2 (0.9504)}$$

$$\frac{T(0, t) - 0}{30 - 0} = 0.233 \longrightarrow T(0, t) = 7.0^\circ\text{C}$$

(b) Repeating the calculations for white cake,

$$Bi = \frac{hL}{k} = \frac{(25 \text{ W/m}^2\cdot^\circ\text{C})(0.05 \text{ m})}{(0.082 \text{ W/m}\cdot^\circ\text{C})} = 15.24 \longrightarrow \lambda_1 = 1.4641 \quad \text{and} \quad A_1 = 1.2661$$

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.10 \times 10^{-6} \text{ m}^2/\text{s})(6 \text{ h} \times 3600 \text{ s/h})}{(0.05 \text{ m})^2} = 0.864 > 0.2$$

$$\theta(0, t)_{\text{wall}} = \frac{T(0, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.2661)e^{-(1.4641)^2 (0.864)}$$

$$\frac{T(0, t) - 0}{30 - 0} = 0.199 \longrightarrow T(0, t) = 6.0^\circ\text{C}$$

(c) Repeating the calculations for chocolate cake,

$$Bi = \frac{hL}{k} = \frac{(25 \text{ W/m}^2\cdot^\circ\text{C})(0.05 \text{ m})}{(0.106 \text{ W/m}\cdot^\circ\text{C})} = 11.79 \longrightarrow \lambda_1 = 1.4356 \quad \text{and} \quad A_1 = 1.2634$$

$$\tau = \frac{\alpha t}{L^2} = \frac{(0.12 \times 10^{-6} \text{ m}^2/\text{s})(6 \text{ h} \times 3600 \text{ s/h})}{(0.05 \text{ m})^2} = 1.0368 > 0.2$$

$$\theta(0, t)_{\text{wall}} = \frac{T(0, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.2634)e^{-(1.4356)^2 (1.0368)}$$

$$\frac{T(0, t) - 0}{30 - 0} = 0.149 \longrightarrow T(0, t) = 4.5^\circ\text{C}$$

4-121 A cold cylindrical concrete column is exposed to warm ambient air during the day. The time it will take for the surface temperature to rise to a specified value, the amounts of heat transfer for specified values of center and surface temperatures are to be determined.

Assumptions 1 Heat conduction in the column is one-dimensional since it is long and it has thermal symmetry about the center line. 2 The thermal properties of the column are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The properties of concrete are given to be $k = 0.79 \text{ W/m} \cdot ^\circ\text{C}$, $\alpha = 5.94 \times 10^{-7} \text{ m}^2/\text{s}$, $\rho = 1600 \text{ kg/m}^3$ and $C_p = 0.84 \text{ kJ/kg} \cdot ^\circ\text{C}$

Analysis (a) The Biot number is

$$Bi = \frac{hr_o}{k} = \frac{(14 \text{ W/m}^2 \cdot ^\circ\text{C})(0.15 \text{ m})}{(0.79 \text{ W/m} \cdot ^\circ\text{C})} = 2.658$$

The constants λ_1 and A_1 corresponding to this

Biot number are, from Table 4-1,

$$\lambda_1 = 1.7240 \quad \text{and} \quad A_1 = 1.3915$$

Once the constant $J_0 = 0.3841$ is determined from Table 4-2

corresponding to the constant λ_1 , the Fourier number is determined to be

$$\frac{T(r_o, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} J_0(\lambda_1 r_o / r_o) \longrightarrow \frac{27 - 28}{16 - 28} = (1.3915) e^{-(1.7240)^2 \tau} (0.3841) \longrightarrow \tau = 0.6253$$

which is above the value of 0.2. Therefore, the one-term approximate solution (or the transient temperature charts) can be used. Then the time it will take for the column surface temperature to rise to 27°C becomes

$$t = \frac{\tau r_o^2}{\alpha} = \frac{(0.6253)(0.15 \text{ m})^2}{(5.94 \times 10^{-7} \text{ m}^2/\text{s})} = 23,685 \text{ s} = \mathbf{6.6 \text{ hours}}$$

(b) The heat transfer to the column will stop when the center temperature of column reaches to the ambient temperature, which is 28°C . That is, we are asked to determine the maximum heat transfer between the ambient air and the column.

$$m = \rho V = \rho \pi r_o^2 L = (1600 \text{ kg/m}^3)[\pi(0.15 \text{ m})^2 (3.5 \text{ m})] = 395.8 \text{ kg}$$

$$Q_{\max} = m C_p [T_\infty - T_i] = (395.8 \text{ kg})(0.84 \text{ kJ/kg} \cdot ^\circ\text{C})(28 - 16)^\circ\text{C} = \mathbf{3990 \text{ kJ}}$$

(c) To determine the amount of heat transfer until the surface temperature reaches to 27°C , we first determine

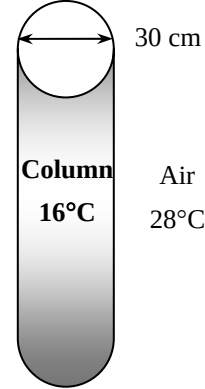
$$\frac{T(0, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.3915) e^{-(1.7240)^2 (0.6253)} = 0.2169$$

Once the constant $J_1 = 0.5787$ is determined from Table 4-2 corresponding to the constant λ_1 , the amount of heat transfer becomes

$$\left(\frac{Q}{Q_{\max}} \right)_{\text{cyl}} = 1 - 2 \left(\frac{T_0 - T_\infty}{T_i - T_\infty} \right) \frac{J_1(\lambda_1)}{\lambda_1} = 1 - 2 \times 0.2169 \times \frac{0.5787}{1.7240} = 0.854$$

$$Q = 0.854 Q_{\max}$$

$$Q = 0.854(3990 \text{ kJ}) = \mathbf{3409 \text{ kJ}}$$



4-122 Long aluminum wires are extruded and exposed to atmospheric air. The time it will take for the wire to cool, the distance the wire travels, and the rate of heat transfer from the wire are to be determined.

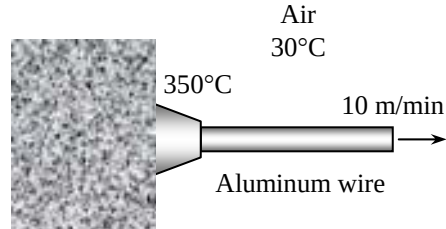
Assumptions 1 Heat conduction in the wires is one-dimensional in the radial direction. 2 The thermal properties of the aluminum are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Biot number is $Bi < 0.1$ so that the lumped system analysis is applicable (this assumption will be verified).

Properties The properties of aluminum are given to be $k = 236 \text{ W/m}\cdot^\circ\text{C}$, $\rho = 2702 \text{ kg/m}^3$, $C_p = 0.896 \text{ kJ/kg}\cdot^\circ\text{C}$, and $\alpha = 9.75 \times 10^{-5} \text{ m}^2/\text{s}$.

Analysis (a) The characteristic length of the wire and the Biot number are

$$L_c = \frac{V}{A_s} = \frac{\pi r_o^2 L}{2\pi r_o L} = \frac{r_o}{2} = \frac{0.0015 \text{ m}}{2} = 0.00075 \text{ m}$$

$$Bi = \frac{hL_c}{k} = \frac{(35 \text{ W/m}^2\cdot^\circ\text{C})(0.00075 \text{ m})}{236 \text{ W/m}\cdot^\circ\text{C}} = 0.00011 < 0.1$$



Since $Bi < 0.1$, the lumped system analysis is applicable. Then,

$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{35 \text{ W/m}^2\cdot^\circ\text{C}}{(2702 \text{ kg/m}^3)(896 \text{ J/kg}\cdot^\circ\text{C})(0.00075 \text{ m})} = 0.0193 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{50 - 30}{350 - 30} = e^{-(0.0193 \text{ s}^{-1})t} \longrightarrow t = \mathbf{144 \text{ s}}$$

(b) The wire travels a distance of

$$\text{velocity} = \frac{\text{length}}{\text{time}} \rightarrow \text{length} = (10 / 60 \text{ m/s})(144 \text{ s}) = \mathbf{24 \text{ m}}$$

This distance can be reduced by cooling the wire in a water or oil bath.

(c) The mass flow rate of the extruded wire through the air is

$$\dot{m} = \rho \dot{V} = \rho(\pi r_o^2 / 4) V = (2702 \text{ kg/m}^3) \pi (0.0015 \text{ m})^2 (10 \text{ m/min}) = 0.191 \text{ kg/min}$$

Then the rate of heat transfer from the wire to the air becomes

$$\dot{Q} = \dot{m} C_p [T(t) - T_\infty] = (0.191 \text{ kg/min})(0.896 \text{ kJ/kg}\cdot^\circ\text{C})(350 - 50)^\circ\text{C} = 51.3 \text{ kJ/min} = \mathbf{856 \text{ W}}$$

4-123 Long copper wires are extruded and exposed to atmospheric air. The time it will take for the wire to cool, the distance the wire travels, and the rate of heat transfer from the wire are to be determined.

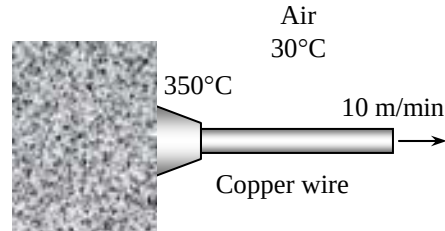
Assumptions 1 Heat conduction in the wires is one-dimensional in the radial direction. 2 The thermal properties of the copper are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Biot number is $Bi < 0.1$ so that the lumped system analysis is applicable (this assumption will be verified).

Properties The properties of copper are given to be $k = 386 \text{ W/m}\cdot^\circ\text{C}$, $\rho = 8950 \text{ kg/m}^3$, $C_p = 0.383 \text{ kJ/kg}\cdot^\circ\text{C}$, and $\alpha = 1.13 \times 10^{-4} \text{ m}^2/\text{s}$.

Analysis (a) The characteristic length of the wire and the Biot number are

$$L_c = \frac{V}{A_s} = \frac{\pi r_o^2 L}{2\pi r_o L} = \frac{r_o}{2} = \frac{0.0015 \text{ m}}{2} = 0.00075 \text{ m}$$

$$Bi = \frac{hL_c}{k} = \frac{(35 \text{ W/m}^2\cdot^\circ\text{C})(0.00075 \text{ m})}{386 \text{ W/m}\cdot^\circ\text{C}} = 0.000068 < 0.1$$



Since $Bi < 0.1$ the lumped system analysis is applicable. Then,

$$b = \frac{hA_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{35 \text{ W/m}^2\cdot^\circ\text{C}}{(8950 \text{ kg/m}^3)(0.383 \text{ kJ/kg}\cdot^\circ\text{C})(0.00075 \text{ m})} = 0.0136 \text{ s}^{-1}$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-bt} \longrightarrow \frac{50 - 30}{350 - 30} = e^{-(0.0136 \text{ s}^{-1})t} \longrightarrow t = \mathbf{204 \text{ s}}$$

(b) The wire travels a distance of

$$\text{velocity} = \frac{\text{length}}{\text{time}} \longrightarrow \text{length} = \left(\frac{10 \text{ m/min}}{60 \text{ s/min}} \right) (204 \text{ s}) = \mathbf{34 \text{ m}}$$

This distance can be reduced by cooling the wire in a water or oil bath.

(c) The mass flow rate of the extruded wire through the air is

$$\dot{m} = \rho \dot{V} = \rho(\pi r_o^2 / 4) V = (8950 \text{ kg/m}^3) \pi (0.0015 \text{ m})^2 (10 \text{ m/min}) = 0.633 \text{ kg/min}$$

Then the rate of heat transfer from the wire to the air becomes

$$\dot{Q} = \dot{m} C_p [T(t) - T_\infty] = (0.633 \text{ kg/min})(0.383 \text{ kJ/kg}\cdot^\circ\text{C})(350 - 50)^\circ\text{C} = 72.7 \text{ kJ/min} = \mathbf{1212 \text{ W}}$$

4-124 A brick house made of brick that was initially cold is exposed to warm atmospheric air at the outer surfaces. The time it will take for the temperature of the inner surfaces of the house to start changing is to be determined.

Assumptions 1 The temperature in the wall is affected by the thermal conditions at outer surfaces only, and thus the wall can be considered to be a semi-infinite medium with a specified outer surface temperature of 18°C. 2 The thermal properties of the brick wall are constant.

Properties The thermal properties of the brick are given to be $k = 0.72 \text{ W/m} \cdot ^\circ\text{C}$ and $\alpha = 0.45 \times 10^{-4} \text{ m}^2/\text{s}$.

Analysis The exact analytical solution to this problem is

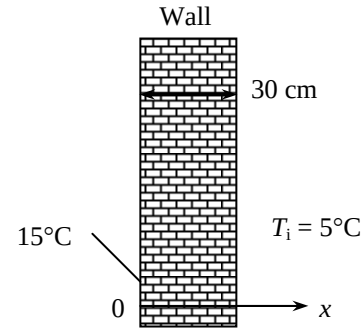
$$\frac{T(x, t) - T_i}{T_s - T_i} = \text{erfc}\left(\frac{x}{\sqrt{\alpha t}}\right)$$

Substituting,

$$\frac{5.1 - 5}{15 - 5} = 0.01 = \text{erfc}\left(\frac{0.3 \text{ m}}{2\sqrt{(0.45 \times 10^{-6} \text{ m}^2/\text{s})t}}\right)$$

Noting from Table 4-3 that $0.01 = \text{erfc}(1.8215)$, the time is determined to be

$$\left(\frac{0.3 \text{ m}}{2\sqrt{(0.45 \times 10^{-6} \text{ m}^2/\text{s})t}}\right) = 1.8215 \longrightarrow t = 15,070 \text{ s} = \mathbf{251 \text{ min}}$$



4-125 A thick wall is exposed to cold outside air. The wall temperatures at distances 15, 30, and 40 cm from the outer surface at the end of 2-hour cooling period are to be determined.

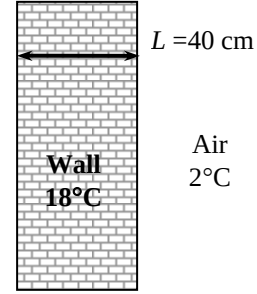
Assumptions 1 The temperature in the wall is affected by the thermal conditions at outer surfaces only. Therefore, the wall can be considered to be a semi-infinite medium 2 The thermal properties of the wall are constant.

Properties The thermal properties of the brick are given to be $k = 0.72 \text{ W/m} \cdot ^\circ\text{C}$ and $\alpha = 1.6 \times 10^{-7} \text{ m}^2/\text{s}$.

Analysis For a 15 cm distance from the outer surface, from Fig. 4-23 we have

$$\left. \begin{aligned} \frac{h\sqrt{\alpha t}}{k} &= \frac{(20 \text{ W/m}^2 \cdot ^\circ\text{C})\sqrt{(1.6 \times 10^{-7} \text{ m}^2/\text{s})(2 \times 3600 \text{ s})}}{0.72 \text{ W/m} \cdot ^\circ\text{C}} = 2.98 \\ \xi &= \frac{x}{2\sqrt{\alpha t}} = \frac{0.15 \text{ m}}{2\sqrt{(1.6 \times 10^{-7} \text{ m}^2/\text{s})(2 \times 3600 \text{ s})}} = 0.70 \end{aligned} \right\} 1 - \frac{T - T_\infty}{T_i - T_\infty} = 0.25$$

$$1 - \frac{T - 2}{18 - 2} = 0.25 \longrightarrow T = 14.0^\circ\text{C}$$



For a 30 cm distance from the outer surface, from Fig. 4-23 we have

$$\left. \begin{aligned} \frac{h\sqrt{\alpha t}}{k} &= \frac{(20 \text{ W/m}^2 \cdot ^\circ\text{C})\sqrt{(1.6 \times 10^{-7} \text{ m}^2/\text{s})(2 \times 3600 \text{ s})}}{0.72 \text{ W/m} \cdot ^\circ\text{C}} = 2.98 \\ \xi &= \frac{x}{2\sqrt{\alpha t}} = \frac{0.3 \text{ m}}{2\sqrt{(1.6 \times 10^{-7} \text{ m}^2/\text{s})(2 \times 3600 \text{ s})}} = 1.40 \end{aligned} \right\} 1 - \frac{T - T_\infty}{T_i - T_\infty} = 0.038$$

$$1 - \frac{T - 2}{18 - 2} = 0.038 \longrightarrow T = 17.4^\circ\text{C}$$

For a 40 cm distance from the outer surface, that is for the inner surface, from Fig. 4-23 we have

$$\left. \begin{aligned} \frac{h\sqrt{\alpha t}}{k} &= \frac{(20 \text{ W/m}^2 \cdot ^\circ\text{C})\sqrt{(1.6 \times 10^{-7} \text{ m}^2/\text{s})(2 \times 3600 \text{ s})}}{0.72 \text{ W/m} \cdot ^\circ\text{C}} = 2.98 \\ \xi &= \frac{x}{2\sqrt{\alpha t}} = \frac{0.4 \text{ m}}{2\sqrt{(1.6 \times 10^{-7} \text{ m}^2/\text{s})(2 \times 3600 \text{ s})}} = 1.87 \end{aligned} \right\} 1 - \frac{T - T_\infty}{T_i - T_\infty} = 0$$

$$1 - \frac{T - 2}{18 - 2} = 0 \longrightarrow T = 18.0^\circ\text{C}$$

Discussion This last result shows that the semi-infinite medium assumption is a valid one.

4-126 The engine block of a car is allowed to cool in atmospheric air. The temperatures at the center of the top surface and at the corner after a specified period of cooling are to be determined.

Assumptions 1 Heat conduction in the block is three-dimensional, and thus the temperature varies in all three directions. 2 The thermal properties of the block are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of cast iron are given to be $k = 52 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 1.7 \times 10^{-5} \text{ m}^2/\text{s}$.

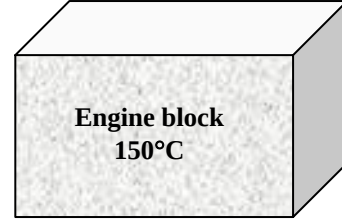
Analysis This rectangular block can physically be formed by the intersection of two infinite plane walls of thickness $2L = 40 \text{ cm}$ (call planes A and B) and an infinite plane wall of thickness $2L = 80 \text{ cm}$ (call plane C). We measure x from the center of the block.

(a) The Biot number is calculated for each of the plane wall to be

$$Bi_A = Bi_B = \frac{hL}{k} = \frac{(6 \text{ W/m}^2 \cdot ^\circ\text{C})(0.2 \text{ m})}{(52 \text{ W/m}\cdot^\circ\text{C})} = 0.0231$$

$$Bi_C = \frac{hL}{k} = \frac{(6 \text{ W/m}^2 \cdot ^\circ\text{C})(0.4 \text{ m})}{(52 \text{ W/m}\cdot^\circ\text{C})} = 0.0462$$

Air
17°C



The constants λ_1 and A_1 corresponding to these Biot numbers are, from Table 4-1,

$$\lambda_{1(A,B)} = 0.150 \quad \text{and} \quad A_{1(A,B)} = 1.0038$$

$$\lambda_{1(C)} = 0.212 \quad \text{and} \quad A_{1(C)} = 1.0076$$

The Fourier numbers are

$$\tau_{A,B} = \frac{\alpha t}{L^2} = \frac{(1.70 \times 10^{-5} \text{ m}^2/\text{s})(45 \text{ min} \times 60 \text{ s/min})}{(0.2 \text{ m})^2} = 1.1475 > 0.2$$

$$\tau_C = \frac{\alpha t}{L^2} = \frac{(1.70 \times 10^{-5} \text{ m}^2/\text{s})(45 \text{ min} \times 60 \text{ s/min})}{(0.4 \text{ m})^2} = 0.2869 > 0.2$$

The center of the top surface of the block (whose sides are 80 cm and 40 cm) is at the center of the plane wall with $2L = 80 \text{ cm}$, at the center of the plane wall with $2L = 40 \text{ cm}$, and at the surface of the plane wall with $2L = 40 \text{ cm}$. The dimensionless temperatures are

$$\theta_{o, \text{wall (A)}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.0038) e^{-(0.150)^2 (1.1475)} = 0.9782$$

$$\theta(L, t)_{\text{wall (B)}} = \frac{T(x, t) - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 L / L) = (1.0038) e^{-(0.150)^2 (1.1475)} \cos(0.150) = 0.9672$$

$$\theta_{o, \text{wall (C)}} = \frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \tau} = (1.0076) e^{-(0.212)^2 (0.2869)} = 0.9947$$

Then the center temperature of the top surface of the cylinder becomes

$$\left[\frac{T(L, 0, 0, t) - T_\infty}{T_i - T_\infty} \right]_{\text{short cylinder}} = \theta(L, t)_{\text{wall (B)}} \times \theta_{o, \text{wall (A)}} \times \theta_{o, \text{wall (C)}} = 0.9672 \times 0.9782 \times 0.9947 = 0.9411$$

$$\frac{T(L, 0, 0, t) - 17}{150 - 17} = 0.9411 \longrightarrow T(L, 0, 0, t) = 142.2^\circ\text{C}$$

(b) The corner of the block is at the surface of each plane wall. The dimensionless temperature for the surface of the plane walls with $2L = 40 \text{ cm}$ is determined in part (a). The dimensionless temperature for the surface of the plane wall with $2L = 80 \text{ cm}$ is determined from

$$\theta(L, t)_{\text{wall (C)}} = \frac{T(x, t) - T_{\infty}}{T_i - T_{\infty}} = A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 L / L) = (1.0076) e^{-(0.212)^2 (0.2869)} \cos(0.212) = 0.9724$$

Then the corner temperature of the block becomes

$$\left[\frac{T(L, L, L, t) - T_{\infty}}{T_i - T_{\infty}} \right]_{\text{short cylinder}} = \theta(L, t)_{\text{wall, C}} \times \theta(L, t)_{\text{wall, B}} \times \theta(L, t)_{\text{wall, A}} = 0.9724 \times 0.9672 \times 0.9672 = 0.9097$$

$$\frac{T(L, L, L, t) - 17}{150 - 17} = 0.9097 \longrightarrow T(L, L, L, t) = \mathbf{138.0^{\circ}\text{C}}$$

4-127 A man is found dead in a room. The time passed since his death is to be estimated.

Assumptions 1 Heat conduction in the body is two-dimensional, and thus the temperature varies in both radial r - and x - directions. 2 The thermal properties of the body are constant. 3 The heat transfer coefficient is constant and uniform over the entire surface. 4 The human body is modeled as a cylinder. 5 The Fourier number is $\tau > 0.2$ so that the one-term approximate solutions (or the transient temperature charts) are applicable (this assumption will be verified).

Properties The thermal properties of body are given to be $k = 0.62 \text{ W/m}\cdot^\circ\text{C}$ and $\alpha = 0.15 \times 10^{-6} \text{ m}^2/\text{s}$.

Analysis A short cylinder can be formed by the intersection of a long cylinder of radius $D/2 = 14 \text{ cm}$ and a plane wall of thickness $2L = 180 \text{ cm}$. We measure x from the midplane. The temperature of the body is specified at a point that is at the center of the plane wall but at the surface of the cylinder. The Biot numbers and the corresponding constants are first determined to be

$$Bi_{\text{wall}} = \frac{hL}{k} = \frac{(9 \text{ W/m}^2\cdot^\circ\text{C})(0.90 \text{ m})}{(0.62 \text{ W/m}\cdot^\circ\text{C})} = 13.06$$

$$\longrightarrow \lambda_1 = 1.4495 \quad \text{and} \quad A_1 = 1.2644$$

$$Bi_{\text{cyl}} = \frac{hr_0}{k} = \frac{(9 \text{ W/m}^2\cdot^\circ\text{C})(0.14 \text{ m})}{(0.62 \text{ W/m}\cdot^\circ\text{C})} = 2.03$$

$$\longrightarrow \lambda_1 = 1.6052 \quad \text{and} \quad A_1 = 1.3408$$

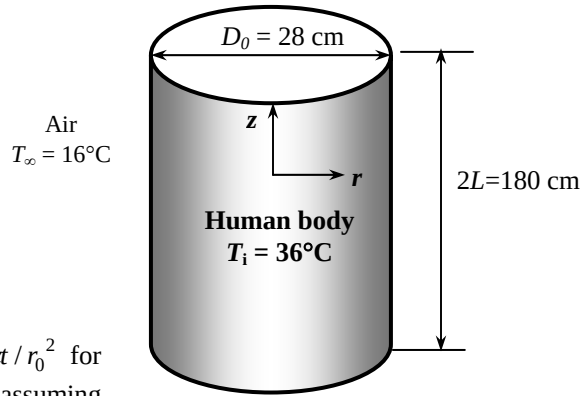
Noting that $\tau = \alpha t / L^2$ for the plane wall and $\tau = \alpha t / r_0^2$ for cylinder and $J_0(1.6052) = 0.4524$ from Table 4-2, and assuming that $\tau > 0.2$ in all dimensions so that the one-term approximate solution for transient heat conduction is applicable, the product solution method can be written for this problem as

$$\theta(0, r_0, t)_{\text{block}} = \theta(0, t)_{\text{wall}} \theta(r_0, t)_{\text{cyl}}$$

$$\frac{23-16}{36-16} = (A_1 e^{-\lambda_1^2 \tau}) \left[A_1 e^{-\lambda_1^2 \tau} J_0(\lambda_1 r / r_0) \right]$$

$$0.40 = \left\{ (1.2644) \exp \left[- (1.4495)^2 \frac{(0.15 \times 10^{-6}) t}{(0.90)^2} \right] \right\} \times \left\{ (1.3408) \exp \left[- (1.6052)^2 \frac{(0.15 \times 10^{-6}) t}{(0.14)^2} \right] (0.4524) \right\}$$

$$\longrightarrow t = 32,404 \text{ s} = \mathbf{9.0 \text{ hours}}$$



4-128 ... 4-131 Design and Essay Problems

